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Investor Presentation | Japan

Electronic Components: Expect Demand for Value-added Products to Expand Further

AI computing innovation to drive further demand expansion for high-value-added electronic components

Key Takeaways

- OW on Murata Mfg: Top Pick. Earnings expansion and utilization rates for high-value-added MLCCs.
- OW on Ibiden: Ibiden’s cutting-edge ABF package substrates remain essential to semiconductor innovation.
- OW on Meiko Electronics: Earnings growth from high-layer-count, high-density PCBs in growth markets is just beginning.
- OW on TDK: Earnings expansion for rechargeable batteries and HDD-related products.
- OW→EW on Alps Alpine (10/5): Achieving the F3/28 medium-term plan ROE target of 10% remains challenging.

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ELECTRONIC COMPONENTS

Japan

Industry View

In-Line

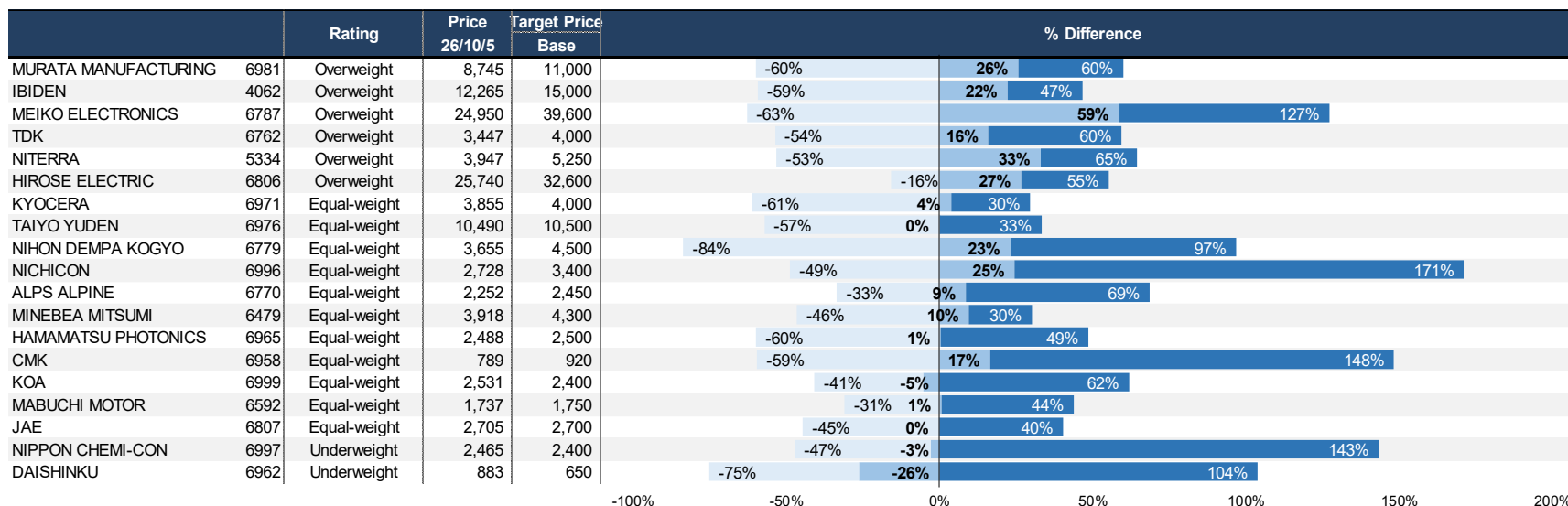
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AI Computing Innovation to Drive Further Demand Expansion for High-value-added Electronic Components

■ Risk-Reward Snapshot: Ranked in order of preference



Note: Iriso Electronics (6908) announced its F3/26 full-year results on September 30; however, this report does not include our forecasts for F3/27 and beyond, as they have not yet been updated

Source: FactSet, Morgan Stanley Research

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Electronic Components: Valuation Comparison and Earnings Forecast Comparison

		Rating	Price	Mkt Capt	Net F.Asset	EV	P/E					Ex. Cash P/E					P/B					ROE				
			26/10/5	(JPY bn)	(JPY bn)	(JPY bn)	F26e	F27e	F28e	F29e	F30e	F26e	F27e	F28e	F29e	F30e	F26e	F27e	F28e	F29e	F30e	F26e	F27e	F28e	F29e	F30e
IBIDEN	4062	Overweight	12,265	6,942.1	135.5	6,806.6	77.9	51.7	37.8	29.6	25.4	73.3	48.7	35.6	27.8	23.9	11.4	9.4	7.7	6.2	5.1	15.7%	20.1%	22.6%	23.5%	22.3%
NITERRA	5334	Overweight	3,947	1,570.2	26.8	1,543.4	12.7	11.6	9.9	8.5	7.6	12.7	12.1	10.5	9.3	8.3	1.9	1.7	1.5	1.4	1.2	15.3%	15.1%	16.1%	16.7%	17.2%
MINEBEA MITSUMI	6479	Equal-weight	3,918	1,586.5	-227.5	1,814.0	17.6	15.5	13.8	12.5	11.4	20.2	17.8	15.9	14.4	13.1	1.6	1.5	1.4	1.3	1.2	9.6%	10.1%	10.5%	10.6%	10.7%
MABUCHI MOTOR	6592	Equal-weight	1,737	447.6	146.8	300.8	17.5	15.4	16.0	14.7	13.4	12.3	11.4	11.9	10.9	9.9	1.3	1.3	1.3	1.4	1.4	7.4%	8.3%	8.2%	9.3%	10.4%
TDK	6762	Overweight	3,447	6,541.4	525.9	6,015.4	23.3	21.9	19.1	17.2	15.2	21.4	20.2	17.5	15.8	14.0	2.7	2.5	2.3	2.1	1.9	12.2%	11.9%	12.5%	12.6%	13.0%
ALPS ALPINE	6770	Equal-weight	2,252	441.4	129.0	312.4	13.4	12.0	10.6	9.0	7.8	9.5	8.5	7.5	6.4	5.6	0.9	0.9	0.8	0.8	0.7	7.2%	7.6%	8.2%	9.1%	9.8%
NIHON DEMP KOGYO	6779	Equal-weight	3,655	80.4	-16.4	96.8	24.4	18.6	15.0	12.9	11.0	28.1	21.5	17.3	14.9	12.7	2.4	2.2	2.0	1.7	1.5	10.3%	12.4%	13.8%	14.3%	14.9%
MEIKO ELECTRONICS	6787	Overweight	24,950	642.0	-92.6	734.6	23.1	17.3	13.5	10.5	8.4	26.5	19.8	15.5	12.1	9.6	4.2	3.5	2.8	2.3	1.9	19.2%	22.0%	23.1%	24.1%	24.5%
HIROSE ELECTRIC	6806	Overweight	25,740	816.0	225.5	590.5	21.7	17.5	15.9	14.4	12.9	15.3	12.7	11.7	10.8	9.9	2.2	2.1	1.9	1.8	1.7	10.2%	12.1%	12.6%	13.1%	13.5%
JAE	6807	Equal-weight	2,705	186.6	10.1	176.5	36.5	24.3	21.2	19.4	17.7	35.3	23.5	20.5	18.8	17.1	1.3	1.2	1.2	1.1	1.1	3.5%	5.1%	5.7%	6.0%	6.3%
CMK	6958	Equal-weight	789	53.2	-22.8	76.1	18.8	18.1	15.2	14.1	11.5	25.5	24.5	20.6	19.0	15.5	0.7	0.7	0.7	0.6	0.6	3.6%	3.7%	4.3%	4.6%	5.5%
DAISHINKU	6962	Underweight	883	28.1	-17.2	45.3	-	46.8	21.6	14.0	10.8	-	75.5	34.8	22.6	17.4	0.8	0.8	0.8	0.8	0.7	0.0%	1.7%	3.6%	5.5%	6.8%
HAMAMATSU PHOTONICS	6965	Equal-weight	2,488	768.7	31.0	737.7	41.3	34.3	28.3	23.1	20.2	42.2	36.0	30.2	25.1	22.5	2.3	2.3	2.3	2.3	2.2	5.6%	6.8%	8.2%	9.9%	11.0%
KYOCERA	6971	Equal-weight	3,855	5,078.1	1,797.8	3,280.3	29.8	29.1	26.2	26.7	22.0	19.6	20.7	20.4	20.8	17.7	1.7	1.7	1.6	1.6	1.5	5.1%	5.1%	5.4%	5.5%	6.2%
TAIYO YUDEN	6976	Equal-weight	10,490	1,394.5	-71.1	1,465.6	48.1	32.9	25.7	21.8	19.8	48.8	33.4	26.1	22.1	20.1	4.0	3.7	3.3	3.0	2.7	8.5%	11.6%	13.6%	14.4%	14.2%
MURATA MANUFACTURING	6981	Overweight	8,745	16,129.0	663.0	15,466.1	41.3	29.0	22.7	18.1	14.8	40.4	28.7	22.7	18.2	15.0	5.6	5.1	4.5	3.8	3.2	13.8%	18.2%	20.8%	22.6%	23.4%
NICHICON	6996	Equal-weight	2,728	184.9	-0.3	185.2	23.6	18.2	16.8	15.8	15.0	24.1	18.9	17.5	16.4	15.6	1.5	1.4	1.3	1.2	1.2	6.4%	7.8%	8.0%	8.0%	7.9%
NIPPON CHEMI-CON	6997	Underweight	2,465	68.9	-48.7	117.6	18.4	14.7	12.3	11.3	11.3	32.0	25.6	21.4	19.6	19.6	1.0	1.0	0.9	0.8	0.8	5.7%	6.7%	7.6%	7.8%	7.3%
KOA	6999	Equal-weight	2,531	86.0	-6.6	92.6	11.8	24.1	18.4	16.8	15.4	11.6	23.7	18.2	16.5	15.2	1.1	1.0	1.0	1.0	0.9	9.0%	4.4%	5.6%	5.9%	6.2%
Average (or total)				43,045.5		39,857.3	27.8	23.8	18.9	16.3	14.3	27.7	25.4	19.8	16.9	14.9	2.5	2.3	2.1	1.9	1.7	8.9%	10.0%	11.1%	11.8%	12.2%

(JPY bn, JPY/Shares)		Sales				OP				Net Profits				EPS				BPS	
		F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe
IBIDEN	4062	562.5	550.0	550.3	701.0	692.3	135.0	127.0	127.7	200.2	185.9	92.9	84.0	88.1	139.9	130.9	157.5	149.7	152.8
NITERRA	5334	793.5	790.0	801.9	816.6	850.4	142.9	150.0	151.0	172.0	181.1	121.6	105.0	130.4	127.8	131.3	311.8	267.5	306.4
MINEBEA MITSUMI	6479	1,733.9	1,690.0	1,719.4	1,780.7	1,804.4	124.6	120.0	123.7	138.2	141.4	89.6	87.0	90.1	101.8	102.9	223.1	216.6	223.4
MABUCHI MOTOR	6592	224.5	225.0	221.8	242.1	237.6	27.5	26.5	25.3	29.4	27.6	24.4	22.2	21.8	26.4	22.4	99.4	92.4	89.8
TDK	6762	3,071.8	2,580.0	2,949.2	3,360.5	3,178.9	360.4	295.0	346.9	400.7	408.0	280.6	225.0	265.8	298.1	309.7	147.8	118.5	139.2
ALPS ALPINE	6770	1,097.4	1,045.0	1,058.7	1,191.1	1,091.4	48.5	48.5	42.3	51.2	49.3	32.8	30.0	29.1	36.7	33.4	167.8	150.4	147.7
NIHON DEMP KOGYO	6779	64.5	62.5	63.0	72.7	70.1	5.5	5.2	5.2	7.5	7.4	3.4	3.2	3.3	4.5	4.5	150.1	139.6	143.3
MEIKO ELECTRONICS	6787	361.5	320.0	348.9	475.4	443.2	39.3	38.0	40.3	59.4	55.0	27.8	27.0	28.5	37.1	38.9	1,081.9	1,052.0	1,110.0
HIROSE ELECTRIC	6806	256.6	250.0	248.7	276.9	267.8	55.3	50.0	54.0	62.4	62.4	38.5	35.0	38.7	46.6	45.0	1,185.7	1,069.6	1,187.2
JAE	6807	248.9	240.0	243.0	256.0	253.4	8.4	9.5	11.4	10.7	14.4	5.0	6.0	7.7	7.5	9.8	74.1	89.0	113.7
CMK	6958	112.3	110.0	115.0	116.2	123.0	5.1	5.0	5.5	5.8	7.0	3.0	3.0	3.3	3.1	4.5	41.9	42.1	46.3
DAISHINKU	6962	43.8	41.0	41.9	46.2	43.6	1.3	1.4	1.8	2.0	2.5	0.0	0.1	0.6	0.6	1.1	-0.4	3.1	17.3
HAMAMATSU PHOTONICS	6965	236.9	212.8	212.1	245.5	234.1	23.4	18.0	16.2	26.2	21.2	17.5	14.0	14.2	20.5	17.2	60.2	46.9	47.3
KYOCERA	6971	2,078.9	2,080.0	2,108.9	2,174.9	2,197.8	165.3	160.0	160.0	171.5	184.8	167.2	160.0	161.0	158.4	173.2	129.5	122.0	125.5
TAIYO YUDEN	6976	428.7	424.0	413.8	486.0	475.8	45.1	45.0	44.9	66.1	79.3	30.1	29.0	30.0	43.9	53.7	218.3	219.0	230.3
MURATA MANUFACTURING	6981	2,194.6	2,110.0	2,155.8	2,628.3	2,549.8	490.8	430.0	466.5	702.6	677.4	383.2	338.0	368.2	539.5	533.6	211.5	185.7	202.6
NICHICON	6996	187.7	185.0	190.3	200.6	206.2	10.7	8.7	12.5	14.6	15.2	7.7	6.7	9.3	9.8	11.1	115.5	99.8	135.6
NIPPON CHEMI-CON	6997	153.8	160.0	159.7	154.9	171.7	7.5	8.0	8.5	8.6	11.5	3.7	4.0	4.6	4.6	6.5	134.1	142.2	168.3
KOA	6999	89.5	87.3	90.0	94.2	95.0	5.7	5.4	6.0	6.8	7.0	8.0	7.7	8.5	3.9	5.3	214.1	208.4	228.9
Average (or total)		13,941.4	13,162.6	13,692.3	15,319.8	14,986.4	1,702.3	1,551.2	1,649.7	2,136.0	2,138.4	1,336.8	1,186.9	1,303.0	1,610.6	1,635.1	4,723.9	4,414.4	4,815.1
					5,821.2	6,018.1													

Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep and one year behind the other companies shown; Fiscal year ends in Mar for all others; Diluted EPS for our forecasts and basic EPS for company and consensus estimates; e = Morgan Stanley Research estimates; FSe = FactSet consensus; Ce = Company estimates; Net Financial Asset is based on current FY end actuals (Net Financial Asset = Cash & cash equivalents + Short-term marketable securities + Investments in securities - Interest-bearing debt)

Source: FactSet, Company data, Morgan Stanley Research

Sales Trends (Quarterly, Semi-annual, Annual)

(JPY bn)	F25 1Q	2Q	3Q	4Q	F26 1Q	2Qe	3Qe	4Qe	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	F26Ce 1He	2He	F26Ce	F26e FSE	F27e FSE	F28e FSE	+1Q FSE	
Sales																										
USD/JPY: Average Rate	144.5	147.5	154.0	156.9	159.5	159.2	155.0	155.0	112.4	135.4	144.5	152.5	150.6	155.0	155.0	155.0	155.0	155.0								
IBIDEN	4062	97.5	98.0	103.1	117.6	123.2	135.7	147.6	156.1	401.1	417.5	370.5	369.4	416.2	562.5	701.0	864.2	1,036.8	1,169.4	253.5	296.5	550.0	550.3	692.3	869.5	131.7
NITERRA	5334	169.9	181.2	174.9	205.2	203.7	194.2	197.9	197.7	491.7	562.6	614.5	653.0	731.2	793.5	816.6	878.3	945.9	1,020.0	391.0	399.0	790.0	801.9	850.4	949.1	200.1
MINEBEA MITSUMI	6479	366.9	411.4	453.9	432.2	426.6	432.0	451.7	423.7	1,124.1	1,292.2	1,402.1	1,522.7	1,664.4	1,733.9	1,780.7	1,855.0	1,927.4	2,003.8	846.5	843.5	1,690.0	1,719.4	1,804.4	1,871.7	448.2
MABUCHI MOTOR	6592	47.0	48.0	52.1	53.4	50.4	56.0	60.4	57.8	134.6	156.7	178.7	196.2	200.4	224.5	242.1	255.2	271.4	288.6	106.4	118.6	225.0	221.8	237.6	243.4	60.3
TDK	6762	535.8	647.6	675.2	646.3	741.0	790.2	785.4	755.2	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	3,071.8	3,360.5	3,667.4	4,007.9	4,384.8	-	-	2,580.0	2,949.2	3,178.9	3,416.4	788.4
ALPS ALPINE	6770	238.9	266.8	255.5	258.3	237.9	289.5	286.0	284.0	802.9	933.1	964.1	990.4	1,019.5	1,097.4	1,191.1	1,231.6	1,311.7	1,399.1	508.0	537.0	1,045.0	1,058.7	1,091.4	1,118.1	288.0
NIHON DEMP KOGYO	6779	12.7	14.0	13.2	14.8	15.3	16.2	16.1	16.9	45.4	52.5	50.3	53.1	54.6	64.5	72.7	78.2	84.6	91.7	31.0	31.5	62.5	63.0	70.1	77.4	15.6
MEIKO ELECTRONICS	6787	53.1	58.4	60.5	68.5	73.2	87.9	97.2	103.2	151.3	167.3	179.5	206.8	240.6	361.5	475.4	555.4	646.8	750.1	-	-	320.0	348.9	443.2	526.3	76.1
HIROSE ELECTRIC	6806	49.0	53.1	54.5	54.7	62.1	64.9	65.6	64.0	163.7	183.2	165.5	189.4	211.3	256.6	276.9	298.6	322.5	348.7	125.0	125.0	250.0	248.7	267.8	288.3	64.3
JAE	6807	51.6	58.8	56.3	61.1	58.3	62.8	61.9	65.9	225.1	235.9	225.8	221.6	227.9	248.9	256.0	272.2	285.1	298.8	120.0	120.0	240.0	243.0	253.4	266.3	62.2
CMK	6958	22.8	24.4	25.8	27.2	27.2	28.0	28.5	28.6	81.5	83.8	90.6	95.5	100.2	112.3	116.2	120.7	125.9	131.6	-	-	110.0	115.0	123.0	130.0	27.8
DAISHINKU	6962	9.4	10.2	10.0	10.0	10.6	10.9	11.2	11.1	41.3	38.4	39.3	38.6	39.6	43.8	46.2	49.5	52.8	56.3	-	-	41.0	41.9	43.6	44.4	10.9
HAMAMATSU PHOTONICS	6965	50.6	56.2	48.7	56.6	51.9	60.6	60.5	63.9	169.0	208.8	221.4	204.0	212.1	236.9	245.5	265.2	287.0	311.1	112.5	126.5	239.0	235.1	252.3	272.2	64.7
KYOCERA	6971	478.0	513.3	530.6	548.2	525.4	520.0	512.5	521.0	1,838.9	2,025.3	2,004.2	2,014.5	2,070.2	2,078.9	2,174.9	2,273.1	2,418.5	2,573.4	-	-	2,080.0	2,108.9	2,197.8	2,319.3	523.2
TAIYO YUDEN	6976	84.8	92.8	88.5	89.2	93.9	110.1	113.0	111.8	349.6	319.5	322.6	341.4	355.3	428.7	486.0	558.4	630.0	686.6	-	-	424.0	413.4	474.7	551.5	107.1
MURATA MANUFACTURING	6981	416.2	486.6	467.5	460.6	502.3	574.5	565.8	552.0	1,812.5	1,686.8	1,640.2	1,743.4	1,830.9	2,194.6	2,628.3	3,104.1	3,682.2	4,348.3	1,040.0	1,070.0	2,110.0	2,155.8	2,549.8	2,982.5	548.5
NICHICON	6996	39.6	41.1	43.5	45.5	44.7	44.8	48.3	49.9	142.2	184.7	181.6	175.8	169.7	187.7	200.6	207.4	214.3	222.0	-	-	185.0	190.3	206.2	227.3	46.2
NIPPON CHEMI-CON	6997	30.9	33.4	35.9	36.7	38.0	38.4	38.5	38.9	140.3	161.9	150.7	122.7	136.8	153.8	154.9	157.5	159.3	161.6	77.0	83.0	160.0	159.7	171.7	180.1	38.0
KOA	6999	16.8	17.8	18.5	19.2	20.9	22.6	23.0	23.0	65.0	75.1	64.8	64.1	72.3	89.5	94.2	100.1	106.4	113.1	43.0	44.3	87.3	90.0	95.0	100.0	23.1
Total		2,771.4	3,113.2	3,168.3	3,205.0	3,306.4	3,539.3	3,571.1	3,524.6	10,082.4	10,966.2	10,970.4	11,407.4	12,257.9	13,941.4	15,319.8	16,792.1	18,516.7	20,359.0	-	-	13,188.8	13,714.9	15,003.5	16,433.9	3,524.5

(JPY bn)	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	
Sales																																
USD/JPY: Average Rate	110.5	124.9	121.9	113.1	107.5	113.3	116.9	114.4	100.7	92.9	85.7	79.0	82.9	100.1	109.8	120.1	108.3	110.8	110.9	108.7	106.1	112.4	135.4	144.5	152.5	150.6	155.0	155.0	155.0	155.0	155.0	
IBIDEN	4062	205.4	212.0	220.5	247.6	319.0	398.6	413.5	309.3	274.2	305.0	300.9	285.9	310.3	318.1	314.1	266.5	300.4	291.1	296.0	323.5	401.1	417.5	370.5	369.4	416.2	562.5	701.0	864.2	1,036.8	1,169.4	
NITERRA	5334	224.3	221.4	228.9	228.8	241.2	284.9	344.9	345.6	292.1	243.9	269.2	284.7	302.8	329.8	347.6	383.3	372.9	409.9	425.0	426.1	427.5	491.7	562.6	614.5	653.0	731.2	793.5	816.6	878.3	945.9	1,020.0
MINEBEA MITSUMI	6479	287.0	279.3	272.2	268.6	294.4	318.4	331.0	334.4	256.2	228.4	269.1	251.4	282.4	371.5	500.7	609.8	638.9	879.1	884.7	978.4	988.4	1,124.1	1,292.2	1,402.1	1,522.7	1,664.4	1,733.9	1,780.7	1,855.0	1,927.4	2,003.8
MABUCHI MOTOR	6592	116.2	105.1	116.4	105.7	99.3	93.9	100.5	107.6	92.6	70.4	82.8	78.9	85.3	108.4	122.5	143.1	140.7	146.9	143.1	131.8	116.4	134.6	156.7	178.7	196.2	200.4	224.5	242.1	255.2	271.4	288.6
TDK	6762	699.9	575.0	608.9	658.9	657.9	795.2	862.0	866.3	727.4	808.9	875.7	814.5	851.6	984.5	1,052.6	1,152.3	1,178.3	1,271.7	1,361.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	3,071.8	3,360.5	3,667.4	4,007.9	4,384.8
ALPS ALPINE	6770	573.1	540.3	601.8	619.7	643.6	709.6	708.1	692.7	539.0	493.6	550.7	526.5	546.4	684.4	748.6	774.0	753.3	858.3	851.3	810.8	718.0	802.9	933.1	964.1	990.4	1,019.5	1,097.4	1,191.1	1,231.6	1,311.7	1,399.1
NIHON DEMP KOGYO	6779	74.1	49.3	49.6	55.9	60.7	63.6	73.3	78.1	59.2	52.6	54.9	50.8	50.6	50.8	47.7	44.9	43.8	44.0	42.5	39.5	39.2	45.4	52.5	50.3	53.1	54.6	64.5	72.7	78.2	84.6	91.7
MEIKO ELECTRONICS	6787	23.0	17.2	21.6	29.2	40.2	50.3	65.3	78.5	75.8	69.8	74.7	63.0	60.7	79.2	90.9	95.3	95.9	108.5	118.9	115.5	119.3	151.3	167.3	179.5	206.8	240.6	361.5	475.4	555.4	646.8	750.1
HIROSE ELECTRIC	6806	83.5	61.1	67.9	79.0	89.6	104.4	115.5	118.0	89.9	84.5	92.4	94.8	95.9	125.0	125.7	120.3	115.1	125.3	124.6	121.8	133.5	163.7	183.2	165.5	189.4	211.3	256.6	276.9	298.6	322.5	348.7
JAE	6807	120.4	102.6	105.6	113.6	115.8	126.3	137.6	156.1	123.0	104.3	110.6	112.4	129.6	159.8	191.2	179.5	209.5	253.9	222.1	208.1	209.7	225.1	235.9	225.8	221.6	227.9	248.9	256.0	272.2	285.1	298.8
CMK	6958	122.3	104.5	111.7	115.4	116.0	119.2	128.2	138.5	107.6	90.1	87.0	80.6	72.7	71.0	73.7	73.1	75.4	86.9	90.2	82.6	70.0	81.5	83.8	90.6	95.5	100.2	112.3	116.2	120.7	125.9	131.6
DAISHINKU	6962	48.0	32.6	31.5	30.2	34.8	35.9	40.9	41.9	36.5	32.0	38.0	35.2	32.9	33.8	31.1	32.2	31.0	30.3	28.5	28.9	33.2	41.3	38.4	39.3	38.6	39.6	43.8	46.2	49.5	52.8	58.3
HAMAMATSU PHOTONICS	6965	51.6	62.6	60.7	60.9	70.5	73.9	87.0	96.3	98.8	72.0	91.0	101.9	98.1	102.2	112.1	120.7	121.9	130.5	144.3	145.9	160.0	208.8	221.4	204.0	212.1	236.9	245.5	265.2	287.0	311.1	
KYOCERA	6971	1,285.1	1,034.6	1,069.8	1,140.8	1,180.7	1,181.5	1,283.9	1,290.4	1,128.6	1,073.8	1,266.9	1,190.9	1,280.1	1,447.4	1,526.5	1,479.6	1,422.8	1,577.0	1,623.7	1,599.1	1,526.9	1,838.9	2,025.3	2,004.2	2,014.5	2,070.2	2,078.9	2,174.9	2,273.1	2,418.5	2,573.4
TAIYO YUDEN	6976	190.0	152.1	153.7	163.3	172.3	186.5	221.2	238.3	185.5	195.7	210.4	183.8	192.9	208.2	227.1	240.4	230.7	244.1	274.3	282.3	300.9	349.6	319.5	322.6	341.4	355.3	428.7	486.0	55		

Operating Profit Trends (Quarterly, Semi-annual, Annual)

(JPY bn)	F25 1Q	2Q	3Q	4Q	F26 1Q	2Qe	3Qe	4Qe	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	F26Ce 1He	2He	F26Ce	F26e FSE	F27e FSE	F28e FSE	+1Q FSE	
OP																										
USD/JPY:Average Rate	144.5	147.5	154.0	156.9	159.5	159.2	155.0	155.0	112.4	135.4	144.5	152.5	150.6	155.0	155.0	155.0	155.0	155.0								
IBIDEN	4062	17.6	14.9	12.0	17.5	26.9	31.0	38.3	70.8	72.4	47.6	47.6	62.0	135.0	200.2	273.8	349.8	407.9	54.5	72.5	127.0	127.7	185.9	266.8	30.0	
NITERRA	5334	33.8	38.1	38.9	27.3	41.8	24.5	40.9	75.5	89.2	107.6	129.7	138.2	142.9	172.0	198.1	224.6	253.4	72.0	78.0	150.0	151.0	181.1	213.5	30.1	
MINEBEA MITSUMI	6479	17.4	27.0	31.3	28.3	26.3	29.7	34.1	90.3	76.5	69.0	91.6	102.7	124.6	138.2	155.1	170.9	187.9	53.0	67.0	120.0	123.7	141.4	158.5	32.5	
MABUCHI MOTOR	6592	6.4	5.7	7.4	5.9	5.4	7.8	6.6	13.8	10.8	15.5	21.6	25.5	27.5	29.4	31.6	34.6	38.3	13.1	13.4	26.5	25.3	27.6	29.7	7.8	
TDK	6762	56.4	91.2	83.1	41.7	86.3	107.1	93.1	166.8	168.8	172.9	224.2	272.4	360.4	400.7	454.0	502.4	565.4	-	-	295.0	346.9	408.0	474.7	107.8	
ALPS ALPINE	6770	3.7	17.5	12.0	8.8	-0.9	10.9	13.1	25.5	35.2	33.6	19.7	34.1	42.0	48.5	51.2	58.1	68.1	78.7	9.0	39.5	48.5	42.3	49.3	55.3	10.6
NIHON DEMP KOGYO	6779	0.8	0.7	0.8	1.2	1.2	1.4	1.3	1.6	5.2	8.3	4.3	4.6	3.4	5.5	7.5	9.1	10.5	12.3	2.5	2.7	5.2	5.2	7.4	9.3	1.3
MEIKO ELECTRONICS	6787	5.6	5.8	6.1	7.1	6.1	9.6	11.2	12.4	13.3	9.6	11.7	19.1	24.6	39.3	59.4	74.6	93.5	115.2	-	-	38.0	40.3	55.0	77.5	8.6
HIROSE ELECTRIC	6806	9.8	10.6	12.0	10.5	13.3	14.4	14.7	12.9	40.8	46.8	34.0	42.7	43.0	55.3	62.4	67.8	74.0	81.1	25.5	24.5	50.0	54.0	62.4	71.6	14.3
JAE	6807	1.5	3.3	1.2	3.0	0.3	2.2	1.8	4.1	18.0	17.6	14.4	15.6	8.9	8.4	10.7	12.2	13.4	14.7	3.8	5.7	9.5	11.4	14.4	17.0	3.4
CMK	6958	0.2	0.3	0.9	1.5	0.4	1.2	1.7	1.8	3.0	2.6	3.5	3.8	2.8	5.1	5.8	6.7	7.1	8.4	-	-	5.0	5.5	7.0	8.0	1.1
DAISHINKU	6962	0.1	0.1	0.3	0.6	0.0	0.2	0.6	0.5	5.2	4.2	2.1	0.9	1.1	1.3	2.0	3.0	3.9	4.8	-	-	1.4	1.8	2.5	3.3	0.2
HAMAMATSU PHOTONICS	6965	4.3	6.5	1.5	3.9	2.4	7.6	6.3	7.0	34.3	57.0	56.7	32.1	16.2	23.4	26.2	31.8	38.8	43.8	10.0	12.0	22.0	21.8	25.6	32.1	6.7
KYOCERA	6971	18.6	23.4	28.7	47.5	49.1	39.4	39.4	148.9	128.5	92.9	27.3	118.1	165.3	171.5	184.5	212.1	249.6	-	-	160.0	160.0	184.8	216.1	27.4	
TAIYO YUDEN	6976	3.1	5.9	7.5	3.5	4.8	12.6	14.1	13.6	68.2	32.0	9.1	10.5	20.0	45.1	66.1	83.8	98.3	107.7	-	-	45.0	44.2	77.7	119.4	12.1
MURATA MANUFACTURING	6981	61.6	103.5	37.9	78.8	98.5	125.0	134.5	132.8	424.1	297.9	215.4	279.7	281.8	490.8	702.6	891.0	1,112.0	1,350.6	202.0	228.0	430.0	466.5	677.4	899.5	128.0
NICHICON	6996	0.7	0.8	2.3	2.6	2.8	2.4	2.5	3.0	6.4	12.7	8.9	5.2	6.5	10.7	14.6	15.8	16.7	17.6	-	-	8.7	12.5	15.2	17.5	2.5
NIPPON CHEMI-CON	6997	0.2	0.8	1.0	1.4	1.7	1.5	1.8	2.5	8.8	12.9	9.4	3.7	3.4	7.5	8.6	9.8	10.4	10.3	3.0	5.0	8.0	8.5	11.5	13.9	1.8
KOA	6999	0.5	0.8	1.8	0.5	0.9	1.5	1.6	1.7	5.7	10.2	3.3	1.2	3.6	5.7	6.8	8.5	9.2	9.9	2.5	2.8	5.4	6.0	7.0	8.0	2.1
Total		242.3	357.0	286.6	291.5	367.2	430.0	458.7	446.3	1,234.3	1,091.6	898.1	995.2	1,176.2	1,702.3	2,136.0	2,569.4	3,050.0	3,557.5	-	-	1,555.2	1,654.5	2,141.2	2,691.6	428.2

(JPY bn)	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	
OP																																
USD/JPY:Average Rate	110.5	124.9	121.9	113.1	107.5	113.3	116.9	114.4	100.7	92.9	85.7	79.0	82.9	100.1	109.8	120.1	108.3	110.8	110.9	108.7	106.1	112.4	135.4	144.5	152.5	150.6	155.0	155.0	155.0	155.0	155.0	
IBIDEN	4062	13.6	12.0	11.1	13.6	20.1	43.6	69.0	67.3	10.6	21.3	33.8	15.5	5.4	23.4	26.0	22.6	7.1	16.7	10.1	19.7	38.6	70.8	72.4	47.6	47.6	62.0	135.0	200.2	273.8	349.8	407.9
NITERRA	5334	25.5	11.8	15.8	20.7	26.1	41.5	52.4	35.0	-5.2	10.7	28.8	24.5	23.8	51.7	62.2	66.3	53.6	67.3	58.7	48.4	47.4	75.5	89.2	107.6	129.7	138.2	142.9	172.0	198.1	224.6	253.4
MINEBEA MITSUMI	6479	33.0	22.0	19.4	18.1	14.1	19.3	26.3	30.8	13.4	12.1	22.2	8.6	10.2	32.2	60.1	51.4	49.0	79.2	72.1	58.9	59.8	90.3	76.5	69.0	91.6	102.7	124.6	138.2	155.1	170.9	187.9
MABUCHI MOTOR	6592	28.6	22.1	29.1	24.6	17.3	8.1	10.7	11.7	5.7	3.0	6.6	3.1	5.1	9.3	16.9	23.0	24.2	24.1	21.2	17.5	12.9	13.8	10.8	15.5	21.6	25.5	27.5	29.4	31.6	34.6	38.3
TDK	6762	56.3	-43.7	22.1	54.3	59.8	60.5	79.6	87.2	-54.3	25.8	63.9	18.7	21.6	36.6	72.5	93.4	208.7	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	360.4	400.7	454.0	502.4	565.4
ALPS ALPINE	6770	24.6	13.2	41.8	36.2	31.1	45.5	22.1	19.9	-26.5	4.7	28.9	15.2	6.9	28.5	53.5	52.3	44.4	71.9	49.6	26.8	13.1	35.2	33.6	19.7	34.1	42.0	48.5	51.2	58.1	68.1	78.7
NIHON DEMP KOGYO	6779	12.6	0.5	0.1	2.5	4.5	5.3	9.4	9.6	-6.9	4.0	2.1	0.1	0.2	0.2	0.4	0.7	-9.6	0.4	-8.3	2.8	5.2	8.3	4.3	4.6	3.4	5.5	7.5	9.1	10.5	12.3	
MEIKO ELECTRONICS	6787	1.3	-0.9	1.7	3.5	5.0	6.5	5.4	7.3	1.8	3.8	3.3	0.8	-0.6	0.9	-2.9	3.3	5.8	7.5	8.9	5.2	6.7	13.3	9.6	11.7	19.1	24.6	39.3	59.4	74.6	93.5	115.2
HIROSE ELECTRIC	6806	26.0	14.4	20.1	26.0	29.7	36.9	38.0	35.4	19.2	20.4	22.0	19.6	21.0	33.0	32.6	28.8	28.5	28.6	23.2	20.4	27.9	40.8	46.8	34.0	42.7	43.0	55.3	62.4	67.8	74.0	81.1
JAE	6807	10.6	5.2	6.6	10.1	7.9	11.0	11.7	14.4	-0.8	2.9	6.3	6.5	8.6	16.5	25.9	17.9	12.1	20.6	17.3	14.2	8.7	18.0	17.6	14.4	15.6	8.9	8.4	10.7	12.2	13.4	14.7
CMK	6958	9.3	-1.1	0.7	9.2	10.4	9.6	6.0	4.8	-2.6	-0.4	-1.0	0.8	0.4	-1.1	0.6	0.4	2.7	4.3	3.8	1.6	-1.7	3.0	2.6	3.5	3.8	2.8	5.1	5.8	6.7	7.1	8.4
DAISHINKU	6962	3.6	-3.5	-2.7	1.7	1.6	2.5	3.8	4.5	-0.4	1.2	1.4	0.3	0.4	0.8	-2.3	0.7	1.4	0.3	0.0	0.3	2.1	5.2	4.2	2.1	0.9	1.1	1.3	2.0	3.0	3.9	4.8
HAMAMATSU PHOTONICS	6965	5.9	8.7	5.9	4.1	9.7	11.7	15.7	19.9	20.1	3.7	15.8	21.8	17.6	16.8	21.7	23.6	20.5	22.8	27.3	25.4	21.8	34.3	57.0	56.7	32.1	16.2	23.4	26.2	31.8	38.8	43.8
KYOCERA	6971	207.2	51.6	83.4	109.0	101.0	103.2	135.1	152.4	43.4	63.9	155.9	97.7	76.9	120.6	93.4	92.7	104.5	95.6	94.8	100.2	70.6	148.9	128.5	92.9	27.3	118.1	165.3	171.5	184.5	212.1	249.6
TAIYO YUDEN	6976	35.4	6.2	8.2	11.0	2.1	6.7	22.0	21.3	-12.8	4.2	8.8	-8.0	5.0	11.4	13.2	23.4	12.4	20.2	35.2	37.2	40.8	68.2	32.0	9.1	10.5	20.0	45.1	66.1	83.8	98.3	107.7
MURATA MANUFACTURING	6981	174.2	51.0	59.2	74.2	69.5	89.8	113.4	115.8	-16.3	26.7	77.5	45.0	58.6	125.9	214.5	275.4	201.2	162.1	266.8	253.2	313.2	424.1	297.9	215.4	279.7	281.8	490.8	702.6	891.0	1,112.0	1,350.6
NICHICON	6996	10.3	5.1	7.7	4.6	6.3	6.0	6.7	4.6	-7.9	-4.5	5.1	2.1	-3.4	4.2	3.9	4.8	3.0	6.2	5.5	2.5	1.6	6.4	12.7	8.9	5.2	6.5	10.7	14.6	15.8	16.7	17.6
NIPPON CHEMI-CON	6997	13.6	-1.0	3.5	3.6	4.7	6.1	9.4	8.7	-6.9	-3.0	8.2	-2.6	-7.																		

Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep and one year behind the other companies shown; Fiscal year ends in Mar for all others; e = Morgan Stanley Research estimates; FSE = FactSet consensus; Ce = Company estimate
Source: Company data, FactSet, Morgan Stanley Research

Capex

	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	F26Ce	
(JPY bn)																																	
Capex																																	
IBIDEN	4062	26.2	19.9	16.9	12.5	23.9	65.8	70.4	42.4	61.8	18.7	60.3	47.3	43.3	37.7	56.4	41.0	21.0	22.4	22.9	58.2	79.4	61.7	131.3	146.6	157.3	64.3	157.6	158.0	208.0	208.0	208.0	210.0
NITERRA	5334	23.5	25.5	10.8	10.4	14.0	26.9	29.3	63.2	24.2	11.0	8.8	14.0	24.0	41.0	36.4	45.3	36.3	37.6	56.6	51.0	33.4	29.6	31.3	41.2	37.5	49.0	47.2	45.0	45.0	45.0	47.3	
MINEBEA MITSUMI	6479	39.9	26.2	16.4	18.8	31.1	21.9	17.0	24.9	19.6	10.8	27.0	26.9	42.8	19.8	35.0	41.6	29.8	53.2	53.1	48.8	44.2	71.9	144.8	81.4	88.7	96.5	111.9	106.8	111.3	115.6	120.2	95.0
MABUCHI MOTOR	6592	4.6	5.1	6.3	5.7	11.6	5.5	3.6	3.7	3.8	3.7	3.2	6.5	6.8	5.8	9.6	16.2	11.2	12.8	16.3	17.9	7.7	9.6	10.6	12.8	14.1	10.3	13.6	15.0	15.0	15.0	13.6	
TDK	6762	99.5	58.8	41.5	44.9	61.0	73.9	70.4	84.3	98.4	64.4	78.6	99.7	85.6	68.6	102.5	160.7	167.6	178.6	173.6	173.4	212.4	291.3	275.7	218.6	225.3	298.6	364.9	350.0	350.0	350.0	370.0	
ALPS ALPINE	6770	31.4	31.8	34.9	47.1	47.1	50.1	45.3	43.2	47.2	18.5	23.5	34.0	31.8	26.6	31.4	41.2	47.7	76.2	52.9	42.4	40.4	53.0	50.8	55.7	51.8	61.6	79.2	60.0	60.0	60.0	75.2	
NIHON DEMP A KOGYO	6779	10.8	3.6	3.5	6.8	11.9	6.9	9.8	13.2	10.9	2.0	3.4	4.9	4.7	2.7	2.0	2.1	6.8	7.1	2.4	3.1	3.2	3.2	4.0	4.6	4.6	3.9	10.0	9.0	9.8	10.6	11.5	-
MEIKO ELECTRONICS	6787	3.5	3.2	2.0	4.2	10.0	10.1	16.6	11.2	13.7	7.0	17.4	9.6	6.6	6.6	4.1	3.1	3.1	9.6	15.8	11.8	10.7	14.4	17.4	19.9	29.8	-	70.0	70.0	70.0	74.4	78.8	70.0
HIROSE ELECTRIC	6806	7.5	4.3	3.5	6.6	5.5	9.0	13.0	10.2	7.3	7.2	9.0	11.3	11.4	11.3	14.6	13.3	15.2	19.6	21.3	18.2	14.6	15.2	19.6	32.8	21.2	19.9	28.2	30.5	32.8	35.5	38.4	25.0
JAE	6807	10.5	7.6	9.1	9.1	11.5	15.2	16.5	17.8	13.8	10.1	10.7	13.5	26.6	22.0	22.2	22.4	21.7	17.6	22.2	24.5	18.2	19.0	22.6	20.4	18.0	23.7	21.0	21.0	21.0	21.4	22.4	21.0
CMK	6958	9.7	10.1	8.5	2.9	4.6	6.2	12.3	14.7	8.0	2.4	1.9	1.8	3.5	2.6	5.0	3.1	4.6	7.0	7.5	8.3	3.7	6.1	6.7	16.0	18.9	9.2	6.3	10.0	10.7	11.2	11.7	6.3
DAISHINKU	6962	3.9	3.4	2.3	1.6	1.5	3.4	4.7	2.6	3.3	2.0	4.9	2.9	1.9	2.6	3.5	3.3	3.6	2.9	2.5	4.0	4.4	7.1	5.9	3.6	7.5	8.2	4.0	4.6	5.0	5.3	5.6	4.8
HAMAMATSU PHOTONICS	6965	9.7	13.7	8.3	6.2	5.8	9.8	11.7	9.7	8.6	9.4	9.2	7.3	8.6	9.2	15.5	14.3	9.3	13.6	14.2	17.4	20.3	13.0	20.4	31.2	30.9	34.8	33.7	15.0	15.0	15.0	15.0	-
KYOCERA	6971	105.9	54.6	40.6	54.9	63.2	90.3	69.9	85.1	63.1	37.9	70.7	66.4	56.7	56.6	56.7	68.9	67.8	86.5	117.0	106.0	117.1	151.8	173.9	161.7	141.9	149.1	222.2	200.0	200.0	160.0	160.0	225.0
TAIYO YUDEN	6976	54.0	16.8	7.4	14.4	22.5	15.8	30.2	44.6	27.0	9.4	17.5	29.1	19.7	16.9	18.8	37.4	31.6	26.5	42.6	44.1	43.9	51.6	63.3	79.9	64.2	40.4	42.0	60.0	60.0	60.0	42.0	
MURATA MANUFACTURING	6981	107.8	34.6	18.2	33.1	48.0	51.0	99.7	125.6	65.4	22.9	56.8	68.4	77.7	68.2	101.2	172.5	158.6	306.6	291.6	281.6	196.7	152.8	208.1	219.5	180.5	247.8	253.0	250.0	250.0	300.0	320.0	255.0
NICHICON	6996	25.9	16.0	8.1	2.6	12.4	11.0	11.9	9.4	7.5	2.2	6.6	15.4	5.4	2.3	2.4	2.6	7.5	6.8	8.6	7.1	6.5	10.1	11.2	16.2	11.1	7.6	12.0	13.0	13.5	13.9	14.4	10.0
NIPPON CHEMI-CON	6997	14.6	11.0	9.4	6.1	7.6	8.4	14.6	14.6	11.9	4.0	9.6	13.5	6.0	3.1	5.2	4.4	4.6	7.5	9.6	5.6	3.5	5.9	7.7	11.2	7.6	5.9	5.1	5.6	5.7	5.8	5.9	5.7
KOA	6999	1.4	2.8	1.9	1.4	1.3	2.8	2.3	2.9	2.7	0.9	2.7	2.4	4.7	3.1	2.3	3.7	2.3	5.1	6.2	4.0	2.2	6.9	11.9	19.1	24.2	5.9	4.1	6.6	7.0	7.4	7.9	4.1
Total	590.1	349.1	249.7	289.3	394.4	484.1	549.1	623.2	498.3	244.3	421.8	474.9	467.8	406.7	524.8	697.1	650.3	897.2	936.7	927.6	862.5	974.1	1,217.2	1,192.3	1,135.1	1,136.7	1,486.0	1,430.1	1,489.8	1,514.1	1,549.8	-	
% of sales																																	
IBIDEN	4062	13%	9%	8%	6%	10%	21%	18%	10%	20%	7%	20%	16%	15%	12%	18%	13%	8%	7%	8%	20%	25%	15%	31%	40%	43%	15%	28%	23%	24%	20%	18%	50%
NITERRA	5334	10%	12%	5%	5%	6%	9%	8%	18%	8%	5%	3%	5%	8%	12%	10%	12%	10%	9%	13%	12%	8%	6%	7%	6%	7%	6%	6%	6%	5%	5%	4%	6%
MINEBEA MITSUMI	6479	14%	9%	6%	7%	11%	7%	5%	7%	8%	5%	10%	11%	15%	5%	7%	7%	5%	6%	6%	5%	4%	6%	11%	6%	6%	6%	6%	6%	6%	6%	6%	6%
MABUCHI MOTOR	6592	4%	5%	5%	5%	12%	6%	4%	3%	4%	5%	4%	8%	8%	5%	8%	11%	8%	9%	11%	14%	7%	7%	7%	7%	7%	5%	6%	6%	6%	5%	7%	
TDK	6762	14%	10%	7%	7%	9%	9%	8%	10%	14%	8%	9%	12%	10%	7%	9%	14%	14%	14%	13%	13%	14%	15%	13%	10%	10%	12%	12%	10%	10%	9%	8%	15%
ALPS ALPINE	6770	5%	6%	6%	8%	7%	7%	6%	6%	9%	4%	4%	6%	6%	4%	4%	5%	6%	9%	6%	5%	6%	7%	5%	6%	5%	6%	7%	5%	5%	4%	7%	
NIHON DEMP A KOGYO	6779	15%	7%	7%	12%	20%	11%	13%	17%	18%	4%	6%	10%	9%	5%	4%	5%	15%	16%	6%	8%	8%	7%	8%	9%	9%	7%	15%	12%	13%	13%	13%	-
MEIKO ELECTRONICS	6787	-	-	-	14%	25%	20%	25%	14%	18%	10%	23%	15%	11%	8%	5%	3%	3%	9%	13%	10%	9%	10%	10%	11%	14%	-	19%	15%	13%	12%	11%	29%
HIROSE ELECTRIC	6806	9%	7%	5%	8%	6%	9%	11%	9%	8%	9%	10%	12%	12%	9%	12%	11%	13%	16%	17%	15%	11%	9%	11%	20%	11%	9%	11%	11%	11%	11%	12%	11%
JAE	6807	9%	7%	9%	8%	10%	12%	12%	11%	11%	10%	10%	12%	21%	14%	12%	12%	10%	7%	10%	12%	9%	8%	10%	9%	8%	10%	8%	8%	8%	8%	7%	9%
CMK	6958	8%	10%	8%	2%	4%	5%	10%	11%	7%	3%	2%	2%	5%	4%	7%	4%	6%	8%	8%	10%	5%	8%	8%	18%	20%	9%	6%	9%	9%	9%	9%	6%
DAISHINKU	6962	8%	10%	7%	5%	4%	10%	11%	6%	9%	6%	13%	8%	6%	8%	11%	10%	12%	10%	9%	14%	13%	17%	15%	9%	19%	21%	9%	10%	10%	10%	10%	12%
HAMAMATSU PHOTONICS	6965	19%	22%	14%	10%	8%	13%	13%	10%	9%	13%	10%	7%	9%	9%	14%	12%	8%	10%	10%	12%	15%	8%	10%	14%	15%	16%	14%	6%	6%	5%	5%	-
KYOCERA	6971	8%	5%	4%	5%	5%	5%	7%	6%	4%	6%	6%	4%	4%	4%	5%	5%	5%	7%	7%	8%	8%	9%	8%	7%	7%	11%	9%	9%	7%	6%	11%	
TAIYO YUDEN	6976	28%	11%	5%	9%	13%	8%	14%	19%	15%	5%	8%	16%	10%	8%	8%	16%	14%	11%	16%	16%	15%	15%	20%	25%	19%	11%	10%	12%	11%	10%	9%	12%
MURATA MANUFACTURING	6981	18%	9%	5%	8%	11%	10%	18%	20%	12%	4%	9%	12%	11%	8%	10%	14%	22%	19%	18%	12%	8%	12%	13%	10%	14%	12%	10%	8%	8%	7%	14%	
NICHICON	6996	21%	15%	7%	3%	12%	10%	10%	8%	8%	3%	6%	14%	6%	2%	2%	2%	7%	6%	7%	6%	6%	7%	6%	9%	6%	4%	6%	6%	7%	6%	6%	6%
NIPPON CHEMI-CON	6997	11%	12%	10%	6%	7%	7%	11%	10%	10%	4%	8%	13%	6%	3%	4%	4%	4%	6%	7%	5%	3%	4%	5%	7%	6%	4%	3%	4%	4%	4%	4%	4%
KOA	6999	2%	5%	3%	2%	3%	6%	4%	6%	7%	3%	6%	6%	12%	7%	5%	8%	5%	10%	11%	8%	4%	11%	16%	29%	38%	8%	5%	7%	7%	7%	6%	6%
Total	12%	8%	6%	6%	8%	9%	10%	10%	10%	5%	8%	9%	9%	7%	8%	10%	9%	11%	11%	11%	10%	10%	11%	11%	10%	9%	11%	9%	9%	8%	8%	8%	-

Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep and one year behind the other companies shown; Fiscal year ends in Mar for all others; e = Morgan Stanley Research estimates; Ce = Company estimates
Source: Company data, FactSet, Morgan Stanley Research

Depreciation

	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	F26Ce		
(JPY bn)																																		
Depreciation																																		
IBIDEN	4062	14.6	17.5	16.6	13.8	14.4	22.3	31.5	39.2	43.5	38.8	40.7	42.4	42.7	35.7	39.4	44.1	33.1	24.6	25.1	24.2	35.4	52.7	54.9	46.0	54.2	66.2	81.4	95.5	110.5	130.5	130.5	81.0	
NITERRA	5334	18.1	20.0	18.5	15.9	14.5	18.0	18.9	25.5	31.8	18.8	16.6	15.4	12.8	11.6	13.3	16.8	19.6	20.9	24.0	32.0	34.7	37.9	39.2	39.8	39.7	45.0	42.1	43.0	43.0	43.0	42.1		
MINEBEA MITSUMI	6479	22.1	24.4	22.8	21.7	22.5	24.0	24.6	26.4	24.0	21.1	20.8	19.6	20.8	23.7	28.8	34.8	28.2	31.6	36.4	46.2	48.6	45.2	53.1	58.4	66.2	70.4	75.2	77.3	80.5	83.6	87.0	75.0	
MABUCHI MOTOR	6592	6.4	6.6	6.3	5.6	5.2	5.7	5.4	4.9	4.4	4.0	3.6	3.3	3.6	4.6	5.1	6.6	6.4	7.4	8.0	8.6	8.9	10.0	11.9	12.8	14.2	13.5	14.9	14.0	14.0	14.0	14.9		
TDK	6762	63.6	61.9	57.8	51.2	52.8	58.5	65.3	71.3	89.6	83.8	77.6	80.2	77.9	83.1	80.2	83.2	87.5	92.2	106.6	125.0	140.3	177.0	206.3	190.5	196.2	204.2	244.3	260.0	280.0	280.0	240.0		
ALPS ALPINE	6770	31.7	32.5	30.0	32.7	40.9	36.9	39.2	37.8	29.9	21.2	18.7	19.8	21.6	24.5	28.0	30.7	33.1	36.0	44.2	46.1	41.3	45.7	46.8	41.5	35.1	34.0	43.0	42.0	42.0	42.0	44.8		
NIHON DEMPMA KOGYO	6779	5.7	5.9	5.9	6.2	6.3	7.1	7.5	7.8	9.0	3.1	3.2	3.0	3.4	3.9	3.8	3.6	3.6	4.1	3.5	3.7	3.1	3.0	3.1	3.3	3.4	3.9	4.4	4.9	5.6	6.3	7.1	4.4	
MEIKO ELECTRONICS	6787	1.0	1.0	1.1	1.4	2.1	3.2	4.7	6.0	7.1	6.8	6.5	6.4	6.3	6.2	7.2	6.5	5.5	5.8	6.3	6.6	7.4	8.0	9.8	11.2	12.5	13.7	19.0	29.0	39.0	46.8	55.4	19.0	
HIROSE ELECTRIC	6806	5.1	5.3	4.3	5.3	6.2	6.9	8.1	10.2	10.7	8.3	7.5	8.9	8.7	11.4	11.4	11.6	8.1	10.8	13.1	15.8	16.8	16.4	16.7	16.8	17.9	19.5	21.0	23.6	26.1	28.6	31.2	18.0	
JAE	6807	6.0	6.4	6.0	6.2	6.6	7.7	8.4	11.2	13.0	10.6	10.0	11.6	14.7	18.9	19.9	17.7	20.3	20.8	19.6	21.1	21.2	20.4	20.6	20.6	20.9	19.7	20.5	20.7	20.8	20.9	21.1	22.0	
CMK	6958	10.3	10.1	9.5	8.2	6.8	6.3	6.2	8.1	9.2	6.8	6.3	5.3	4.1	4.7	5.0	4.9	4.1	4.3	4.7	4.5	4.2	4.8	5.3	5.9	6.5	9.0	9.5	10.0	10.6	11.1	8.5		
DAISHINKU	6962	4.0	3.9	3.4	3.1	2.8	2.7	3.1	3.4	3.6	3.2	3.7	3.9	3.6	3.5	3.6	2.5	2.7	2.9	2.9	2.7	3.0	3.6	4.1	4.0	4.1	4.0	4.1	4.3	5.0	5.2	5.5	6.0	5.2
HAMAMATSU PHOTONICS	6965	5.2	6.6	7.1	7.1	7.1	6.7	7.1	8.0	9.2	9.7	8.7	8.1	8.7	8.7	8.0	9.5	10.5	10.0	11.3	12.1	12.7	13.6	13.5	14.5	16.7	18.9	23.4	27.7	26.7	25.4	24.1	-	
KYOCERA	6971	74.2	88.5	75.3	70.3	65.9	73.2	82.2	87.0	97.6	72.8	71.5	73.1	73.6	76.2	73.0	76.6	77.4	82.8	63.9	92.7	109.1	129.0	149.6	157.0	160.9	158.2	120.0	130.0	135.0	120.0	120.0	120.0	
TAIYO YUDEN	6976	17.1	20.7	20.3	18.3	18.0	17.1	18.4	23.3	27.9	23.9	19.3	19.3	19.8	20.8	21.8	23.8	24.9	25.6	26.5	27.0	29.3	31.3	34.9	39.4	46.3	49.1	50.0	52.0	52.0	52.0	50.0		
MURATA MANUFACTURING	6981	54.0	56.5	50.8	44.6	42.4	45.1	49.8	65.1	81.0	69.9	61.8	61.0	72.3	76.9	84.9	99.1	113.5	141.6	124.4	140.3	143.1	155.6	161.3	175.9	173.3	178.2	180.6	190.0	200.0	220.0	240.0	182.0	
NICHICON	6996	13.0	14.6	13.2	9.7	8.8	8.9	9.4	10.5	11.1	8.4	7.7	9.2	7.6	5.1	4.3	4.4	3.4	3.1	4.5	5.3	5.2	6.1	7.5	6.9	8.5	8.1	8.5	9.4	10.3	11.1	11.9	8.5	
NIPPON CHEMI-CON	6997	8.8	8.6	8.8	8.2	7.2	7.5	8.8	11.1	11.6	9.5	9.1	9.2	9.5	8.6	7.9	7.6	6.7	6.6	7.0	7.7	6.0	6.3	6.3	6.8	7.1	7.3	7.2	6.9	6.9	6.7	6.7	6.6	
KOA	6999	2.0	2.5	2.8	2.2	1.9	2.0	2.3	2.4	2.6	2.4	2.0	1.9	1.9	2.1	2.2	2.4	2.3	2.5	2.9	3.4	3.6	3.5	4.1	4.9	5.9	7.1	7.2	7.3	7.5	7.7	7.9	7.2	
Total	363.0	393.6	360.6	331.8	332.5	360.0	400.8	459.4	516.7	423.3	395.5	401.6	413.5	430.1	447.8	486.3	491.0	533.4	534.5	625.2	674.3	769.4	848.5	855.6	889.1	927.9	976.8	1,048.0	1,115.4	1,154.9	1,190.9	-		
% of sales																																		
IBIDEN	4062	7%	8%	8%	6%	6%	7%	8%	9%	14%	14%	13%	14%	15%	12%	12%	14%	12%	8%	9%	8%	11%	13%	13%	12%	15%	16%	14%	14%	13%	13%	11%	19%	
NITERRA	5334	8%	9%	8%	8%	7%	6%	5%	7%	11%	8%	6%	5%	4%	4%	4%	5%	5%	6%	7%	8%	8%	7%	6%	7%	6%	6%	6%	5%	5%	4%	6%		
MINEBEA MITSUMI	6479	8%	9%	8%	8%	8%	8%	7%	8%	9%	9%	8%	8%	7%	6%	6%	6%	4%	4%	4%	5%	5%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	5%	
MABUCHI MOTOR	6592	6%	6%	5%	5%	5%	6%	5%	5%	5%	6%	4%	4%	4%	4%	5%	5%	5%	6%	7%	8%	7%	8%	7%	7%	7%	7%	7%	6%	5%	5%	5%	7%	
TDK	6762	9%	11%	9%	8%	8%	7%	8%	8%	12%	10%	9%	10%	9%	8%	7%	7%	7%	8%	9%	9%	9%	9%	9%	9%	9%	9%	8%	8%	8%	7%	6%	10%	
ALPS ALPINE	6770	6%	6%	5%	5%	6%	5%	6%	5%	6%	4%	3%	4%	4%	4%	4%	4%	4%	5%	6%	6%	6%	6%	5%	4%	4%	3%	4%	3%	3%	3%	4%		
NIHON DEMPMA KOGYO	6779	8%	12%	12%	11%	10%	11%	10%	10%	15%	6%	6%	6%	7%	8%	8%	8%	8%	9%	8%	9%	8%	9%	8%	7%	6%	7%	7%	7%	7%	8%	8%		
MEIKO ELECTRONICS	6787	4%	6%	5%	5%	5%	6%	7%	8%	9%	10%	9%	10%	10%	8%	8%	7%	6%	5%	6%	6%	6%	6%	6%	6%	6%	5%	6%	7%	7%	7%	8%		
HIROSE ELECTRIC	6806	6%	6%	6%	7%	7%	7%	7%	9%	12%	10%	8%	9%	9%	9%	10%	7%	9%	11%	13%	13%	10%	9%	10%	9%	9%	9%	9%	9%	9%	9%	9%	9%	
JAE	6807	5%	6%	6%	5%	6%	6%	6%	7%	11%	10%	9%	10%	11%	12%	10%	10%	10%	8%	9%	10%	10%	9%	9%	9%	9%	9%	8%	8%	8%	7%	7%	10%	
CMK	6958	8%	10%	8%	7%	6%	5%	5%	6%	9%	8%	7%	7%	6%	7%	7%	5%	5%	5%	6%	6%	5%	6%	6%	6%	7%	8%	8%	8%	8%	8%	8%	8%	
DAISHINKU	6962	8%	12%	11%	10%	8%	8%	7%	8%	10%	10%	10%	11%	11%	10%	12%	8%	9%	10%	10%	9%	9%	9%	11%	10%	11%	11%	11%	11%	11%	11%	13%		
HAMAMATSU PHOTONICS	6965	10%	11%	12%	12%	10%	9%	8%	8%	9%	14%	10%	8%	8%	7%	8%	9%	8%	8%	8%	8%	8%	8%	6%	7%	8%	9%	10%	11%	10%	9%	8%	-	
KYOCERA	6971	6%	9%	7%	6%	6%	6%	6%	7%	7%	6%	6%	6%	6%	5%	5%	5%	5%	4%	6%	7%	7%	7%	7%	8%	8%	8%	8%	6%	6%	5%	5%	6%	
TAIYO YUDEN	6976	9%	14%	13%	11%	10%	9%	8%	10%	15%	12%	9%	10%	10%	10%	10%	11%	10%	10%	10%	10%	9%	11%	12%	14%	14%	12%	11%	9%	8%	8%	8%	14%	
MURATA MANUFACTURING	6981	9%	14%	13%	11%	10%	9%	9%	10%	15%	13%	10%	11%	9%	8%	8%	10%	10%	8%	9%	9%	9%	10%	11%	10%	10%	10%	10%	7%	6%	6%	6%	10%	
NICHICON	6996	11%	14%	12%	10%	8%	8%	8%	9%	12%	10%	7%	9%	8%	5%	4%	4%	3%	3%	4%	4%	5%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	5%	
NIPPON CHEMI-CON	6997	7%	9%	9%	8%	7%	6%	7%	8%	10%	9%	7%	9%	10%	8%	6%	6%	5%	5%	7%	5%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	5%	
KOA	6999	3%	5%	5%	4%	4%	4%	4%	5%	6%	7%	5%	5%	5%	5%	5%	5%	5%	5%	5%	7%	7%	5%	5%	8%	9%	10%	8%	7%	7%	7%	10%		
Total	7.3%	9.4%	8.3%	7%	7%	7%	7%	8%	11%	9%	7%	8%	8%	7%	7%	7%	7%	7%	6%	7%	8%	8%	8%	8%	8%	8%	7%	7%	7%	6%	6%	-		

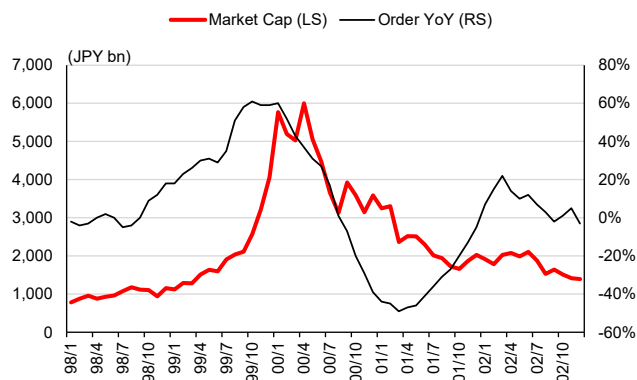
Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep and one year behind the other companies shown; Fiscal year ends in Mar for all others; e = Morgan Stanley Research estimates; Ce = Company estimates
Source: Company data, FactSet, Morgan Stanley Research

Inventory and Inventory Days

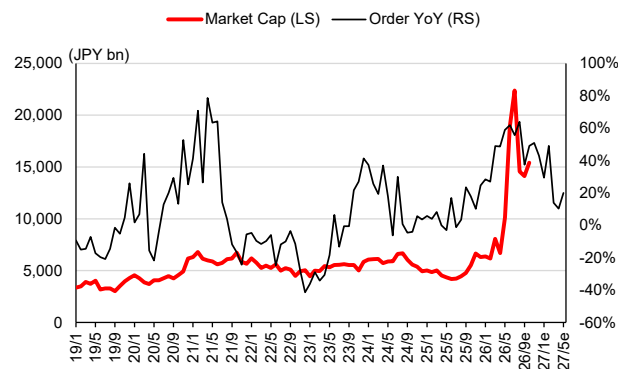
(JPY bn)	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	
Inventory																																
IBIDEN	4062	20.3	16.7	15.4	20.0	20.9	32.0	37.4	42.2	31.8	25.1	29.8	35.8	38.7	38.6	38.0	37.1	33.6	40.0	48.8	45.3	57.0	74.6	79.2	65.3	67.2	71.0	73.2	83.1	95.7	108.7	117.5
NITERRA	5334	51.7	51.1	46.0	42.2	45.4	51.1	65.4	76.5	52.7	59.7	62.1	67.8	71.1	78.4	82.1	86.9	95.0	113.1	116.7	126.8	156.8	192.3	189.6	183.9	217.7	231.1	213.8	220.9	229.2	238.0	
MINEBEA MITSUMI	6479	52.8	49.9	43.2	41.5	47.0	48.9	45.9	42.4	38.7	35.9	44.0	49.0	58.2	63.7	52.2	103.0	120.4	150.1	141.4	169.1	171.4	219.3	263.1	294.9	350.9	391.3	387.3	392.6	407.9	423.0	438.9
MABUCHI MOTOR	6592	26.5	23.5	17.6	14.7	16.4	17.1	17.8	18.0	17.1	13.5	16.1	18.6	20.5	24.2	29.0	31.0	26.7	33.0	35.7	33.9	32.5	50.9	61.4	60.4	59.8	57.8	63.9	65.4	64.4	63.5	62.0
TDK	6762	116.4	91.1	73.9	77.3	74.9	89.0	89.8	88.8	105.7	105.1	121.7	137.2	137.9	136.4	151.0	157.1	154.5	207.5	226.9	236.5	288.9	437.0	443.0	406.1	410.0	585.4	521.4	542.1	560.5	613.7	629.8
ALPS ALPINE	6770	93.1	79.8	82.3	78.6	85.0	83.7	77.4	72.5	52.1	48.1	56.6	69.1	74.3	73.1	85.4	87.5	89.6	96.1	95.8	100.5	95.8	133.4	165.4	163.7	138.6	138.3	148.4	160.3	164.9	174.6	185.2
NIHON DENPA KOGYO	6779	11.6	6.8	6.9	6.6	7.3	8.6	8.5	9.8	4.8	9.4	9.0	11.7	11.3	11.0	12.5	10.6	12.2	13.7	14.4	12.9	10.1	11.8	12.8	11.1	10.5	12.4	12.7	13.9	14.6	15.3	16.1
MEIKO ELECTRONICS	6787	3.1	2.8	4.1	4.6	4.3	5.9	8.8	9.8	8.6	8.9	11.3	9.1	10.5	11.6	13.7	11.6	12.5	14.3	16.0	19.0	22.2	29.8	32.6	35.6	38.5	49.5	65.4	82.1	91.3	101.0	111.0
HIROSE ELECTRIC	6806	7.6	5.0	4.7	5.7	5.5	6.6	6.7	6.6	5.6	6.3	7.3	7.2	7.9	8.4	9.1	8.2	8.9	11.7	13.9	14.5	16.9	21.7	27.7	24.4	25.1	30.2	35.1	37.9	40.9	44.2	47.8
JAE	6807	17.6	14.0	13.8	12.5	11.7	12.4	12.4	12.6	10.3	9.1	9.9	9.8	11.3	12.4	12.7	12.9	16.1	18.9	18.0	20.3	21.1	29.0	28.6	28.3	29.2	32.8	35.2	35.5	37.0	37.9	38.9
CMK	6958	9.9	10.3	9.9	9.6	8.5	9.5	10.3	12.4	7.9	9.6	9.0	8.0	6.7	7.9	9.2	8.5	8.5	9.8	10.3	10.6	10.3	16.1	17.3	17.3	18.6	18.4	20.4	20.8	21.2	21.8	22.4
DAISHINKU	6962	14.0	12.4	6.2	6.7	7.3	8.6	9.2	8.7	7.2	7.4	8.1	8.0	8.3	9.1	9.8	8.4	9.4	10.7	12.8	13.1	14.8	16.5	17.2	16.4	18.0	24.6	27.0	27.8	29.2	30.4	31.6
HAMAMATSU PHOTONICS	6965	13.0	13.0	12.8	12.8	13.3	13.7	15.4	17.8	20.2	20.0	20.7	22.3	24.4	25.1	26.9	29.5	30.0	31.8	34.5	37.5	43.5	44.4	59.1	75.6	80.6	77.0	81.2	77.1	76.8	79.7	83.9
KYOCERA	6971	252.8	205.8	183.2	197.2	213.4	190.6	209.2	205.2	199.6	177.4	232.9	270.3	296.5	335.8	354.5	327.9	331.2	364.9	343.9	344.3	345.4	452.5	539.4	540.2	521.8	522.0	557.8	576.1	730.7	619.0	615.9
TAIYO YUDEN	6976	36.9	28.3	25.1	23.5	25.5	27.3	32.2	40.5	30.3	28.6	32.5	30.5	35.9	37.6	44.1	48.7	48.8	52.1	60.9	62.6	67.9	96.8	104.8	102.3	110.0	125.8	120.1	128.4	142.1	174.0	181.7
MURATA MANUFACTURING	6981	92.0	59.2	53.3	54.8	59.0	66.4	86.1	114.6	94.1	89.2	110.5	132.0	160.9	170.6	186.3	217.5	211.4	290.3	349.3	334.4	361.3	464.7	575.0	513.0	482.8	520.5	567.2	642.7	740.6	847.5	972.0
NICHICON	6996	21.1	18.4	18.9	16.0	17.3	18.2	20.6	20.1	16.0	14.5	17.6	18.6	16.9	17.6	17.7	17.5	17.8	18.8	22.1	21.7	21.6	31.1	37.2	36.7	33.2	32.3	35.0	36.3	36.4	36.4	36.5
NIPPON CHEMI-CON	6997	23.1	18.2	18.9	18.8	18.7	21.9	23.5	24.4	22.4	19.5	21.0	27.1	21.8	20.9	22.7	20.8	20.9	23.0	27.9	26.5	25.5	31.6	33.0	29.2	33.1	31.2	35.8	35.2	35.0	34.5	34.1
KOA	6999	9.4	7.5	5.7	5.4	5.1	5.4	5.8	5.6	3.8	3.7	4.6	4.4	4.9	4.6	5.2	4.9	5.0	5.9	7.6	7.3	7.8	10.8	13.9	14.0	13.4	13.5	14.2	14.5	14.8	15.2	15.5
Total		873.0	713.8	641.8	648.6	686.5	717.0	782.5	828.5	728.9	690.9	824.5	930.9	1,014.7	1,079.7	1,198.6	1,224.8	1,244.6	1,487.5	1,593.4	1,627.3	1,740.8	2,328.8	2,703.1	2,624.1	2,625.3	2,951.9	3,032.3	3,185.5	3,524.8	3,669.5	3,878.8
Days in inventory (days)																																
IBIDEN	4062	38.6	38.2	33.6	36.1	38.1	41.0	44.7	49.6	54.2	49.8	45.1	50.8	58.0	59.5	59.0	57.5	61.3	57.3	69.6	75.3	78.4	85.4	96.8	98.4	94.6	88.6	81.7	76.1	73.8	72.3	72.1
NITERRA	5334	99.9	106.5	99.6	93.9	89.9	86.6	85.6	98.7	93.6	106.4	113.7	106.2	103.3	114.0	122.1	118.7	123.3	123.7	135.1	143.5	149.5	154.3	174.7	179.5	172.6	162.8	161.7	162.8	149.1	143.8	138.9
MINEBEA MITSUMI	6479	89.6	90.9	83.5	76.1	69.6	70.0	67.2	63.5	75.1	77.7	72.1	85.5	87.0	77.8	74.7	73.2	79.5	70.4	74.1	70.0	75.9	78.5	82.2	87.0	94.1	98.8	104.7	103.4	102.2	102.3	102.3
MABUCHI MOTOR	6592	122.7	137.7	107.6	93.0	89.6	91.9	88.5	84.8	93.1	107.8	90.0	105.8	111.1	99.9	111.7	110.0	111.4	109.4	126.9	139.9	148.5	155.1	171.9	165.3	153.1	152.6	140.8	136.3	128.9	118.8	109.2
TDK	6762	74.0	81.5	65.5	58.2	57.4	51.1	52.4	51.3	59.6	62.3	62.9	74.5	74.3	65.5	65.4	67.7	66.4	71.0	80.5	88.1	91.8	99.0	100.6	103.2	98.2	105.5	95.8	83.7	79.0	72.0	69.4
ALPS ALPINE	6770	64.6	69.9	61.9	58.4	56.4	53.0	49.3	46.6	47.0	44.1	43.3	53.0	57.1	48.2	48.7	50.7	53.7	50.6	50.8	53.4	59.3	63.0	71.2	75.3	67.7	60.3	58.3	58.1	59.5	58.7	58.6
NIHON DENPA KOGYO	6779	72.9	82.9	62.0	56.3	55.6	61.4	59.6	58.9	49.1	68.1	78.8	97.1	102.4	99.3	110.9	117.0	120.2	125.3	144.7	139.8	129.1	121.6	124.5	121.4	106.3	107.5	103.3	98.9	99.5	96.8	94.6
MEIKO ELECTRONICS	6787	55.1	69.5	73.3	71.3	52.2	47.5	49.6	52.1	50.0	54.0	58.2	67.9	67.1	58.4	55.5	56.2	55.2	54.3	55.9	64.6	73.8	76.5	80.6	83.6	80.9	84.5	73.5	73.1	74.4	71.7	68.9
HIROSE ELECTRIC	6806	59.5	65.6	48.2	48.4	46.3	44.1	41.7	37.9	41.9	45.4	47.0	47.0	49.7	43.1	46.3	47.9	50.9	55.7	66.9	73.8	74.6	79.4	94.9	102.2	86.7	82.5	81.7	85.9	85.9	86.0	86.3
JAE	6807	76.2	73.6	63.4	57.4	50.5	46.5	43.4	38.1	39.8	41.4	39.6	40.3	37.0	35.0	31.8	33.5	30.1	30.3	37.0	40.8	42.0	49.9	54.2	55.8	58.4	59.1	59.8	61.2	59.5	59.3	58.8
CMK	6958	36.7	39.0	37.0	37.3	34.7	33.1	32.3	34.0	37.1	39.1	42.4	43.2	41.7	41.5	48.0	49.9	47.7	45.2	47.6	53.1	60.4	69.5	85.6	82.9	81.9	79.2	74.6	76.9	76.0	74.9	74.2
DAISHINKU	6962	125.0	159.5	119.4	100.5	93.1	107.1	107.7	107.6	96.8	106.2	93.0	102.0	111.5	119.1	129.0	131.6	139.1	155.7	192.0	201.2	204.5	196.6	229.6	207.4	214.6	256.2	281.1	285.9	281.9	278.6	274.6
HAMAMATSU PHOTONICS	6965	144.4	157.6	152.4	150.9	139.8	141.2	128.3	136.5	149.8	186.8	163.0	160.2	178.2	177.1	177.5	178.6	178.6	171.7	171.7	182.7	206.1	187.4	195.9	242.2	284.7	259.8	233.5	225.6	200.1	186.4	175.3
KYOCERA	6971	83.8	105.3	89.1	80.7	87.6	87.9	81.0	85.6	88.3	87.3	84.2	105.5	108.6	108.0	110.8	113.9	114.6	105.8	111.5	108.5	112.4	109.9	124.0	135.8	133.2	130.2	137.3	136.6	121.5	144.8	130.1
TAIYO YUDEN	6976	96.9	104.2	85.2	73.2	64.1	64.7	65.8	73.9	79.0	67.6	67.8	73.2	78.2	83.5	85.5	95.7	99.2	101.1	107.4	114.5	112.4	133.8	158.5	147.0	143.6	157.4	142.1	129.5	122.6	112.5	116.3
MURATA MANUFACTURING	6981	96.1	108.9	83.8	76.6	81.2	77.3	82.9	94.4	95.7	87.4	87.8	106.9	111.7	109.2	103.5	103.5	111.3	98.7	119.7	131.0	126.0	144.4	187.7	197.9	177.2	173.4	167.7	159.9	154.0	149.8	145.0
NICHICON	6996	68.5	81.5	75.2	75.5	70.4	73.4	71.8	73.1	75.4	71.1	65.6	70.2	78.1	71.1	71.5	71.3	77.9	71.8	73.8	78.8	79.7	81.7	82.6								

Market Cap/Order Monthly Trend for Murata Mfg.

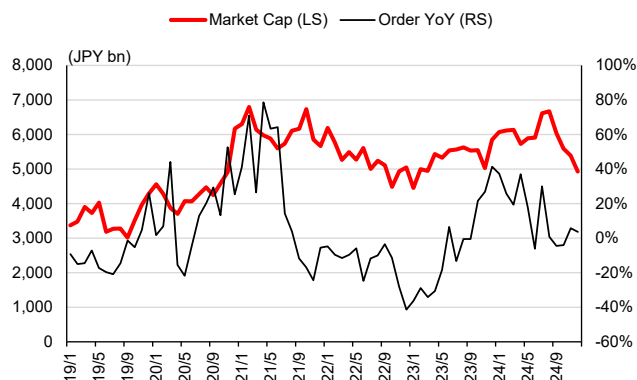
■ For dot-com bubble era (1998/1~2002/12)



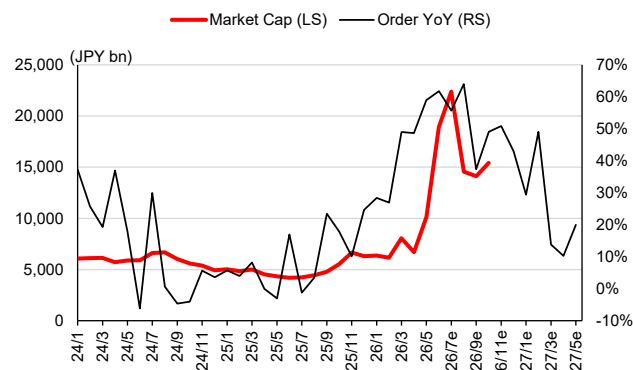
■ From 2019



■ For COVID-19 era (2019/1~2024/12)



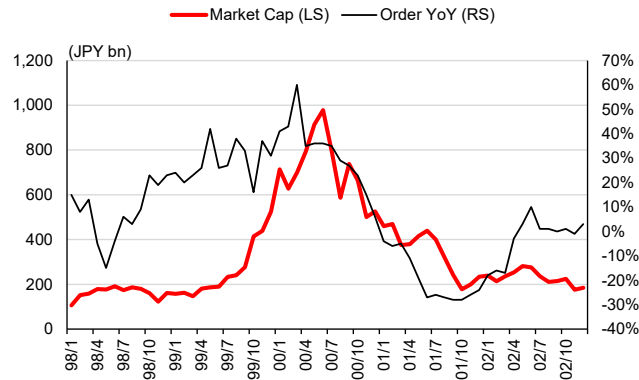
■ After COVID-19 era



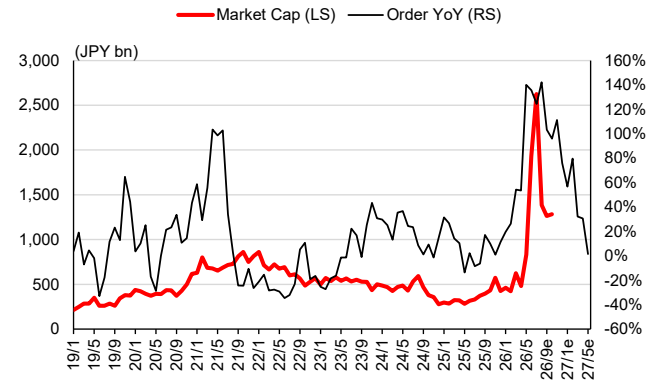
Note: For July 26 and thereafter, we estimate the same order level as June 26
 Source: FactSet, Company data, Morgan Stanley Research; e = Morgan Stanley Research estimates

Market Cap/Order Monthly Trend for Taiyo Yuden

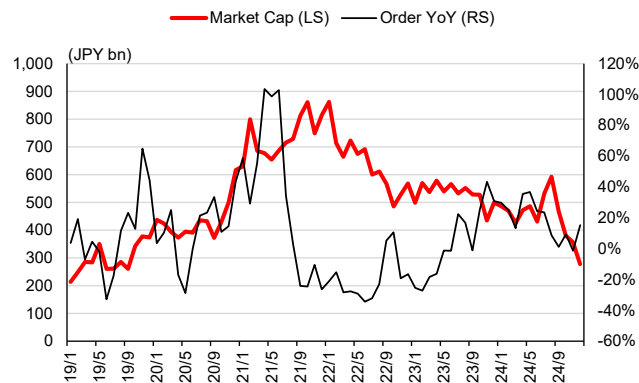
■ For dot-com bubble era (1998/1~2002/12)



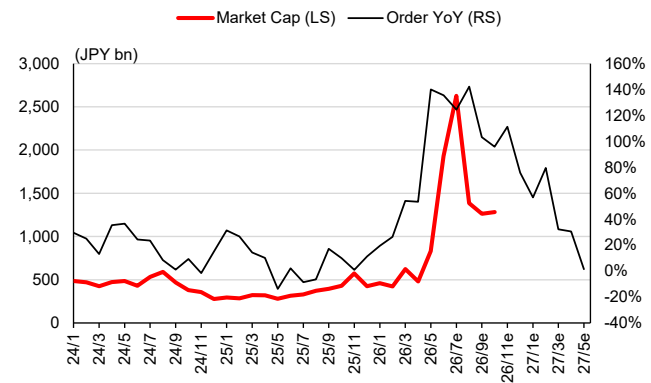
■ From 2019



■ For COVID-19 era (2019/1~2024/12)



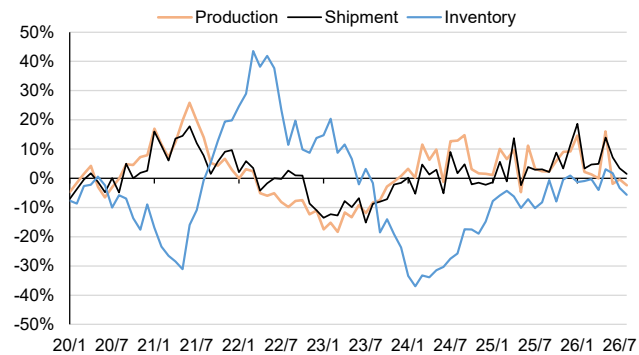
■ After COVID-19 era



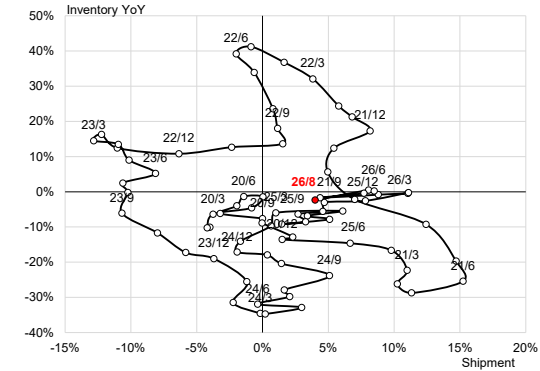
Note: Order data after Jan 19 are capacitor only; For July 26 and thereafter, we estimate the same order level as June 26
Source: FactSet, Company data, Morgan Stanley Research; e = Morgan Stanley Research estimates

Production & Inventory

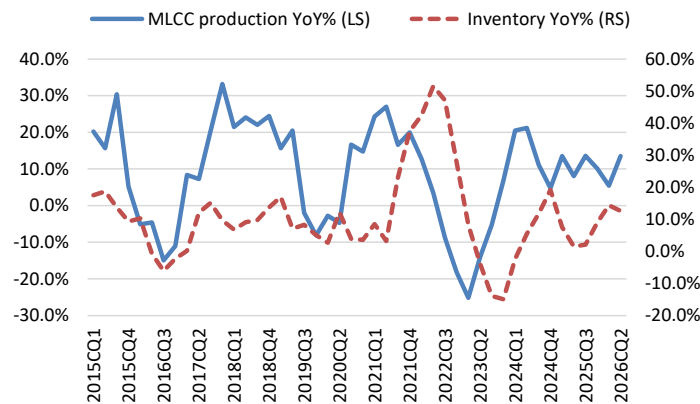
■ Industrial production: Production, shipments, inventories for Electronic parts and devices* (1)



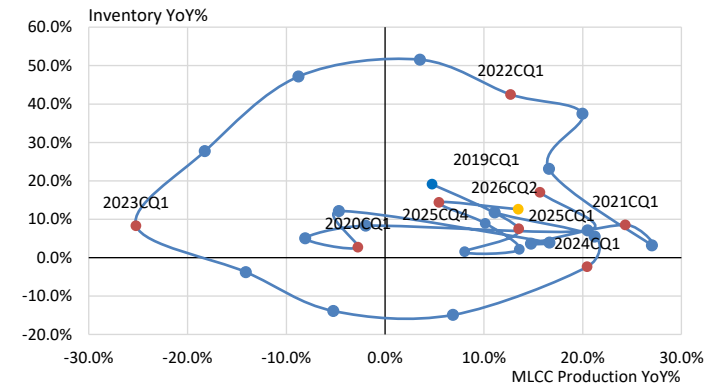
■ Industrial production: Production, shipments, inventories for Electronic parts and devices* (2)



■ Taiyo Yuden: MLCC production & inventory (1)



■ Taiyo Yuden: MLCC Production & inventory (2)

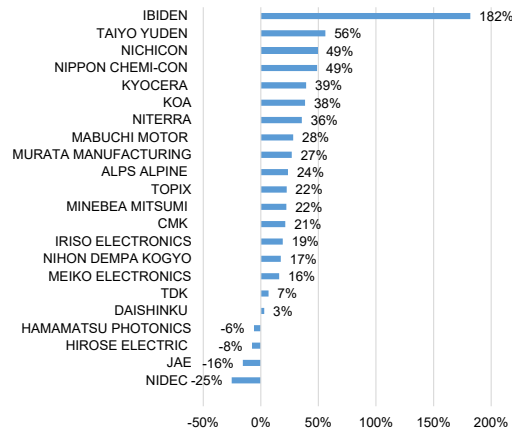


Note: *3-month average
Source: METI, Company data, Morgan Stanley Research

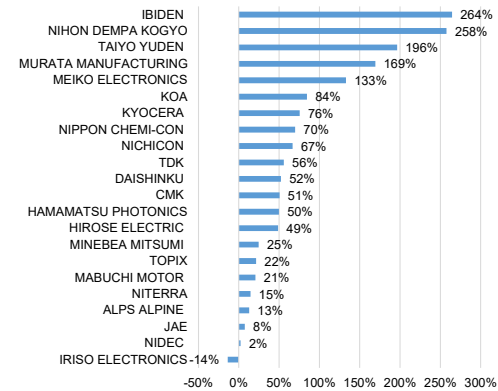
Stock Price Performance

Rank	2020	2021	2022	2023	2024	2025	2026 Based on 10/5 Closing Price
1	IBIDEN 85%	NIHON DEMP A KOGYO 184%	NITERRA 22%	CMK 80%	MEIKO ELECTRONICS 122%	IBIDEN 182%	IBIDEN 264%
2	NIDEC 73%	MEIKO ELECTRONICS 147%	KOA 17%	MEIKO ELECTRONICS 73%	TDK 54%	TAIYO YUDEN 56%	NIHON DEMP A KOGYO 258%
3	DAISHINKU 51%	DAISHINKU 135%	ALPS ALPINE 10%	IBIDEN 63%	NITERRA 52%	NICHICON 49%	TAIYO YUDEN 196%
4	TAIYO YUDEN 44%	CMK 72%	JAE 6%	TDK 55%	ALPS ALPINE 31%	NIPPON CHEMI-CON 49%	MURATA MANUFACTURING 169%
5	MURATA MANUFACTURING 38%	MINEBEA MITSUMI 59%	MABUCHI MOTOR -2%	JAE 53%	TOPIX 18%	KYOCERA 39%	MEIKO ELECTRONICS 133%
6	HAMAMATSU PHOTONICS 31%	IBIDEN 42%	IRISO ELECTRONICS -2%	MINEBEA MITSUMI 47%	HIROSE ELECTRIC 17%	KOA 38%	KOA 84%
7	TDK 25%	TAIYO YUDEN 37%	NICHICON -3%	NITERRA 37%	NIDEC 0%	NITERRA 36%	KYOCERA 76%
8	NIHON DEMP A KOGYO 19%	JAE 26%	TDK -3%	MURATA MANUFACTURING 36%	MABUCHI MOTOR -4%	MABUCHI MOTOR 28%	NIPPON CHEMI-CON 70%
9	NICHICON 14%	HAMAMATSU PHOTONICS 24%	TOPIX -5%	KYOCERA 26%	JAE -8%	MURATA MANUFACTURING 27%	NICHICON 67%
10	HIROSE ELECTRIC 12%	HIROSE ELECTRIC 24%	KYOCERA -9%	MABUCHI MOTOR 25%	MINEBEA MITSUMI -11%	ALPS ALPINE 24%	TDK 56%
11	MABUCHI MOTOR 8%	NITERRA 14%	NIPPON CHEMI-CON -13%	TOPIX 25%	MURATA MANUFACTURING -14%	TOPIX 22%	DAISHINKU 52%
12	KOA 8%	KYOCERA 14%	NIHON DEMP A KOGYO -13%	DAISHINKU 24%	NICHICON -15%	MINEBEA MITSUMI 22%	CMK 51%
13	TOPIX 5%	TOPIX 10%	HAMAMATSU PHOTONICS -14%	NICHICON 6%	KYOCERA -23%	CMK 21%	HAMAMATSU PHOTONICS 50%
14	IRISO ELECTRONICS -8%	KOA 9%	HIROSE ELECTRIC -14%	ALPS ALPINE 3%	NIPPON CHEMI-CON -26%	IRISO ELECTRONICS 19%	HIROSE ELECTRIC 49%
15	MINEBEA MITSUMI -10%	NIDEC 4%	MURATA MANUFACTURING -28%	TAIYO YUDEN -3%	IRISO ELECTRONICS -27%	NIHON DEMP A KOGYO 17%	MINEBEA MITSUMI 25%
16	NIPPON CHEMI-CON -13%	MURATA MANUFACTURING -2%	IBIDEN -30%	HIROSE ELECTRIC -4%	NIHON DEMP A KOGYO -31%	MEIKO ELECTRONICS 16%	TOPIX 22%
17	KYOCERA -15%	NIPPON CHEMI-CON -2%	CMK -34%	HAMAMATSU PHOTONICS -8%	DAISHINKU -35%	TDK 7%	MABUCHI MOTOR 21%
18	NITERRA -17%	IRISO ELECTRONICS -3%	MINEBEA MITSUMI -40%	IRISO ELECTRONICS -13%	KOA -36%	DAISHINKU 3%	NITERRA 15%
19	MEIKO ELECTRONICS -22%	NICHICON -3%	TAIYO YUDEN -42%	NIPPON CHEMI-CON -16%	IBIDEN -39%	HAMAMATSU PHOTONICS -6%	ALPS ALPINE 13%
20	JAE -29%	TDK -13%	NIDEC -49%	NIDEC -17%	HAMAMATSU PHOTONICS -39%	HIROSE ELECTRIC -8%	JAE 8%
21	CMK -40%	MABUCHI MOTOR -15%	MEIKO ELECTRONICS -49%	KOA -18%	TAIYO YUDEN -39%	JAE -16%	NIDEC 2%
22	ALPS ALPINE -46%	ALPS ALPINE -20%	DAISHINKU -55%	NIHON DEMP A KOGYO -19%	CMK -48%	NIDEC -25%	IRISO ELECTRONICS -14%

■ Stock price performance (CY2025)



■ Stock price performance (CY2026 YTD)



Source: FactSet

FX Sensitivities of Electronic Components Makers/Changes in the Competitiveness of Core Products Under a Weak Yen Environment

		F26e		FX assumption (by company)		FX sensitivity (JPY bn/year)	Assumption:¥150/USD		Assumption:¥160/USD	
		OP (JPY bn)		USD/JPY	USD/JPY		OP (JPY bn)	Change (%)	OP (JPY bn)	Change (%)
IBIDEN	4062	135.0		155.0	150.0	0.8	131.0	-3%	139.0	3%
NITERRA	5334	142.9		155.0	150.0	1.0	137.9	-3%	147.9	3%
MINEBEA MITSUMI	6479	124.6		155.0	155.0	1.5	117.1	-6%	132.1	6%
MABUCHI MOTOR	6592	27.5		155.0	150.0	0.3	26.0	-5%	29.0	5%
TDK	6762	360.4		155.0	150.0	2.0	350.4	-3%	370.4	3%
ALPS ALPINE	6770	48.5		155.0	150.0	1.2	42.5	-12%	54.5	12%
NIHON DEMP A KOGYO	6779	5.5		155.0	150.0	0.1	5.1	-7%	5.9	7%
MEIKO ELECTRONICS	6787	39.3		155.0	150.0	0.1	38.8	-1%	39.8	1%
HIROSE ELECTRIC	6806	55.3		155.0	150.0	0.3	53.7	-3%	57.0	3%
JAE	6807	8.4		155.0	153.0	0.3	6.9	-18%	9.9	18%
CMK	6958	5.1		155.0	150.0	0.0	5.1	0%	5.1	0%
DAISHINKU	6962	1.3		155.0	150.0	0.1	1.0	-19%	1.5	19%
HAMAMATSU PHOTONICS	6965	23.4		155.0	151.7	0.3	21.9	-6%	24.9	6%
KYOCERA	6971	165.3		155.0	150.0	1.0	160.3	-3%	170.3	3%
TAIYO YUDEN	6976	45.1		155.0	149.8	0.9	40.6	-10%	49.6	10%
MURATA MANUFACTURING	6981	490.8		155.0	149.1	4.5	468.3	-5%	513.3	5%
NICHICON	6996	10.7		155.0	145.0	0.1	10.2	-5%	11.3	5%
NIPPON CHEM-CON	6997	7.5		155.0	153.0	0.3	6.2	-17%	8.8	17%
KOA	6999	5.7		150.0	157.0	0.1	5.7	0%	6.9	21%
Total		1,702.3				14.9	1,628.6	-4%	1,777.1	4%

			Changes in the competitiveness of core products under a weak-yen environment
IBIDEN	4062	↑	Flip-chip package costs rising vs. peers in S. Korea, Taiwan, Austria; profits highly likely to grow
NITERRA	5334	→	Expecting profit growth from weak yen as core parts for automotive plugs/sensors made in Japan, but likely no change to competitiveness given large number of Japanese peers
MINEBEA MITSUMI	6479	→	Potential sales decline with over 70% of sales \$-denominated, but expecting THB/USD, CNY/USD rates to have big impact on profits
MABUCHI MOTOR	6592	→	Forecasting profit growth, but expecting no major shift in competitiveness for mainstay products given high overseas production weighting
TDK	6762	↑	Cost competitiveness for passive components may improve versus foreign peers, but limited impact for rechargeable batteries, HDD heads given lack of ¥-denominated costs
ALPS ALPINE	6770	→	Forecasting profit growth, but expecting no major shift in product competitiveness given limited rivalry with foreign firms for mainstay products
NIHON DEMP A KOGYO	6779	↑	Expecting increase in competitiveness on high foreign sales exposure, limited use of imported materials
MEIKO ELECTRONICS	6787	→	Expecting no major change in product competitiveness given high foreign production exposure, some use of imported materials
HIROSE ELECTRIC	6806	→	Expecting no major change in product competitiveness as US micro connector peers also undertaking production development/mass production in Japan
JAE	6807	→	Expecting no major change in product competitiveness as US micro connector peers also undertaking production development/mass production in Japan
CMK	6958	→	Expecting no major shift in competitiveness given high sales exposure to domestic customers
DAISHINKU	6962	↑	Expecting increase in competitiveness on high foreign sales exposure, limited use of imported materials
HAMAMATSU PHOTONICS	6965	↑	Yen weakness positive for earnings as 95% of production in Japan & 75% of sales abroad, although still certain level of ¥-denominated sales overseas
KYOCERA	6971	→	Forecasting growth in ¥-converted profits, but expecting no major shift in competitiveness of mainstay products
TAIYO YUDEN	6976	↑	Cost competitiveness of mainstay MLCCs rising vs. peers in S. Korea & Taiwan; potential growth in ¥-converted profits on high foreign sales exposure
MURATA MANUFACTURING	6981	↑	Cost competitiveness of mainstay MLCCs rising vs. peers in S. Korea & Taiwan; potential growth in ¥-converted profits on high foreign sales exposure
NICHICON	6996	→	Positive on aluminum electrolytic capacitor earnings, but worries over pressure on NECST product earnings given rising costs of imported parts/materials
NIPPON CHEM-CON	6997	↑	Cost competitiveness of mainstay aluminum electrolytic capacitors rising vs. peers in China; potential growth in ¥-converted profits on high foreign sales exposure
KOA	6999	→	Expecting substantial profit growth but no major change in product competitiveness as domestic peers with high market share also getting same benefits

Note: e = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

Sales and Production by Region

■ Sales by region (FY25)

Code	Company Name	Japan	US/America	Europe	China	Others
4062	IBIDEN	27.1%	11.8%	6.0%	2.7%	52.3%
5334	NITERRA	21.9%	23.6%	18.3%	9.4%	26.8%
6479	MINEBEA MITSUMI	28.7%	22.0%	11.1%	17.5%	20.7%
6592	MABUCHI MOTOR	12.1%	17.9%	23.0%	0.0%	46.9%
6762	TDK	7.3%	5.8%	7.2%	55.0%	24.6%
6770	ALPS ALPINE	11.7%	18.2%	-	20.2%	49.9%
6779	NIHON DEMPA KOGYO	16.7%	14.6%	8.1%	34.0%	26.6%
6787	MEIKO ELECTRONICS	32.9%	14.5%	1.5%	16.4%	34.8%
6806	HIROSE ELECTRIC	17.2%	-	-	36.6%	46.2%
6807	JAE	38.1%	9.9%	-	35.1%	16.9%
6958	CMK	60.4%	-	9.0%	17.5%	13.1%
6962	DAISHINKU	12.9%	6.8%	11.0%	32.1%	37.2%
6965	HAMAMATSU PHOTONICS	22.6%	24.2%	23.4%	0.0%	29.9%
6971	KYOCERA	29.2%	19.2%	20.2%	12.3%	19.1%
6976	TAIYO YUDEN	5.8%	7.6%	7.8%	29.5%	49.4%
6981	MURATA MANUFACTURING	7.3%	13.9%	8.4%	47.2%	23.2%
6996	NICHICON	49.3%	8.7%	5.6%	26.6%	10.0%
6997	NIPPON CHEMI-CON	19.2%	10.0%	10.5%	33.6%	26.7%
6999	KOA	25.3%	16.0%	19.1%	-	39.6%
Total		18.7%	14.6%	10.4%	28.7%	27.6%

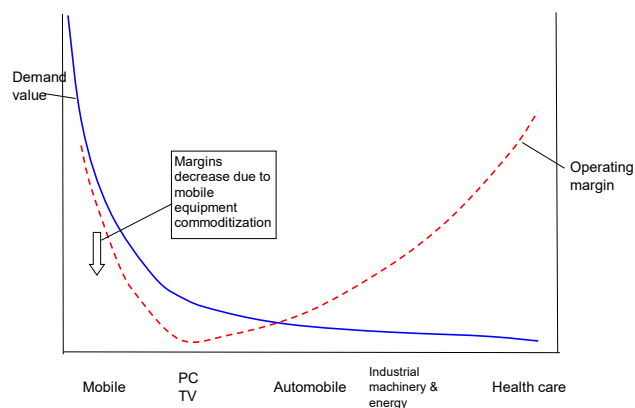
■ Production by region (FY25e)

Code	Company Name	Japan	US/America	Europe	China	Others
4062	IBIDEN	75.0%	4.0%	3.0%	2.0%	16.0%
6479	MINEBEA MITSUMI	22.0%	26.0%	11.0%	19.0%	22.0%
6592	MABUCHI MOTOR	0.0%	7.0%	3.0%	35.0%	55.0%
6762	TDK	12.5%	5.0%	10.0%	60.0%	12.5%
6770	ALPS ALPINE	30.0%	10.0%	5.0%	25.0%	30.0%
6779	NIHON DEMPA KOGYO	70.0%	0.0%	0.0%	20.0%	10.0%
6787	MEIKO ELECTRONICS	10.0%	0.0%	0.0%	45.0%	45.0%
6806	HIROSE ELECTRIC	50.0%	0.0%	0.0%	10.0%	40.0%
6807	JAE	60.0%	5.0%	0.0%	20.0%	15.0%
6908	IRISO ELECTRONICS	13.0%	0.0%	0.0%	49.0%	38.0%
6958	CMK	20.0%	0.0%	0.0%	35.0%	45.0%
6962	DAISHINKU	60.0%	0.0%	0.0%	30.0%	10.0%
6965	HAMAMATSU PHOTONICS	80.0%	5.0%	10.0%	5.0%	0.0%
6971	KYOCERA	60.0%	10.0%	10.0%	10.0%	10.0%
6976	TAIYO YUDEN	35.0%	0.0%	0.0%	20.0%	45.0%
6981	MURATA MANUFACTURING	70.0%	3.0%	3.0%	15.0%	9.0%
6996	NICHICON	75.0%	0.0%	0.0%	15.0%	10.0%
6997	NIPPON CHEMI-CON	65.0%	5.0%	0.0%	15.0%	15.0%
6999	KOA	70.0%	0.0%	0.0%	5.0%	25.0%
Total		39.9%	8.6%	7.9%	25.1%	18.6%

Note: Production data based on Morgan Stanley Research estimation
Source: Company data, Morgan Stanley Research

Preparing for Smartphone Price Compression and Slowdown in Technological Innovation

■ Potential for smartphone commoditization to knock down margins, but also greater earnings opportunities from computerization of autos and other areas



■ Electronic component firms: Sales breakdown by applications (F25)

	Sales in F25 (JPY bn)					
	Smartphone	Computers	AV/Office	Automotive	Industrial	Others
Ibiden	416.2	7.3	268.6	0.0	55.9	16.6
Nittera	731.2	6.7	0.0	0.0	587.5	40.7
Minebea Mitsumi	1,664.4	217.9	141.1	303.1	567.8	99.9
Mabuchi Motor	200.4	0.0	0.0	18.3	154.5	9.2
TDK	2,504.8	1,007.5	465.0	79.4	459.4	402.8
Alps Alpine	1,019.5	295.6	41.1	56.3	555.1	24.7
NDK	54.6	9.8	0.3	0.3	29.9	5.8
Meiko Electronics	240.6	29.4	16.5	0.0	97.5	4.4
Hirose Electric	211.3	44.3	8.9	6.8	54.3	62.0
JAE	227.9	55.5	18.0	18.0	95.1	24.1
CMK	100.2	0.0	0.4	0.5	89.5	0.0
Daishinku	39.6	10.2	5.2	4.0	14.8	4.2
Hamamatsu Photonics	204.0	0.0	0.0	1.0	5.6	66.3
Kyocera	2,070.2	139.4	154.8	491.9	303.7	414.0
Taiyo Yuden	355.3	85.3	64.0	28.4	106.6	71.1
Murata Mfg.	1,830.9	653.0	310.4	56.0	474.5	250.3
Nichicon	169.7	0.0	15.3	17.0	40.7	61.1
Nippon Chemi-Con	136.8	0.0	48.1	6.9	40.1	28.2
KOA	72.3	0.0	8.0	4.0	37.4	8.5
Total	12,249.8	2,561.8	1,565.8	1,092.1	3,770.0	1,593.9

	Smartphone	Computers	AV/Office	Automotive	Industrial	Others
Ibiden	100%	2%	65%	0%	13%	4%
Nittera	100%	1%	0%	0%	80%	6%
Minebea Mitsumi	100%	13%	8%	18%	34%	6%
Mabuchi Motor	100%	0%	0%	9%	77%	5%
TDK	100%	40%	19%	3%	18%	16%
Alps Alpine	100%	29%	4%	6%	54%	2%
NDK	100%	18%	1%	1%	55%	11%
Meiko Electronics	100%	12%	7%	0%	41%	2%
Hirose Electric	100%	21%	4%	3%	26%	29%
JAE	100%	24%	8%	8%	42%	11%
CMK	100%	0%	0%	1%	89%	0%
Daishinku	100%	26%	13%	10%	37%	11%
Hamamatsu Photonics	100%	0%	0%	1%	3%	33%
Kyocera	100%	7%	7%	24%	15%	20%
Taiyo Yuden	100%	24%	18%	8%	30%	20%
Murata Mfg.	100%	36%	17%	3%	26%	14%
Nichicon	100%	0%	9%	10%	24%	36%
Nippon Chemi-Con	100%	0%	35%	5%	29%	21%
KOA	100%	0%	11%	6%	52%	12%
Total	100%	21%	13%	9%	31%	13%

Source: Company data, Morgan Stanley Research

Memory Price Outlook

	1Q26	2Q26	3Q26E - Old	3Q26E - New	4Q26E - Old	4Q26E - New	1Q27F
PC DRAM	DDR4: up 120-125% DDR5: up 110-115% Blended ASP: up 110-115%	DDR4: up 35-40% DDR5: up 45-50% Blended ASP: up 45-50%	Blended ASP: up 18-23%	DDR4: up 15-20% DDR5: up 18-23% Blended ASP: up 18-23%	up 3-8%	DDR4: up 13-18% DDR5: up 13-18% Blended ASP: up 13-18%	up 3-8%
Server DRAM	DDR4: up 95-100% DDR5: up 93-98% Blended ASP: up 93-98%	DDR4: up 50-55% DDR5: up 53-58% Blended ASP: up 53-58%	DDR4: up 13-18% DDR5: up 13-18% Blended ASP: up 13-18%	DDR4: up 13-18% DDR5: up 13-18% Blended ASP: up 13-18%	up 3-8%	DDR4: up 10-15% DDR5: up 10-15% Blended ASP: up 10-15%	up 3-8%
Mobile DRAM	LPDDR4X: up 58-63% LPDDR5(x): up 58-63%	LPDDR4X: up 73-78% LPDDR5(x): up 80-85%	LPDDR4X: up 8-13% LPDDR5(x): up 8-13%	LPDDR4X: up 8-13% LPDDR5(x): up 8-13%	up 0-5%	LPDDR4X: up 0-5% LPDDR5(x): up 0-5%	LPDDR4X: up 5-10% LPDDR5(x): up 13-18%
Graphics DRAM	GDDR6: up 95-100% GDDR7: up 95-100%	GDDR6: up 30-35% GDDR7: up 40-45%	GDDR6: up 15-20% GDDR7: up 15-20%	GDDR6: up 15-20% GDDR7: up 15-20%	GDDR6: up 0-5% GDDR7: up 3-8%	GDDR6: up 5-10% GDDR7: up 8-13%	up 3-8%
Consumer DRAM	DDR3: up 75-80% DDR4: up 75-80%	DDR3: up 55-60% DDR4: up 55-60%	DDR3: up 35-40% DDR4: up 28-33%	DDR3: up 35-40% DDR4: up 28-33%	up 3-8%	DDR3: up 3-8% DDR4: up 3-8%	up 3-8%
Total DRAM	Conventional DRAM: up 93-98% HBM Blended: up 83-88%	Conventional DRAM: up 58-63% HBM Blended: 53-58%	Conventional DRAM: up 13-18% HBM Blended: 8-13%	Conventional DRAM: up 13-18% HBM Blended: 8-13%	Conventional DRAM: up 3-8%	Conventional DRAM: up 10-15% HBM Blended: up 15-20%	Conventional DRAM: up 5-10%
	1Q26	2Q26	3Q26E - Old	3Q26E - New	4Q26E - Old	4Q26E - New	1Q27F
eMMC	up 95-100%	up 75-80%	up 5-10%	up 18-23%	mostly flat	up 5-10%	up 0-5%
UFS							
Enterprise SSD	up 75-80%	up 48-53%	up 18-23%	up 20-25%	up 3-8%	up 23-28%	up 13-18%
Client SSD	up 105-110%	up 70-75%	up 13-18%	up 20-25%	mostly flat	up 5-10%	up 0-5%
3D NAND Wafers (TLC & QLC)	up 75-80%	up 28-33%	up 5-10%	up 5-10%	mostly flat	up 3-8%	mostly flat
Total NAND Flash	up 85-90%	up 55-60%	up 10-15%	up 18-23%	up 0- 5%	up 15-20%	up 13-18%

Note: E/F = Morgan Stanley Research estimates based on published note on Sep 16
Source: TrendForce, Morgan Stanley Research

End-product Market and Major Components Market (Calendar Year)

(mn units)	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e
Server	8.7	9.0	9.2	9.7	9.6	10.2	11.8	11.8	12.7	13.5	14.9	12.4	14.5	16.4	19.9	21.9
YoY % Change	0%	3%	3%	5%	-2%	7%	16%	0%	7%	7%	10%	-17%	17%	13%	21%	10%
PC	346.1	311.8	304.7	272.1	256.2	261.3	260.2	267.5	306.6	353.6	293.8	253.2	261.6	288.5	254.8	245.8
YoY % Change	-4%	-10%	-2%	-11%	-6%	2%	0%	3%	15%	15%	-17%	-14%	3%	10%	-12%	-4%
Desktop	146.0	134.2	131.3	110.7	100.8	100.8	96.0	95.9	85.8	95.8	85.7	72.2	73.0	83.6	74.5	72.7
YoY % Change	-4%	-8%	-2%	-16%	-9%	0%	-5%	0%	-11%	12%	-11%	-16%	1%	15%	-11%	-2%
Notebook	200.1	177.5	173.4	161.4	155.4	160.5	164.1	171.6	220.8	257.7	208.1	181.0	188.6	204.9	180.3	173.0
YoY % Change	-4%	-11%	-2%	-7%	-4%	3%	2%	5%	29%	17%	-19%	-13%	4%	9%	-12%	-4%
Smartphone	726.7	1,018.7	1,301.5	1,437.6	1,469.5	1,465.3	1,402.5	1,372.6	1,281.2	1,359.8	1,206.2	1,165.5	1,237.5	1,267.7	1,104.2	1,143.7
YoY % Change	47%	40%	28%	10%	2%	0%	-4%	-2%	-7%	6%	-11%	-3%	6%	2%	-13%	4%
5G smartphone	0.0	0.0	0.0	0.0	0.0	0.0	0.0	16.1	255.3	559.5	636.8	703.4	814.6	893.1	838.3	910.1
YoY % Change	-	-	-	-	-	-	-	-	1485%	119%	14%	10%	16%	10%	-6%	9%
iPhone	135.9	153.5	192.7	231.5	215.4	215.8	204.9	199.3	203.4	235.8	226.7	235.6	234.3	251.7	245.4	259.0
YoY % Change	46%	13%	26%	20%	-7%	0%	-5%	-3%	2%	16%	-4%	4%	-1%	7%	-2%	6%
HDD	578.0	551.9	564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	130.4	125.2
YoY % Change	-7%	-5%	2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-34%	-29%	1%	0%	5%	-4%
Enterprise	33.2	31.8	32.4	32.1	24.8	22.6	21.5	17.4	12.9	13.4	8.6	5.8	4.4	3.2	2.7	1.9
NL	28.1	35.5	36.9	37.2	41.0	42.4	51.3	52.7	59.1	73.5	61.5	39.4	58.6	67.6	76.4	78.6
3.5"ATA	216.2	203.2	202.6	161.0	142.9	135.4	120.5	103.9	87.0	89.9	55.0	45.2	35.6	30.9	31.5	27.5
2.5"	298.7	280.7	291.7	238.6	215.1	202.7	182.0	143.6	101.0	82.5	46.3	32.0	25.3	22.0	19.8	17.3
Global Light vehicle sales	75.0	78.2	81.4	83.5	92.3	94.4	93.9	89.9	77.2	80.3	79.0	86.7	88.7	91.9	90.5	93.6
YoY % Change	5%	4%	4%	3%	11%	2%	-1%	-4%	-14%	4%	-2%	10%	2%	4%	-2%	3%
ICE				83.5	91.8	93.7	92.6	88.5	75.1	75.2	71.0	76.5	77.3	77.8	74.7	75.1
YoY % Change					10%	2%	-1%	-4%	-15%	0%	-6%	8%	1%	1%	-4%	1%
EV				-	0.5	0.7	1.3	1.4	2.1	5.1	8.0	10.2	11.4	14.1	15.8	18.5
YoY % Change						40%	86%	8%	50%	143%	57%	28%	12%	24%	12%	17%

Note: e = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

End-product Market and Major Components Market (Quarterly)

	CY2021				CY2022				CY2023				CY2024				CY2025				CY2026			
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q
Server	2.9	3.4	3.5	3.7	3.4	3.8	3.9	3.8	3.0	3.1	3.1	3.2	3.2	3.7	3.7	4.0	3.9	4.2	4.2	4.4	4.1	4.9		
YoY % Change	9%	3%	8%	8%	14%	13%	12%	1%	-10%	-19%	-22%	-16%	7%	18%	20%	26%	22%	15%	15%	10%	3%	15%		
Sequential % Change	-15%	15%	4%	7%	-10%	14%	3%	-4%	-20%	2%	0%	3%	2%	13%	1%	8%	-1%	7%	0%	4%	-7%	19%		
PC	85.7	85.7	88.3	93.8	80.6	71.6	74.4	67.2	57.3	61.4	68.7	65.9	59.5	63.4	69.1	69.6	64.0	69.7	76.5	78.3	65.6	67.0		
YoY % Change	58%	14%	5%	1%	-6%	-16%	-16%	-28%	-29%	-14%	-8%	-2%	4%	3%	1%	6%	8%	10%	11%	12%	3%	-4%		
Sequential % Change	-8%	0%	3%	6%	-14%	-11%	4%	-10%	-15%	7%	12%	-4%	-10%	7%	9%	1%	-8%	9%	10%	2%	-16%	2%		
Desktop	22.3	23.6	24.3	25.6	22.7	21.5	21.8	19.7	17.5	17.4	18.6	18.7	17.3	17.3	18.8	19.6	18.5	19.5	22.3	23.4	18.2	18.8		
YoY % Change	14%	19%	10%	5%	2%	-9%	-10%	-23%	-23%	-19%	-15%	-5%	-1%	0%	1%	5%	7%	12%	19%	19%	-2%	-4%		
Sequential % Change	-9%	6%	3%	5%	-11%	-5%	2%	-10%	-11%	-1%	7%	0%	-7%	0%	8%	4%	-6%	5%	15%	5%	-22%	3%		
Notebook (incl. Netbook)	63.5	62.1	64.0	68.1	57.9	50.1	52.6	47.5	39.8	44.0	50.0	47.2	42.2	46.0	50.3	50.0	45.5	50.3	54.2	54.9	47.4	48.3		
YoY % Change	83%	12%	3%	0%	-9%	-19%	-18%	-30%	-31%	-12%	-5%	-1%	6%	5%	1%	6%	8%	9%	8%	10%	4%	-4%		
Sequential % Change	-7%	-2%	3%	6%	-15%	-13%	5%	-10%	-16%	11%	14%	-6%	-11%	9%	9%	-1%	-9%	10%	8%	1%	-14%	2%		
Smartphone	344.7	313.4	334.2	367.6	314.5	287.6	303.2	300.6	268.4	267.9	303.7	324.0	300.4	292.3	314.7	328.8	302.6	297.4	326.5	337.5	294.8	276.3		
YoY % Change	25%	13%	-6%	-2%	-9%	-8%	-9%	-18%	-15%	-7%	0%	8%	12%	9%	4%	1%	1%	2%	4%	3%	-3%	-7%		
Sequential % Change	-8%	-9%	7%	10%	-14%	-9%	5%	-1%	-11%	0%	13%	7%	-7%	-3%	8%	4%	-8%	-2%	10%	3%	-13%	-6%		
5G smartphone	125.1	118.1	136.5	179.8	155.5	146.3	161.6	173.4	160.5	149.3	182.6	211.1	187.9	186.0	211.4	229.3	213.6	203.9	228.0	247.6	219.0	206.2		
YoY % Change	491%	139%	110%	50%	24%	24%	18%	-4%	3%	2%	13%	22%	17%	25%	16%	9%	14%	10%	8%	8%	3%	1%		
Sequential % Change	4%	-6%	16%	32%	-14%	-6%	10%	7%	-7%	-7%	22%	16%	-11%	-1%	14%	8%	-7%	-5%	12%	9%	-12%	-6%		
iPhone	55.3	44.4	51.1	85.0	56.5	45.4	52.3	72.5	55.8	44.8	54.4	80.6	53.0	46.1	57.4	77.9	59.1	48.4	59.6	84.7	61.8	55.7		
YoY % Change	50%	18%	23%	-3%	2%	2%	2%	-15%	-1%	-1%	4%	11%	-5%	3%	6%	-3%	12%	5%	4%	9%	4%	15%		
Sequential % Change	-37%	-20%	15%	66%	-34%	-20%	15%	39%	-23%	-20%	21%	48%	-34%	-13%	25%	36%	-24%	-18%	23%	42%	-27%	-10%		
HDD	64.1	67.6	66.3	61.2	52.8	44.6	38.5	36.3	33.5	31.3	28.6	28.9	29.7	30.3	32.1	31.7	28.7	30.4	31.6	33.0	33.0	33.0		
YoY % Change	-5%	15%	4%	-13%	-18%	-34%	-42%	-41%	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%	9%		
Sequential % Change	-9%	5%	-2%	-8%	-14%	-15%	-14%	-6%	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%	0%		
Enterprise	3.2	3.5	3.5	3.3	2.7	2.4	1.6	1.7	1.4	1.6	1.4	1.4	1.5	1.0	1.0	0.9	0.8	0.8	0.7	0.8	0.8	0.6		
NL	16.0	19.3	19.7	18.5	18.7	18.0	14.2	10.6	11.4	9.2	9.0	9.9	12.3	13.8	15.9	16.5	14.6	16.5	17.3	19.2	18.7	19.1		
3.5"ATA	22.2	23.6	23.2	20.8	16.5	14.1	13.0	12.8	11.2	12.1	11.2	10.7	9.3	9.4	9.0	7.9	7.6	7.9	8.0	7.6	8.3	8.7		
2.5"	22.7	21.2	19.9	18.7	14.9	10.0	9.7	11.3	9.5	8.4	7.1	6.9	6.6	6.1	6.2	6.4	5.7	5.3	5.6	5.5	5.3	4.7		
HDD/PC	0.7	0.8	0.8	0.7	0.7	0.6	0.5	0.5	0.6	0.5	0.4	0.4	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.5	0.5		

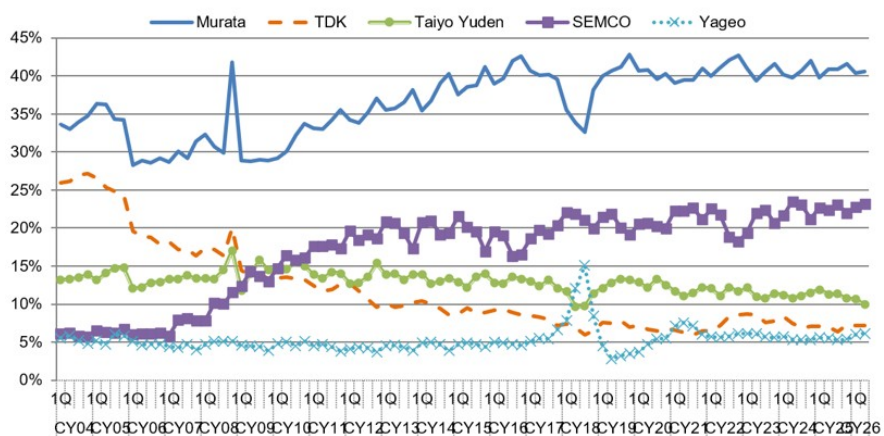
Source: Company data, Morgan Stanley Research

MLCC Market Data

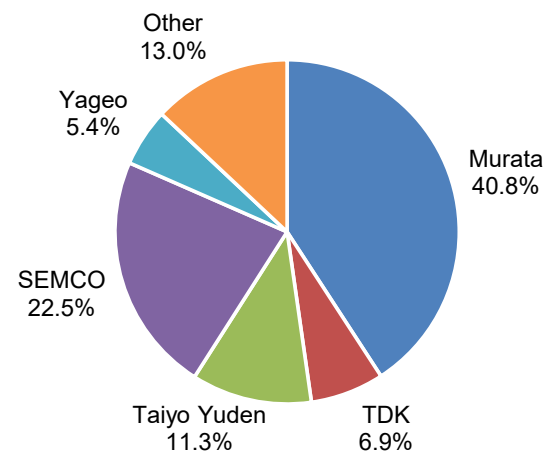
■ MLCC market forecasts

Base Case	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e	2029e	2030e	2031e
Total worldwide shipment quantity (Bil pcs)	1,504	1,480	1,640	2,050	2,029	2,167	2,293	2,605	2,735	3,080	3,480	3,828	3,254	3,695	4,446	3,572	3,187	3,386	3,773	4,037	4,360	4,709	5,086	5,493	5,932
YoY (%)	14.9%	-1.6%	10.8%	24.9%	-1.0%	6.8%	5.8%	13.6%	5.0%	12.6%	13.0%	10.0%	-15.0%	13.5%	20.3%	-19.7%	-10.8%	6.2%	11.4%	7.0%	8.0%	8.0%	8.0%	8.0%	8.0%
CAGR (1991~)	14.9%	13.8%	13.7%	14.2%	13.4%	13.1%	12.8%	12.8%	12.5%	12.5%	12.4%	11.3%	11.4%	11.6%	10.5%	9.7%	9.6%	9.7%	9.6%	9.6%	9.5%	9.5%	9.4%	9.4%	9.4%
ASP in worldwide shipment (Cent)	0.47	0.42	0.36	0.38	0.39	0.36	0.33	0.30	0.29	0.26	0.27	0.37	0.39	0.38	0.39	0.40	0.40	0.39	0.39	0.46	0.56	0.61	0.66	0.72	0.75
YoY (%)	-0.2%	-10.1%	-14.0%	6.8%	1.2%	-7.7%	-7.2%	-10.6%	-4.1%	-7.2%	3.8%	34.3%	4.6%	-2.8%	4.7%	2.0%	-1.0%	-1.0%	-1.0%	17.9%	22.6%	8.0%	9.0%	8.2%	4.6%
CAGR (1991~)	-8.6%	-8.7%	-9.0%	-8.2%	-7.8%	-7.8%	-7.7%	-7.9%	-7.7%	-7.7%	-7.3%	-6.0%	-5.6%	-5.5%	-5.2%	-5.0%	-4.9%	-4.8%	-4.6%	-4.1%	-3.4%	-3.1%	-2.8%	-2.5%	-2.4%
Total worldwide shipment value (Bil \$)	7.00	6.19	5.90	7.87	7.89	7.77	7.63	7.75	7.80	8.15	9.56	14.13	12.56	13.86	17.47	14.32	12.64	13.30	14.67	18.51	24.52	28.60	33.68	39.37	44.45
YoY (%)	14.6%	-11.6%	-4.7%	33.4%	0.2%	-1.5%	-1.8%	1.6%	0.7%	4.5%	17.3%	47.7%	-11.1%	10.3%	26.0%	-18.1%	-11.7%	5.2%	10.3%	26.2%	32.4%	16.7%	17.8%	16.9%	12.9%
CAGR (1991~)	5.0%	4.0%	3.5%	4.8%	4.6%	4.3%	4.0%	3.9%	3.8%	3.8%	4.3%	5.7%	5.0%	5.2%	5.8%	5.0%	4.4%	4.4%	4.6%	5.1%	5.8%	6.1%	6.4%	6.6%	6.8%

■ Global share of the top 5 ceramic capacitor firms (USD base)



■ Global share of the top 5 ceramic capacitor firms (USD base, CY25)



Note: e = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

Capacitor Market: 016008 MLCCs are 75% Smaller than 0201 MLCCs

■ MLCC size comparison

	4532	3216	2012	1608	1005	0603	0402	0201	016008
Length	4.5mm	3.2mm	2.0mm	1.6mm	1.0mm	0.6mm	0.4mm	0.25mm	0.16mm
Width	3.2mm	1.6mm	1.25mm	0.8mm	0.5mm	0.3mm	0.2mm	0.125mm	0.08mm
Thickness	2.0mm	0.85mm	0.6mm	0.8mm	0.5mm	0.3mm	0.2mm	0.125mm	0.08mm
Volume	28.8mm ³	4.35mm ³	1.5mm ³	1.024mm ³	0.25mm ³	0.054mm ³	0.016mm ³	0.004mm ³	0.001mm ³
Volume ratio	7,373	1,114	384	262	64.0	13.8	4.1	1.0	0.26

■ Number of MLCCs per vehicle

(units: pcs)	GAS	HV	BEV	PHEV
Power Train	300-500	1,000-1,600	2,000-2,500	2,500-3,000
ADAS(Lv2-5)	1,500-5,000			
Safety	300-1,000			
Infotainment	500-2,500			
Other	500-2,500			

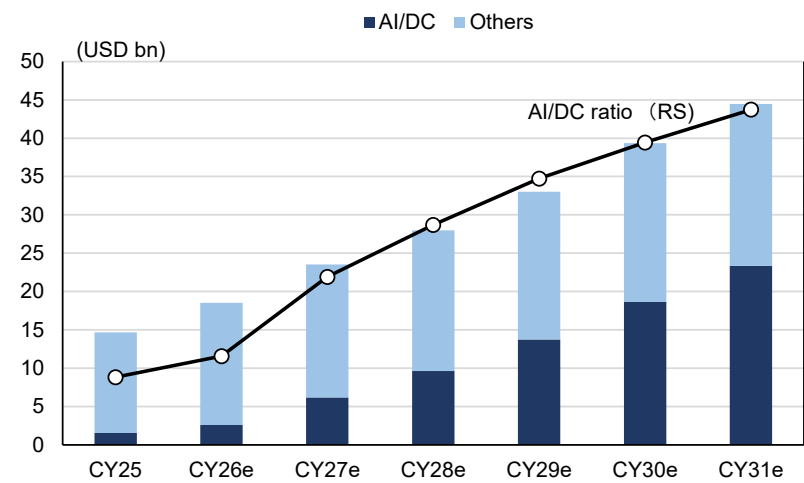
■ MLCC maximum capacitance by size

<=1000	220						
<=100		100	100	47			
<=10					10		
<=1						1	0.1
(mm)	3216	2012	1608	1005	0603	0402	0201
Murata Mfg	○	○	○	○	○	○	○

Source: Company data, Morgan Stanley Research

AI/DC-use MLCC Shipment Value / Situations around Barium Titanate Production

AI/DC-use MLCC shipment value



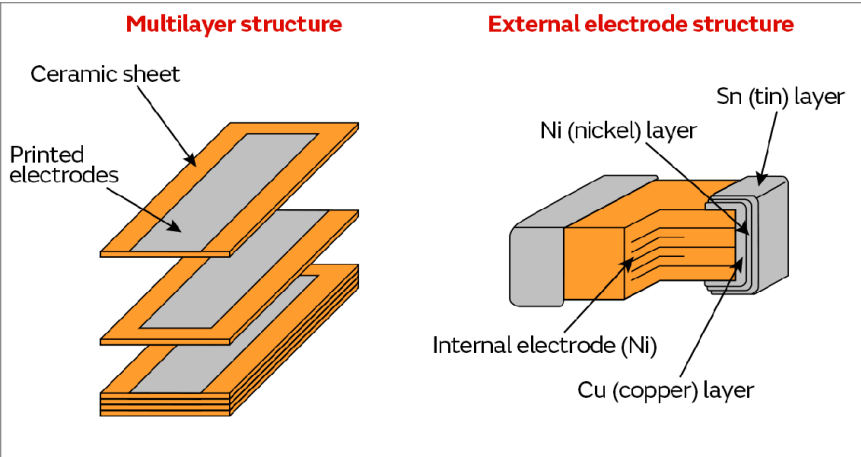
Situations around Barium Titanate production

		Solid-phase method	Oxalate method	Hydrothermal synthesis
Internal production	Murata Mfg.	○		
	Taiyo Yuden	○		
	TDK	○		
	SEMCO	○		
	Yageo	○		
External sourcing	Sakai Chemical			○
	Nippon Chem. Industrial		○	
	Ishihara Chemical		○	
	Noritake	○		
	Toda Kogyo			○
	Sinocera	○		○

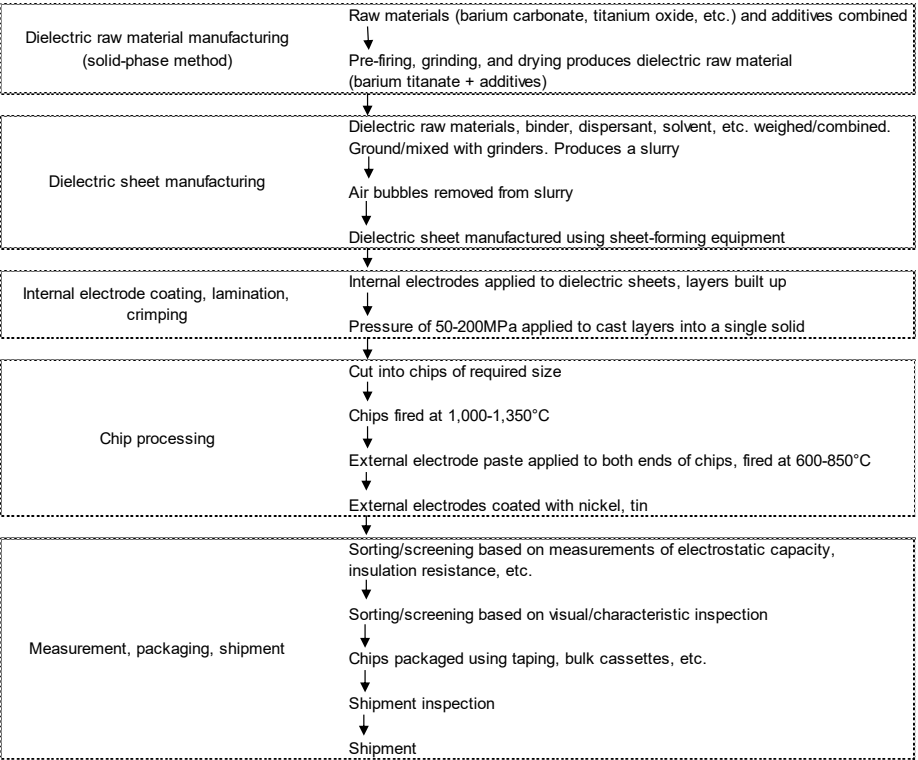
Note: e = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

MLCC Basics

MLCC composition



MLCC production process

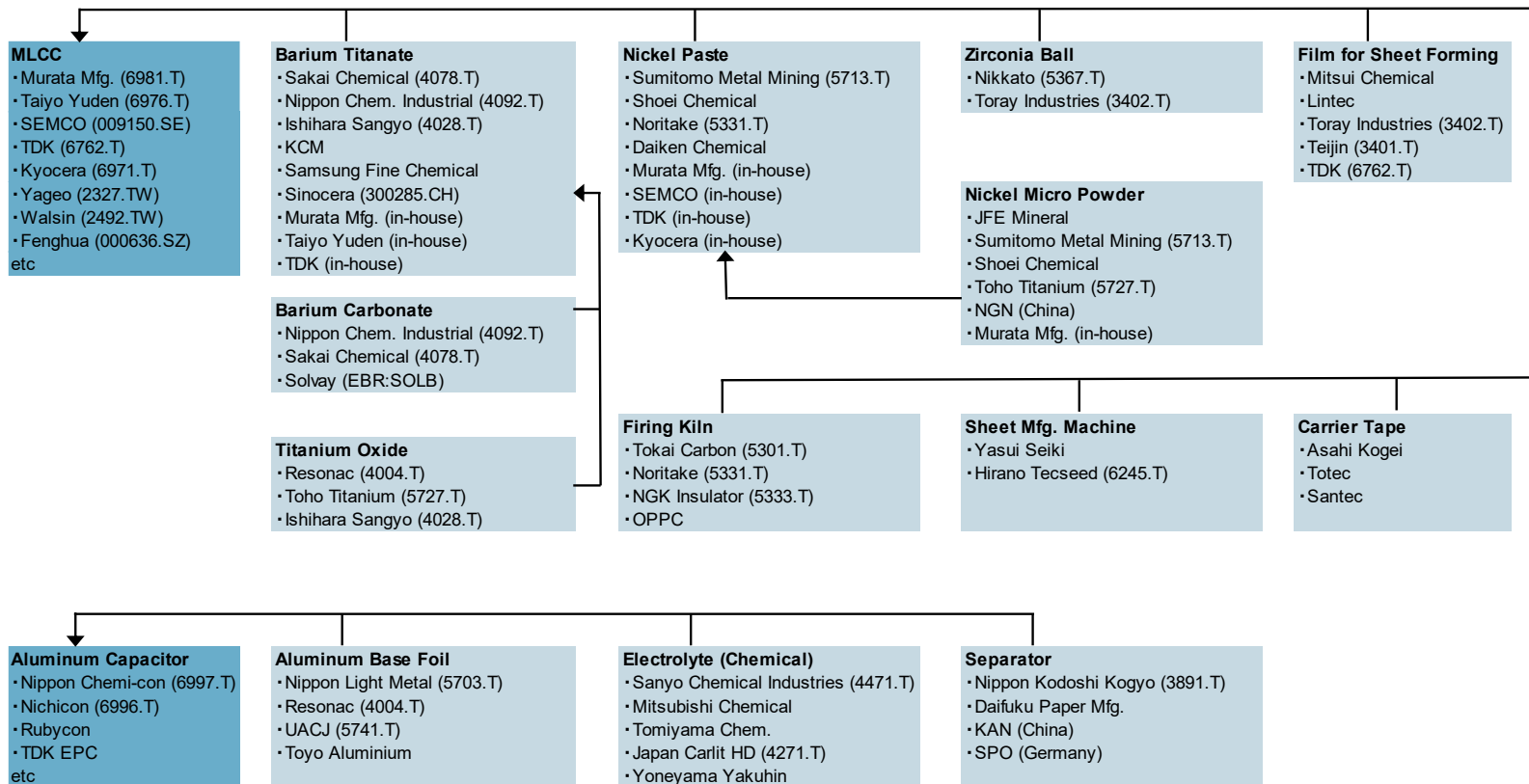


Source: Murata Mfg., Morgan Stanley Research

MLCC Sales, Market Share, OP

	CY23				CY24				CY25				CY26			2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q																		
Total (USD mn)	2,940	3,131	3,354	3,221	3,171	3,276	3,518	3,334	3,321	3,682	3,937	3,731	3,825	4,367	5,917	7,868	7,885	7,769	7,626	7,731	7,802	8,167	9,564	14,129	12,562	13,861	17,311	14,316	12,645	13,299	14,672	
Major 5	2,567	2,724	2,918	2,802	2,759	2,850	3,061	2,900	2,889	3,203	3,426	3,246	3,328	3,799	4,411	6,308	6,462	6,444	6,405	6,586	6,660	7,000	8,215	11,903	10,744	11,798	15,013	12,467	11,010	11,571	12,764	
Murata	1,205	1,233	1,362	1,340	1,273	1,305	1,432	1,399	1,320	1,504	1,611	1,553	1,545	1,773	1,708	2,467	2,677	2,730	2,784	2,933	3,043	3,349	3,834	4,934	5,168	5,591	6,945	5,921	5,140	5,409	5,988	
TDK	256	268	256	251	265	247	241	236	234	256	253	271	277	314	816	1,053	957	864	757	738	701	730	753	931	934	939	1,105	1,095	1,031	988	1,015	
Taiyo Yuden	357	343	359	367	354	354	390	383	395	416	447	401	408	433	843	1,177	1,093	1,061	1,047	1,020	1,028	1,073	1,209	1,484	1,613	1,767	2,026	1,683	1,425	1,481	1,659	
SEMCO	569	689	750	664	685	770	811	705	754	823	907	821	869	1,010	791	1,241	1,389	1,471	1,488	1,542	1,525	1,455	1,871	2,987	2,590	2,820	3,860	2,943	2,672	2,972	3,305	
Yageo	180	191	190	180	181	175	187	177	185	204	207	200	229	269	254	370	345	318	329	353	363	393	549	1,567	439	681	1,077	825	742	720	797	
Other	373	407	436	419	412	426	457	433	432	479	512	485	497	568	1,505	1,548	1,423	1,326	1,222	1,145	1,142	1,162	1,438	2,400	1,815	2,063	2,298	1,848	1,635	1,729	1,907	
Market Share																																
Major 5	87.3%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	74.6%	80.2%	81.9%	82.9%	84.0%	85.2%	85.4%	85.7%	85.9%	84.2%	85.5%	85.1%	86.7%	87.1%	87.1%	87.0%	87.0%	
Murata	41.0%	39.4%	40.6%	41.7%	40.2%	39.8%	40.7%	42.0%	39.8%	40.8%	40.9%	41.6%	40.4%	40.6%	28.9%	31.4%	33.9%	35.1%	36.5%	37.9%	39.0%	41.0%	40.1%	34.9%	41.1%	40.3%	40.1%	41.4%	40.6%	40.7%	40.8%	
TDK	8.7%	8.6%	7.6%	7.8%	8.4%	7.5%	6.8%	7.1%	7.1%	7.0%	6.4%	7.3%	7.2%	7.2%	13.8%	13.4%	12.1%	11.1%	9.9%	9.5%	9.0%	8.9%	7.9%	6.6%	7.4%	6.8%	6.4%	7.6%	8.1%	7.4%	6.9%	
Taiyo Yuden	12.1%	10.9%	10.7%	11.4%	11.2%	10.8%	11.1%	11.5%	11.9%	11.3%	11.4%	10.7%	10.7%	9.9%	14.2%	15.0%	13.9%	13.7%	13.7%	13.2%	13.2%	13.1%	12.6%	10.5%	12.8%	12.7%	11.7%	11.8%	11.3%	11.1%	11.3%	
SEMCO	19.4%	22.0%	22.4%	20.7%	21.6%	23.5%	23.1%	21.2%	22.7%	22.3%	23.0%	22.0%	22.7%	23.1%	13.4%	15.8%	17.6%	18.9%	19.5%	19.9%	19.5%	17.8%	19.6%	21.1%	20.6%	20.3%	22.3%	20.6%	21.1%	22.3%	22.5%	
Yageo	6.1%	6.1%	5.7%	5.3%	5.7%	5.3%	5.3%	5.3%	5.3%	5.5%	5.3%	5.4%	6.0%	6.2%	4.3%	4.7%	4.4%	4.1%	4.3%	4.6%	4.7%	4.8%	5.7%	11.1%	3.5%	4.9%	6.2%	5.8%	5.9%	5.4%	5.4%	
Other	12.7%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	25.4%	19.7%	18.1%	17.1%	16.0%	14.8%	14.6%	14.2%	15.0%	17.0%	14.4%	14.9%	13.3%	12.9%	12.9%	13.0%	13.0%	
Market Share in Major 5																																
Murata	47.0%	45.3%	46.7%	47.8%	46.2%	45.8%	46.8%	48.2%	45.7%	46.9%	47.0%	47.8%	46.4%	46.7%	38.7%	39.1%	41.4%	42.4%	43.5%	44.5%	45.7%	47.8%	46.7%	41.5%	48.1%	47.4%	46.3%	47.5%	46.7%	46.8%	46.9%	
TDK	10.0%	9.9%	8.8%	9.0%	8.6%	8.7%	7.9%	8.1%	8.1%	8.0%	7.4%	8.4%	8.3%	8.3%	18.5%	16.7%	14.8%	13.4%	11.8%	11.2%	10.5%	10.4%	9.2%	7.8%	8.7%	8.0%	7.4%	8.8%	9.4%	8.5%	7.9%	
Taiyo Yuden	13.9%	12.6%	12.3%	13.1%	12.8%	12.4%	12.7%	13.2%	13.7%	13.0%	13.1%	12.3%	12.3%	11.4%	19.1%	18.7%	16.9%	16.5%	16.4%	15.5%	15.4%	15.3%	14.7%	12.5%	15.0%	15.0%	13.5%	13.5%	12.9%	12.8%	13.0%	
SEMCO	22.2%	25.3%	25.7%	23.7%	24.8%	27.0%	26.5%	24.3%	26.1%	25.7%	26.5%	25.3%	26.1%	26.6%	17.9%	19.7%	21.5%	22.8%	23.2%	23.4%	22.9%	20.8%	22.8%	25.1%	24.1%	23.9%	25.7%	23.6%	24.3%	25.7%	25.9%	
Yageo	7.0%	7.0%	6.5%	6.4%	6.6%	6.1%	6.1%	6.1%	6.4%	6.4%	6.1%	6.2%	6.9%	7.1%	5.7%	5.9%	5.3%	4.9%	5.1%	5.4%	5.5%	5.6%	6.7%	13.2%	4.1%	5.8%	7.2%	6.6%	6.7%	6.2%	6.2%	
YoY Growth	-29.4%	-17.1%	0.3%	6.2%	7.9%	4.7%	6.1%	3.5%	4.7%	12.4%	11.9%	11.9%	15.2%	18.6%	-28.9%	-17.2%	-0.1%	5.9%	7.5%	4.7%	4.9%	11.9%	15.2%	18.6%	18.5%	16.7%	14.8%	13.4%	11.8%	11.2%	10.5%	10.4%
Major 5	-27.6%	-20.8%	-3.0%	3.6%	5.5%	5.9%	5.1%	4.4%	3.7%	15.2%	12.5%	11.0%	17.0%	17.9%	-5.6%	-3.1%	-9.9%	-4.9%	3.7%	-8.1%	-5.8%	-6.0%	-11.5%	4.0%	4.9%	14.9%	18.1%	22.5%				
TDK	-28.9%	-18.5%	-11.9%	3.8%	-0.7%	3.3%	8.6%	4.5%	11.6%	17.6%	14.6%	4.6%	3.2%	4.0%	-28.9%	-18.5%	-11.9%	3.8%	-0.7%	3.3%	8.6%	4.5%	11.6%	17.6%	14.6%	4.6%	3.2%	4.0%				
Taiyo Yuden	-39.3%	-16.3%	19.0%	20.3%	20.4%	11.9%	8.1%	6.2%	10.0%	6.8%	11.9%	16.4%	15.3%	22.8%	-39.3%	-16.3%	19.0%	20.3%	20.4%	11.9%	8.1%	6.2%	10.0%	6.8%	11.9%	16.4%	15.3%	22.8%				
SEMCO	-23.9%	-9.9%	-1.2%	-1.9%	0.7%	-8.7%	-1.8%	-1.8%	2.2%	16.7%	11.2%	13.1%	23.7%	31.7%	-23.9%	-9.9%	-1.2%	-1.9%	0.7%	-8.7%	-1.8%	-1.8%	2.2%	16.7%	11.2%	13.1%	23.7%	31.7%				
Yageo	-32.0%	-16.8%	2.7%	8.7%	10.4%	4.7%	4.9%	3.5%	4.7%	12.4%	11.9%	11.9%	15.2%	18.6%	-32.0%	-16.8%	2.7%	8.7%	10.4%	4.7%	4.9%	3.5%	4.7%	12.4%	11.9%	11.9%	15.2%	18.6%				
Other	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%				
QoQ Growth	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%				
Major 5	-6.8%	2.3%	10.5%	-1.7%	-5.0%	2.5%	9.8%	-2.3%	-5.6%	13.9%	7.2%	-3.6%	-0.5%	14.7%	-6.8%	2.3%	10.5%	-1.7%	-5.0%	2.5%	9.8%	-2.3%	-5.6%	13.9%	7.2%	-3.6%	-0.5%	14.7%				
TDK	-3.2%	5.0%	-4.7%	-1.7%	5.5%	-6.9%	-2.4%	-1.9%	-0.7%	9.4%	-1.5%	7.4%	2.1%	14.7%	-3.2%	5.0%	-4.7%	-1.7%	5.5%	-6.9%	-2.4%	-1.9%	-0.7%	9.4%	-1.5%	7.4%	2.1%	14.7%				
Taiyo Yuden	1.1%	-3.9%	4.8%	2.0%	-3.4%	0.0%	10.1%	-1.8%	3.2%	5.4%	7.3%	-10.4%	1.8%	6.2%	1.1%	-3.9%	4.8%	2.0%	-3.4%	0.0%	10.1%	-1.8%	3.2%	5.4%	7.3%	-10.4%	1.8%	6.2%				
SEMCO	3.1%	21.0%	9.0%	-11.5%	3.2%	12.4%	5.3%	-13.0%	6.9%	9.2%	10.3%	-9.5%	5.8%	16.3%	3.1%	21.0%	9.0%	-11.5%	3.2%	12.4%	5.3%	-13.0%	6.9%	9.2%	10.3%	-9.5%	5.8%	16.3%				
Yageo	-2.1%	6.3%	-0.8%	-5.1%	0.5%	-3.6%	6.7%	-5.0%	4.6%	10.0%	1.7%	-3.4%	14.4%	17.2%	-2.1%	6.3%	-0.8%	-5.1%	0.5%	-3.6%	6.7%	-5.0%	4.6%	10.0%	1.7%	-3.4%	14.4%	17.2%				
Other	-3.0%	9.0%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%	-3.0%	9.0%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%				
OP																																
Murata	35.8	42.8	52.0	50.8	47.4	63.7	64.8	58.8	47.9	61.2	73.0	66.4	70.1	98.6	-8.8	31.9	22.3	19.3	50.6	72.2	115.6	73.3	105.8	176.1	190.5	196.6	312.1	270.5	181.4	234.7	248.4	
TDK	7.9	9.7	8.2	10.7	8.8	8.1	6.4	4.7	2.4	1.8	3.0	6.6	6.0	7.8	-20.3	-3.9	-9.4	-15.4	-0.2	3.4	5.9	3.3	7.4	19.0	20.4	16.1	27.6	43.3	36.5	27.9	13.8	
Taiyo Yuden	-2.9	0.0	0.8	2.0	1.3	1.6	2.4	0.5	2.9	3.5	4.5	5.8	3.1	3.8	-0.2	14.7	-0.6	0.4	9.1	6.7	15.9	6.9	15.4	33.0	41.8	37.5	55.4	43.4	-0.1	5.8	16.7	
SEMCO	7.5	12.3	10.8	7.9	8.9	14.5	17.0	14.0	11.5	14.3	18.1	15.2	22.6	28.8	12.2	19.1	12.3	8.6	11.1	8.1	20.5	11.9	26.1	46.1	44.5	48.7	96.4	58.9	38.5	54.4	59.0	
- SEMCO (KRW bn)	72.6	117.8	98.6	70.2	80.1	127.9	154.2	128.0	109.5	138.3	169.7	142.5	211.0	271.6	160.2	252.5	168.8	146.9	126.3	78.0	190.4	126.7	261.2	459.7	470.3	536.0	1,005.3	585.8	359.2	490.2	560.0	
OP (USD mn)																																
- Murata	270.8	311.1	359.4	343.5	319.5	408.5	434.0	385.9	313.7	423.2	495.1	431.1	446.6	618.4	-90.8	367.2	277.5	241.4	515.0	672.0	953.7	680.8	942.8	1,587.4	1,746.							

Supply Chains of MLCC and Aluminium Capacitors



Source: Company data, Morgan Stanley Research

MLCC Coverages: Medium-term Results and Forecasts

■ Murata Mfg. (6981)

(bn JPY, JPY/share)	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe
Sales	1,811.2	1,686.8	1,640.2	1,743.4	1,830.9	2,194.6	2,628.3	3,104.1	3,682.2	4,348.3
Components	984.3	911.2	933.8	1,033.1	1,159.7	1,526.1	1,838.7	2,156.5	2,528.1	2,918.2
Capacitors	788.5	738.8	753.5	831.8	936.4	1,274.5	1,577.0	1,884.3	2,245.0	2,623.8
Inductors and EMI Filters	195.8	172.3	180.3	201.3	223.3	251.6	261.7	272.2	283.1	294.4
Devices/Modules	815.0	761.0	695.2	697.2	656.0	653.2	776.7	934.7	1,141.1	1,417.0
High Frequency Devices/Communications Module	528.2	453.6	440.1	443.6	394.8	377.9	453.5	544.2	653.0	783.6
Battery/Power supply	180.4	214.6	164.4	155.7	154.1	162.3	203.4	263.5	353.5	490.7
Functional Devices	106.4	92.8	90.7	97.8	107.1	113.0	119.8	127.0	134.6	142.7
Others	11.9	14.7	11.2	13.1	15.2	15.3	13.0	13.0	13.0	13.0
OP	424.1	297.9	215.4	279.7	281.8	490.8	702.6	891.0	1,112.0	1,350.6
Components	355.5	280.1	234.2	275.2	315.1	501.4	671.4	834.5	1,029.9	1,236.3
Devices/Modules	69.7	20.6	-13.0	10.0	-26.5	-2.4	37.2	62.5	88.0	120.3
Others	-1.2	-2.8	-5.7	-5.4	-6.8	-8.2	-6.0	-6.0	-6.0	-6.0
OPM	23.4%	17.7%	13.1%	16.0%	15.4%	22.4%	26.7%	28.7%	30.2%	31.1%
ROE	15.0%	10.9%	7.3%	9.1%	8.8%	13.8%	18.2%	20.8%	22.6%	23.4%
EPS	163.7	133.8	95.7	125.1	127.7	211.5	302.0	385.3	484.1	592.5
FCF	209.2	118.4	288.1	243.8	231.4	208.5	341.2	468.1	580.4	731.8

■ Taiyo Yuden (6976)

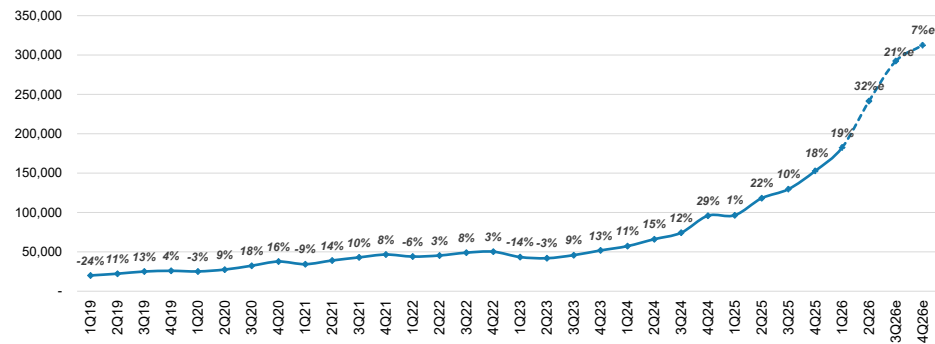
(bn JPY, JPY/share)	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe
Sales	282.3	300.9	349.6	319.5	322.6	341.4	355.3	428.7	486.0	558.4	630.0	686.6
Capacitor	176.5	195.2	230.4	208.1	205.8	232.1	251.8	320.8	374.3	441.6	507.9	558.7
Inductor	38.8	41.6	48.9	52.9	55.6	61.5	64.3	70.2	72.2	77.3	82.7	88.4
Complex devices	49.8	46.9	48.8	32.6	34.9	23.0	14.8	12.1	14.5	14.5	14.5	14.5
Others	17.3	17.2	21.5	25.9	26.3	24.8	24.5	25.6	25.0	25.0	25.0	25.0
OP	37.2	40.8	68.2	32.0	9.1	10.5	20.0	45.1	66.1	83.8	98.3	107.7
OPM	13.2%	13.5%	19.5%	10.0%	2.8%	3.1%	5.6%	10.5%	13.6%	15.0%	15.6%	15.7%
ROE	8.7%	12.6%	20.0%	7.5%	2.6%	0.7%	4.5%	8.5%	11.6%	13.6%	14.4%	14.2%
EPS	142.7	227.3	432.2	185.7	63.5	16.1	107.6	218.3	318.5	408.6	482.0	530.1
FCF	11.6	10.7	16.7	2.1	-31.7	-29.6	32.4	36.7	19.1	24.8	23.2	51.1

Note: MSe = Morgan Stanley Research estimates

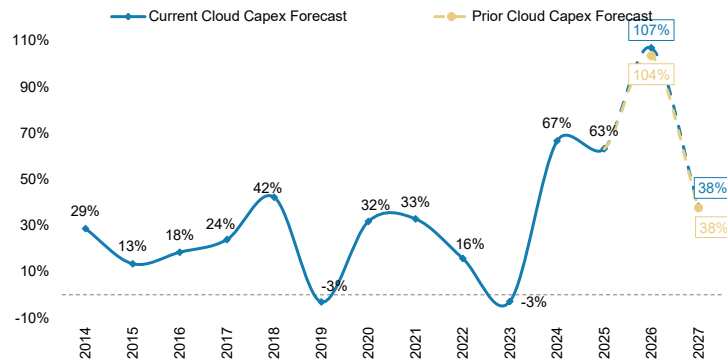
Source: Company data, Morgan Stanley Research

Capex by Hyperscalers Has Increased Significantly

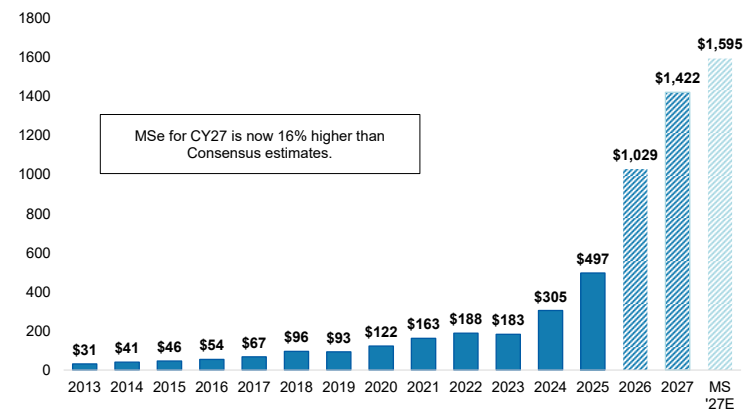
■ Total cash capex by 15 cloud suppliers and QoQ change



■ Total cash capex by 15 cloud suppliers and YoY change



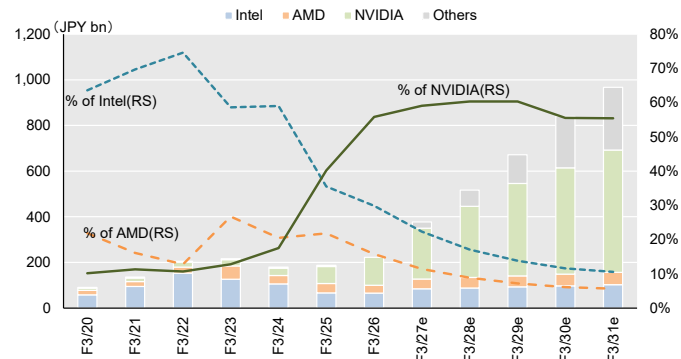
■ Total cash capex by 15 cloud suppliers



Note: MS'27E and e = Morgan Stanley Research estimates, Consensus is based on FactSet
Source: Company data, FactSet, Morgan Stanley Research

We Expect Ibiden Will Maintain Nearly 100% Market Share in FC Packages for NVIDIA

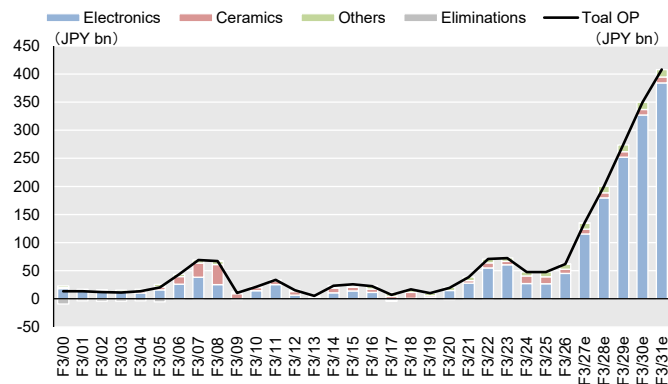
■ ABF package substrate sales by customers*



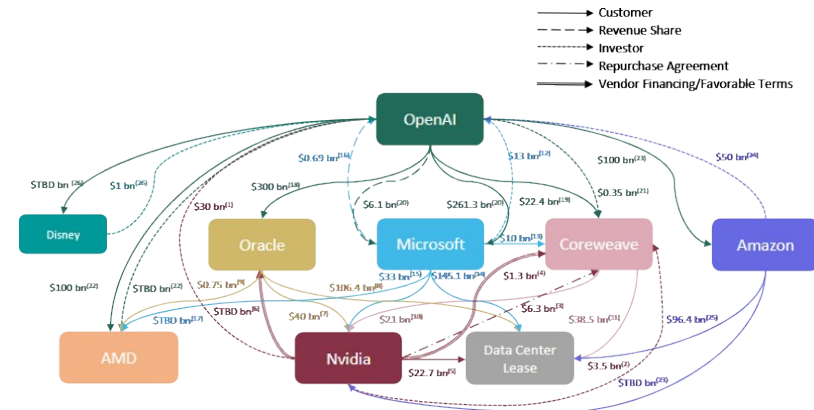
■ Size and # of layers for NVIDIA ABF packages

Nvidia	Ampere	Hopper	Blackwell	Rubin	Feynman
Package size	36,300	38,280	82,782	144,918	187,860
Substrate L (mm)	55	55	73	83	93
Substrate W (mm)	55	58	81	97	101
Layer count (L)	12	12	14	18	20
1 = 50,000	0.726	0.7656	1.65564	2.89836	3.7572

■ OP by segment and total OP



■ AI ecosystem capital flows



Note: *Morgan Stanley Research assumption based on company data, e = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

Ibiden: Medium-term Results and Forecasts

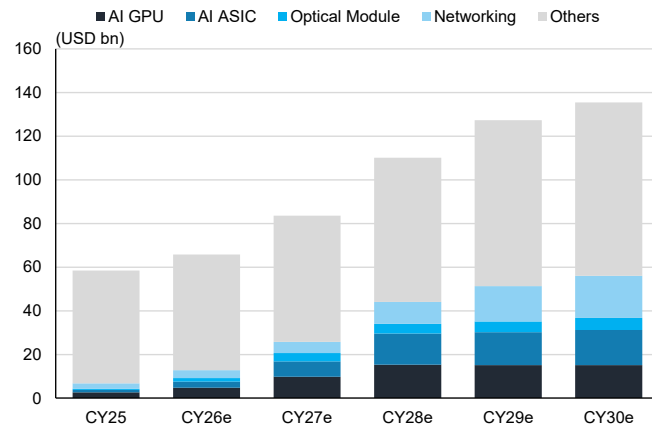
(bn JPY, JPY/share)	F3/07	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe	CAGR F3/25- F3/31e
Sales	398.6	291.1	296.0	323.5	401.1	417.5	370.5	369.4	416.2	562.5	701.0	864.2	1,036.8	1,169.4	21.2%
Electronics	205.0	116.0	132.2	166.1	237.0	250.7	190.7	197.2	243.3	385.7	525.5	680.4	848.3	975.9	30.5%
Ceramics	96.2	102.5	88.4	87.4	90.7	89.9	96.5	84.1	82.6	86.0	81.5	87.0	88.8	90.8	1.3%
Others	97.4	72.7	75.4	70.0	73.5	76.9	83.3	88.1	90.3	90.8	94.0	96.8	99.7	102.7	2.6%
OP	69.0	10.1	19.7	38.6	70.8	72.4	47.6	47.6	62.0	135.0	200.2	273.8	349.8	407.9	43.0%
Electronics	38.6	2.5	14.9	27.8	55.1	60.6	27.3	26.8	45.2	115.3	179.7	252.2	327.0	384.1	55.8%
Ceramics	24.9	3.0	-1.0	4.6	8.7	6.1	13.4	12.2	7.6	8.7	9.0	9.5	10.0	10.5	-2.5%
Others	5.9	4.7	5.8	6.6	7.1	6.1	7.1	8.7	9.0	10.9	11.6	12.1	12.7	13.3	7.9%
Eliminations	-0.4	-0.0	-0.1	-0.4	-0.1	-0.6	-0.1	-0.2	0.2	0.0	0.0	0.0	0.0	0.0	
OPM	17.3%	3.5%	6.7%	11.9%	17.7%	17.3%	12.8%	12.9%	14.9%	24.0%	28.6%	31.7%	33.7%	34.9%	
ROE	19.0%	1.2%	4.2%	8.8%	12.1%	13.3%	6.9%	6.8%	12.2%	15.7%	20.1%	22.6%	23.5%	22.3%	
EPS	80.9	5.9	20.3	46.0	73.8	93.4	56.4	60.3	107.5	157.5	237.2	324.5	414.6	483.7	
CapEx	70.4	22.9	57.1	78.2	60.6	131.3	146.6	157.3	64.3	157.6	158.0	208.0	208.0	208.0	
Electronics	54.1	14.2	48.7	70.4	54.6	121.5	136.9	146.8	58.1	147.2	150.0	200.0	200.0	200.0	
Ceramics	9.3	5.4	6.1	3.1	3.0	3.1	5.2	7.1	1.8	1.9	3.0	3.0	3.0	3.0	
Others	6.9	3.4	2.3	4.7	3.0	6.6	4.5	3.4	4.5	8.5	5.0	5.0	5.0	5.0	
Depreciation	31.5	25.1	24.2	35.4	52.7	54.9	46.0	54.2	66.2	81.4	95.5	110.5	130.5	130.5	
Electronics	16.6	13.8	12.3	25.5	43.1	45.4	36.7	44.3	55.2	70.9	86.0	101.0	121.0	121.0	
Ceramics	12.5	9.6	8.4	6.8	6.1	5.6	5.0	5.8	6.2	5.4	5.5	5.5	5.5	5.5	
Others	2.4	3.0	3.5	3.1	3.5	3.9	4.3	4.1	4.8	5.1	4.0	4.0	4.0	4.0	
FCF	7.7	0.5	-12.0	-43.4	40.7	21.7	68.0	-45.3	54.0	5.8	51.6	63.6	135.5	184.8	

Note: MSe = Morgan Stanley Research estimates

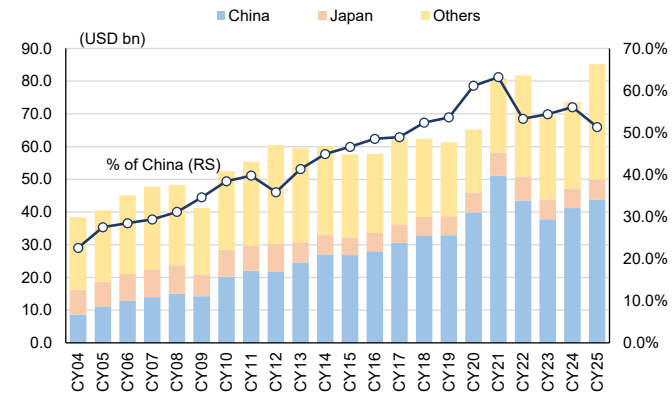
Source: Company data, Morgan Stanley Research

PCB Market Overview

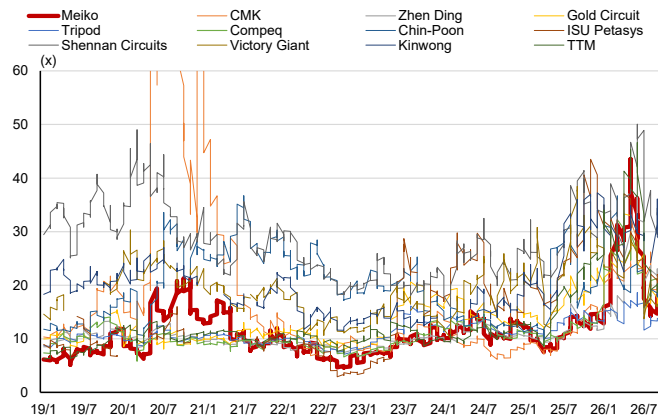
■ PCB market value



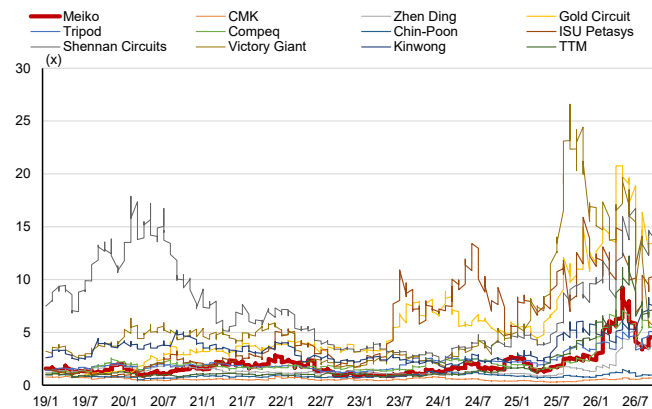
■ PCB/ABF Package production value by region



■ 1-year forward P/E for PCB suppliers



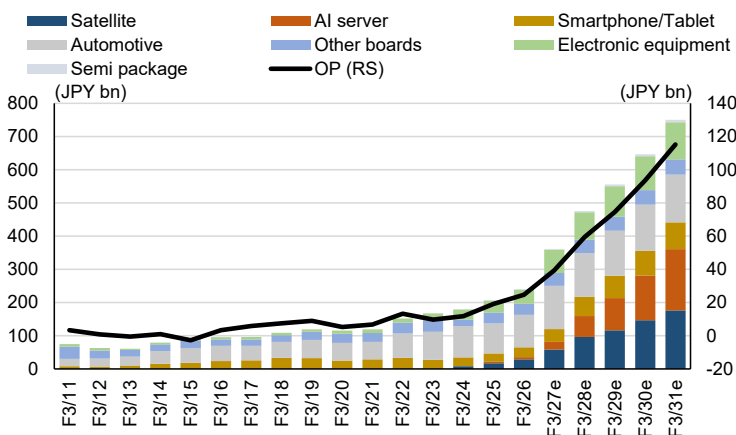
■ Actual P/B for PCB suppliers



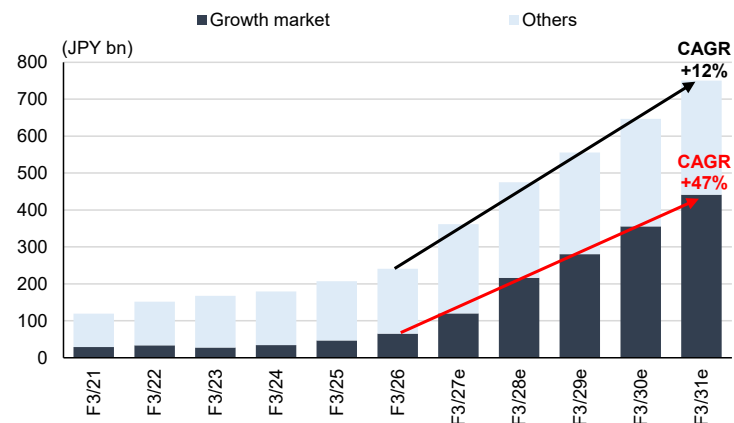
Note: e = Morgan Stanley Research estimates
Source: Prismark, Factset, Morgan Stanley Research

Meiko Electronics: Forecasts Overview

Sales by application/Total OP



Sales of growth market (LEO satellite, AI server, Smartphone/Tablet)



Mid-term forecasts

(JPY bn, JPY/share)	F3/24	F3/25	F3/26	F3/27 MSe	F3/27 Ce	F3/28 MSe	F3/28 Ce	F3/29 MSe	F3/29 Ce	F3/30 MSe	F3/31 MSe
Sales	179.5	206.8	240.6	361.5	320.0	475.4	390.0	555.4	460.0	646.8	750.1
Automotive	94.7	90.4	97.5	130.8	100.0	132.2	101.0	135.5	104.0	139.4	143.6
Smartphone/Tablet	23.6	25.5	29.4	38.2	37.0	57.8	44.0	67.8	54.0	74.1	81.0
Satellite	7.6	16.6	28.2	57.9	55.0	96.2	79.0	116.2	94.0	146.3	176.3
AI server	-	4.2	7.3	23.5	17.0	62.5	38.0	96.5	60.0	135.0	184.2
Moludes	-	9.2	9.2	13.7	13.0	16.0	16.0	17.4	18.0	19.0	20.7
Amusement/Smart appliance/Others	14.1	23.7	24.4	24.9	29.0	24.7	28.0	24.8	29.0	25.0	25.1
Electronic equipment	30.3	36.2	42.6	70.0	66.0	82.1	76.0	92.1	86.0	102.1	112.1
IC Package	0.7	1.0	2.0	2.5	3.0	4.0	8.0	5.0	15.0	6.0	7.0
OP	11.7	19.1	24.6	39.3	38.0	59.4	51.0	74.6	65.0	93.5	115.2
OPM	6.5%	9.2%	10.2%	10.9%	11.9%	12.5%	13.1%	13.4%	14.1%	14.5%	15.4%
ROE	12.9%	14.5%	16.2%	19.2%	17.1%	22.0%	20.1%	23.1%	21.1%	24.1%	24.5%
EPS	428.7	569.5	758.6	1,081.9	1,039.7	1,445.5	1,362.0	1,850.6	1,749.0	2,364.9	2,968.8
FCF	1.8	-2.7	-27.9	-46.6	-	-28.8	-	1.6	-	16.9	35.5

Note: e/MSe=Morgan Stanley Research estimates, Ce=Company estimates

Source: Company data, Morgan Stanley Research

Meiko Electronics: Overviews and Mid-term Guidances for Major Sites

Base		Characteristics	Key items produced	Plant area (K㎡)	Capacity (K㎡/M)
Japan	Kanagawa	Test production/development/small-lot production. A pilot base for development	MLB, HDI, FPC, F/R, Metal	9.8	5.0
	Fukushima	Medium-lot mass production. Mainly focused on amusement business	MLB	5.7	20.0
	Kahoku	Autos	MLB, HDI	17.3	30.0
	Tendo	A smart factory. HDI production including for the domestic autos business	MLB, HDI	65.0	30.0
	Ishinomaki 1	Advanced substrates (MSAP, ultra-thick copper, built-in parts, etc)	HDI, IC Package	19.0	2.0
	Ishinomaki 2	Production of consumer electronics/autos legacy FC-BGA; SAP only	IC Package		
China	Guangzhou	Mass production of automotive MLB/HDI	MLB, HDI	75.0	165.0
	Wuhan 1	Mass production of automotive products, HDD/SSD, OA MLB	MLB, Metal	120.0	230.0
	Wuhan 2	Mass production of high-end HDI for smartphones	HDI		
Vietnam	Vietnam 1	Suited for the production of FPC, F/R and some EMS	FPC, F/R	191.0	195.0
	Vietnam 2	Mainly automotive MLB	MLB, HDI, FPC, Metal		
	Vietnam 3	Production of HDI for high-end smartphones, FCBGA for memories/sensors; MSAP compatible	HDI, IC Package		
	Vietnam 4	Securing capacity for No.2 plant, production of super-multilayer, high-density HDI, module substrates for memories	HDI, IC Package		
	Thang Long	HDI for high-end smartphones, module substrates	HDI, IC Package	31.0	20.0
Sites under construction		Purpose	Key items produced	Plant area (K㎡)	Capacity (K㎡/M)
Vietnam	Hoa Binh 1	Production of mid-high layer, high-density HDI for new smartphone customers and opt. transceivers; MSAP	HDI	93.0	-
	Quang Minh 3	Production of high-density HDI for existing high-end smartphone customers	HDI	63.0	-
	Yen Quang 1	Production of HDI for satellite communications	HDI	110.0	-
	Hoa Binh 2 (JV)	Production of super-multilayer MLB for AI servers	MLB	(60 for floor area)	-

(JPY bn)	Major applications	CapEx	F3/27 Ce	F3/28 Ce	F3/29 Ce	F3/30 Ce
Vietnam 4	Satellites, Accelerator boards, eSSD	45	50	70	75	80
Hoa Binh 1	High-end smartphones, Opt. transceivers	50	7	20	30	50
Quang Ming 3	High density HDI incl. high-end smartphones	40	-	20	30	50
Yen Quang	Satellites	60	-	20	40	80
Hoa Binh 2	Main board for AI servers (JV with ACCL in TW)	-	Undisclosed, sales will be recorded from F3/28			

Note: Ce=Company guidance

Source: Company data, Morgan Stanley Research

HDD Market by Company (Quarterly, Fiscal Year)

(mn units)	CY23				CY24				CY25				CY26																			
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25	CY26e	CY27e	CY28e	
HDD Shipment (by Size)																																
Total	33.5	31.3	28.6	28.9	29.7	30.3	32.1	31.7	28.7	30.4	31.6	33.0	33.0	33.0		564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	130.4	125.2	124.9		
Enterprise	1.4	1.6	1.4	1.4	1.5	1.0	1.0	0.9	0.8	0.7	0.8	0.8	0.6			32.4	32.1	24.8	22.6	21.5	17.4	12.9	13.4	8.4	5.8	4.4	3.2	2.7	1.9	1.1		
NL	11.4	9.2	9.0	9.9	12.3	13.8	15.9	16.5	14.6	16.5	17.3	19.20	18.7	19.1		36.9	37.2	41.0	42.4	51.3	52.7	59.1	73.5	61.5	39.4	58.6	67.6	76.4	78.6	81.8		
3.5"ATA	11.2	12.1	11.2	10.7	9.3	9.4	9.0	7.9	7.6	7.9	8.0	7.6	8.3	8.7		202.6	161.0	142.9	135.4	120.5	103.9	87.0	89.9	55.0	45.2	35.6	30.9	31.5	27.5	26.0		
2.5"	9.5	8.4	7.1	6.9	6.6	6.1	6.2	6.4	5.7	5.3	5.6	5.5	5.3	4.7		291.7	238.6	215.1	202.7	182.0	143.6	101.0	82.5	46.3	32.0	25.3	22.0	19.8	17.3	16.0		
1.8"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
YoY % Change	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%	9%		2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-33.6%	-29%	1%	0%	5%	-4%	0%		
Enterprise	-48%	-34%	-15%	-16%	6%	-36%	-30%	-33%	-45%	-21%	-23%	-16%	7%	-26%		2%	-1%	-23%	-9%	-5%	-19%	-26%	3%	-37%	-32%	-23%	-29%	-15%	-30%	-41%		
NL	-39%	-49%	-37%	-7%	8%	51%	78%	67%	19%	19%	9%	16%	28%	16%		4%	1%	10%	3%	21%	3%	12%	24%	-16.3%	-36%	49%	15.5%	12.9%	2.9%	4.1%		
3.5"ATA	-33%	-4%	-14%	-16%	-16%	-22%	-19%	-26%	-19%	-17%	-12%	-4%	9%	11%		0%	-21%	-11%	-5%	-11%	-14%	-16%	3%	-39%	-18%	-21%	-13%	2%	-13%	-5%		
2.5"	-36%	-20%	-26%	-39%	-30%	-28%	-13%	-7%	-14%	-13%	-10%	-14%	-7%	-11%		4%	-18%	-10%	-6%	-10%	-21%	-30%	-18%	-44%	-31%	-21%	-10%	-13%	-8%			
1.8"	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-6%	-98%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Sequential % Change	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%	0%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Enterprise	-16%	13%	-16%	4%	7%	-32%	-7%	-1%	-13%	-1%	-10%	8%	-4%	-21%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
NL	8%	-20%	2%	11%	24%	12%	15%	4%	-12%	13%	5%	11%	-3%	2%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
3.5"ATA	-13%	9%	-7%	-4%	-13%	1%	-4%	-13%	-4%	4%	2%	-5%	10%	5%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
2.5"	-16%	-11%	-16%	-3%	-4%	-8%	2%	3%	-11%	-8%	6%	-1%	-3%	-12%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
1.8"	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
HDD Shipment (by Company)																																
Total	33.5	31.3	28.6	28.9	29.7	30.3	32.2	31.7	28.7	30.4	31.6	33.0	33.0	33.0		564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	130.4	125.2	124.9		
Seagate	14.5	14.3	12.0	11.6	11.4	12.3	13.0	13.0	11.4	12.5	12.5	13.1	13.1	13.3		224.1	188.6	155.2	148.2	150.0	129.1	126.7	110.3	74.1	52.4	49.7	49.4	-	-	-	-	
Western Digital	12.6	11.8	10.4	10.8	11.7	12.1	13.2	13.5	12.1	12.9	13.9	14.3	14.4	14.6		249.3	215.6	191.4	161.8	139.7	114.2	110.8	94.3	63.9	45.6	50.5	53.2	-	-	-	-	
Toshiba	6.4	5.2	6.2	6.5	6.7	5.9	6.0	5.2	5.2	5.0	5.2	5.7	5.6	5.2		90.7	75.8	93.5	93.0	85.7	74.4	70.0	54.7	34.3	24.3	23.8	21.1	-	-	-	-	
Market Share																																
Seagate	43%	46%	42%	40%	38%	41%	40%	41%	40%	41%	39%	39%	40%	40%		40%	40%	37%	37%	40%	41%	49%	43%	43%	43%	40%	40%	-	-	-	-	
Western Digital	38%	38%	36%	37%	39%	40%	41%	43%	42%	42%	44%	43%	44%	44%		44%	46%	45%	40%	37%	36%	43%	36%	37%	37%	41%	43%	-	-	-	-	
Toshiba	19%	17%	22%	22%	23%	20%	19%	17%	18%	16%	17%	17%	17%	16%		16%	16%	22%	23%	23%	23%	27%	21%	20%	19%	17%	17%	-	-	-	-	
YoY % Change	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%	9%		2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-34%	-29%	1%	0%	5%	-4%	0%		
Seagate	-37%	-28%	-24%	-25%	-21%	-14%	8%	12%	0%	2%	-4%	0%	15%	6%		1%	-16%	-18%	-4%	1%	-14%	-2%	-13%	-33%	-29%	-5%	0%	-	-	-	-	
Western Digital	-36%	-28%	-29%	-16%	-7%	3%	27%	25%	3%	7%	5%	6%	19%	13%		1%	-14%	-11%	-15%	-14%	-18%	-3%	-15%	-32%	-29%	11%	5%	-	-	-	-	
Toshiba	-37%	-37%	-22%	-19%	5%	14%	-4%	-19%	-22%	-16%	-12%	8%	7%	5%		8%	-16%	23%	-1%	-8%	-13%	-6%	-22%	-37%	-29%	-2%	-11%	-	-	-	-	
Sequential % Change	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%	0%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Seagate	-6%	-1%	-16%	-3%	-2%	8%	5%	0%	-12%	10%	0%	5%	0%	1%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Western Digital	-2%	-6%	-12%	4%	8%	3%	9%	2%	-10%	7%	8%	3%	0%	1%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toshiba	-21%	-19%	20%	5%	3%	-11%	0%	-12%	-1%	-4%	5%	8%	-2%	-6%		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
HDD Shipment Capacity (EB, by Company)																																
Total	236.0	189.8	196.1	214.3	264.6	289.4	340.3	362.3	346.6	391.8	432.5	457.0	477.2	504.3		534.9	535.8	622.6	720.7	856.3	902.5	1,019.1	1,344.8	1,163.5	836.2	1,256.6	1,627.9	2,043.1	2,371.3	2,883.9		
Seagate	118.7	91.2	89.6	95.1	99.1	114.2	137.5	150.8	143.6	162.5	181.7	190.1	199.0	218.0		221.6	222.7	252.2	287.2	366.5	367.0	480.6	614.2	539.5	394.6	501.6	677.9	-	-	-	-	
Western Digital	96.5	78.8	79.1	90.3	127.3	142.6	163.8	172.5	162.0	187.0	204.1	215.0	222.0	231.0		240.1	250.1	286.1	342.7	390.0	422.7	422.3	544.0	476.3	344.7	606.2	768.1	-	-	-	-	
Toshiba	20.8	19.8	27.4	28.9	38.2	32.6	39.2	39.0	41.0	42.3	46.7	51.9	56.1	55.3		73.2	63.0	84.3	90.8	98.8	112.9	116.2	187.5	147.7	96.9	149.0	181.9	-	-	-	-	
Market Share																																
Seagate	50%	48%	46%	44%	37%	39%	40%	42%	41%	41%	42%	42%	42%	43%		41%	42%	41%	40%	43%	41%	47%	46%	46%	47%	40%	42%	-	-	-	-	
Western Digital	41%	42%	40%	42%	48%	49%	48%	48%	47%	48%	47%	47%	47%	46%		45%	47%	46%	48%	46%	47%	41%	40%	41%	41%	48%	47%	-	-	-	-	
Toshiba	9%	10%	14%	13%	14%	11%	12%	11%	11%	12%	11%	11%	12%	11%		14%	12%	14%	13%	12%	13%	11%	14%	13%	12%	12%	11%	-	-	-	-	
YoY % Change	-28%	-41%	-31%	-6%	12%	52%	74%	69%	31%	35%	27%	26%	38%	29%		16%	0%	16%	16%	19%	6%	13%	32%	-13%	-28%	50%	30%	26%	16%	22%		
Seagate	-23%	-41%	-24%	-15%	-17%	25%	53%	59%	45%	42%	32%	26%	39%	34%		14%	0%	13%	14%	28%	0%	31%	28%	-12%	-27%	27%	35%	-	-	-	-	
Western Digital	-27%	-39%	-39%	8%	32%	81%	107%	91%	27%	31%	25%	25%	37%	24%		19%	4%	14%	20%	14%	8%	0%	29%	-12%	-28%	76%	27%	-	-	-	-	
Toshiba	-51%	-45%	-26%	-10%	84%	65%	43%	35%	7%	30%	19%	33%	37%	31%		14%	-14%	34%	8%	9%	14%	3%	61%	-21%	-34%	54%	22%	-	-	-	-	
Sequential % Change	3%	-20%	3%																													

HDD Head Market

	CY23		CY24		CY25		CY26			CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25					
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q												
HDD Head Shipment																										
Total	262.2	202.9	217.1	237.7	262.5	297.1	327.3	332.4	330.0	348.7	377.6	397.8	401.9	414.8	1,931.6	1,756.1	1,702.8	1,714.4	1,715.4	1,496.2	1,517.9	1,747.9	1,349.0	919.9	1,219.3	1,454.1
TDK	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8	60.6	471.6	376.0	375.0	361.9	330.2	299.6	254.4	338.2	230.8	141.3	183.4	193.8
Seagate	118.8	92.1	86.9	101.7	103.3	116.1	132.8	140.7	128.3	140.8	155.1	165.2	165.4	166.9	634.0	616.2	617.1	636.6	683.5	594.1	652.2	755.2	579.1	399.5	492.9	589.4
Western Digital	103.8	84.4	95.2	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	183.7	187.3	826.0	763.9	710.7	715.9	701.7	602.5	611.3	654.5	539.1	379.1	543.0	670.9
Market Share																										
TDK	15%	13%	16%	17%	16%	16%	15%	13%	15%	13%	13%	13%	13%	15%	24%	21%	22%	21%	19%	20%	17%	19%	17%	15%	15%	13%
Seagate	45%	45%	40%	43%	39%	39%	41%	42%	39%	40%	41%	42%	41%	40%	33%	35%	36%	37%	40%	40%	43%	43%	43%	43%	40%	41%
Western Digital	40%	42%	44%	40%	44%	45%	44%	45%	47%	47%	46%	46%	46%	45%	43%	43%	42%	42%	41%	40%	40%	37%	40%	41%	45%	46%
YoY % Change	-38%	-47%	-33%	9%	0%	46%	51%	40%	26%	17%	15%	20%	22%	19%	0%	-9%	-3%	1%	0%	-13%	1%	15%	-23%	-32%	33%	19%
TDK	-45%	-53%	-41%	-5%	9%	80%	41%	8%	11%	-5%	1%	18%	10%	34%	-3%	-20%	0%	-3%	-9%	-9%	-15%	33%	-32%	-39%	30%	6%
Seagate	-40%	-47%	-32%	26%	-13%	26%	53%	38%	24%	21%	17%	17%	29%	19%	1%	-3%	0%	3%	7%	-13%	10%	16%	-23%	-31%	23%	20%
Western Digital	-33%	-44%	-31%	2%	12%	58%	53%	55%	33%	22%	19%	22%	19%	15%	2%	-8%	-7%	1%	-2%	-14%	1%	7%	-18%	-30%	43%	24%
HDD Head/HDD	7.84	6.49	7.59	8.22	8.83	9.80	10.20	10.47	11.50	11.47	11.95	12.04	12.18	12.56	3.42	3.75	4.02	4.25	4.57	4.71	5.84	6.74	7.83	7.52	9.84	11.75
TDK's Head Shipment by Customer																										
Total	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8	60.6	471.6	376.0	375.0	361.9	330.2	299.6	254.4	338.2	230.8	141.3	183.4	193.8
Seagate	4.3	3.7	1.0	0.3	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.2	6.4	17.4	14.0	25.7	24.0	41.3	51.7	43.1	45.5	28.1	9.3	0.3	0.0
Maxtor	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
HGST	2.9	2.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7	4.4	13.1	16.0	11.9	21.5	22.6	1.4	0.0	4.8	7.6	5.1	0.0	0.0
Western Digital	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	43.7	17.5	16.2	12.2	5.7	1.4	0.0	0.0	0.0	0.0	0.0	0.0
Samsung	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	130.2	86.0	21.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Toshiba	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9	49.8	267.2	242.5	299.8	304.2	260.6	245.1	211.3	287.9	195.1	126.9	183.1	193.8
Fujitsu	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Others	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
TDK's Shipment Share by Customer																										
Seagate	4%	4%	1%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	4%	3%	2%	4%	4%	6%	8%	6%	6%	5%	2%	0%	0%
Maxtor	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
HGST	3%	3%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	3%	4%	5%	3%	6%	6%	0%	0%	1%	2%	2%	0%	0%
Western Digital	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	8%	4%	4%	3%	2%	1%	0%	0%	0%	0%	0%	0%
Samsung	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toshiba	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Fujitsu	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Others	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
YoY % Change	-45%	-53%	-41%	-5%	9%	80%	41%	8%	11%	-5%	1%	18%	10%	34%	-3%	-20%	0%	-3%	-9%	-9%	-15%	33%	-32%	-39%	30%	6%
Sequential % Change	-7%	-33%	33%	15%	7%	9%	4%	-12%	10%	-6%	10%	3%	4%	15%	-	-	-	-	-	-	-	-	-	-	-	-

Source: TSR, Morgan Stanley Research

HDD Motor Market

	CY23			CY24			CY25			CY26																										
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q																						
HDD Motor Shipment																																				
Total	34.2	27.2	31.6	29.5	30.5	32.7	33.2	32.6	30.8	31.7	32.5	34.3	34.0	34.1					575.2	476.5	431.7	407.3	381.6	307.7	271.8	248.8	146.1	122.5	129.0	129.3						
Nidec	19.2	13.2	19.1	17.4	18.6	18.9	19.3	19.2	18.8	18.8	20.0	20.2	20.4	20.5					408.6	380.3	362.6	341.9	317.6	250.1	210.9	133.7	82.1	68.9	76.0	77.8						
Minebea Mitsumi	15.0	14.0	12.5	12.1	11.9	13.8	13.9	13.4	12.0	12.9	12.5	14.1	13.6	13.6					70.9	67.7	69.1	65.4	63.8	57.5	60.9	115.1	64.0	53.6	53.0	51.5						
Market Share																																				
Nidec	56%	49%	60%	59%	61%	58%	58%	59%	61%	59%	62%	59%	60%	60%					71%	80%	84%	84%	83%	81%	78%	54%	56%	56%	59%	60%						
Minebea Mitsumi	44%	51%	40%	41%	39%	42%	42%	41%	39%	41%	38%	41%	40%	40%					12%	14%	16%	16%	17%	19%	22%	46%	44%	44%	41%	40%						
YoY % Change																																				
Nidec	-32%	-27%	2%	8%	-11%	20%	5%	11%	1%	-3%	-2%	5%	10%	8%					3%	-17%	-9%	-6%	-6%	-19%	-12%	-8%	-41%	-16%	5%	0%						
Nidec	-25%	-30%	3%	-9%	-3%	43%	1%	10%	1%	-1%	4%	5%	9%	9%					4%	-7%	-5%	-6%	-7%	-21%	-16%	-37%	-39%	-16%	10%	2%						
Minebea Mitsumi	-40%	-25%	1%	49%	-21%	-1%	11%	11%	1%	-7%	-10%	5%	13%	5%					-4%	-5%	2%	-5%	-2%	-10%	6%	89%	-44%	-16%	-1%	-3%						
Sequential % Change																																				
Nidec	25%	-20%	16%	-7%	3%	7%	2%	2%	-2%	-6%	3%	6%	-1%	0%					-	-	-	-	-	-	-	-	-	-	-	-						
Nidec	0%	-31%	45%	-9%	7%	2%	2%	-1%	-2%	0%	6%	1%	1%	0%					-	-	-	-	-	-	-	-	-	-	-	-						
Minebea Mitsumi	85%	-7%	-11%	-3%	-2%	16%	1%	-4%	-10%	8%	-3%	13%	-4%	0%					-	-	-	-	-	-	-	-	-	-	-	-						
HDD Motor/HDD																																				
	1.02	0.87	1.10	1.02	1.03	1.08	1.03	1.03	1.07	1.04	1.03	1.04	1.03	1.03					1.02	1.02	1.02	1.01	1.02	0.97	1.05	0.96	0.85	1.00	1.04	1.04						
Nidec's Motor Shipment by Size																																				
Total	19.2	13.2	19.1	17.4	18.6	18.9	19.3	19.2	18.8	18.8	20.0	20.2	20.4	20.5					408.6	380.3	362.6	341.9	317.6	250.1	210.9	133.7	82.1	68.9	76.0	77.8						
Enterprise	0.6	0.5	0.7	0.8	0.8	0.4	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.3					23.1	22.4	15.0	14.9	13.0	7.3	4.6	3.4	2.9	2.6	1.9	1.6						
NL	7.0	4.0	7.1	6.3	8.6	9.5	10.4	10.9	11.1	11.4	11.3	11.8	12.1	12.4					28.6	31.9	32.5	39.2	50.0	48.6	59.7	41.8	31.0	24.4	39.4	45.6						
3.5"ATA	5.9	4.6	6.0	5.4	5.2	4.7	4.3	3.7	3.4	3.6	4.3	4.3	4.5	4.5					146.4	143.1	134.2	124.6	108.2	87.6	76.7	45.2	26.2	21.9	17.9	15.6						
2.5"	5.7	4.1	5.3	4.9	4.0	4.3	4.3	4.2	3.9	3.4	4.0	3.7	3.5	3.3					210.2	209.2	180.9	163.2	146.5	106.9	69.9	43.5	22.2	20.0	16.8	15.0						
1.8"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0					0.6	0.4	-	-	-	-	-	-	-	-	-	-						
1"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0					0.0	0.0	-	-	-	-	-	-	-	-	-	-						
As % of Total																																				
Enterprise	3%	4%	4%	5%	4%	2%	2%	2%	2%	2%	2%	2%	2%	1%					6%	6%	4%	4%	4%	3%	2%	3%	4%	4%	3%	2%						
NL	36%	30%	37%	36%	46%	50%	54%	57%	59%	61%	57%	58%	59%	60%					7%	8%	9%	11%	16%	19%	28%	31%	38%	35%	52%	59%						
3.5"ATA	31%	35%	31%	31%	28%	25%	22%	19%	18%	19%	22%	21%	22%	22%					36%	38%	37%	36%	34%	35%	36%	34%	32%	32%	24%	20%						
2.5"	30%	31%	28%	28%	22%	23%	22%	22%	21%	18%	20%	18%	17%	16%					51%	55%	50%	48%	46%	43%	33%	33%	27%	29%	22%	19%						
1.8"	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%					0%	0%	-	-	-	-	-	-	-	-	-	-						
1"	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%					0%	0%	-	-	-	-	-	-	-	-	-	-						
YoY % Change																																				
Enterprise	-25%	-30%	3%	-9%	-3%	43%	1%	10%	1%	-1%	4%	5%	9%	9%					4%	-7%	-5%	-6%	-7%	-21%	-16%	-37%	-39%	-16%	10%	2%						
Enterprise	-33%	0%	40%	-20%	33%	-20%	-57%	-50%	-50%	0%	33%	0%	0%	-25%					-3%	-3%	-33%	-1%	-13%	-44%	-37%	-26%	-15%	-10%	-27%	-16%						
NL	-24%	-57%	6%	9%	23%	138%	46%	73%	29%	20%	9%	8%	9%	9%					20%	12%	2%	21%	28%	-3%	23%	-30%	-26%	-21%	61%	16%						
3.5"ATA	-24%	-57%	6%	9%	23%	138%	46%	73%	29%	20%	9%	8%	9%	9%					0%	-2%	-6%	-7%	-13%	-19%	-12%	-41%	-42%	-16%	-18%	-13%						
2.5"	-34%	-10%	3%	-14%	-12%	2%	-28%	-31%	-35%	-23%	0%	16%	32%	25%					6%	0%	-14%	-10%	-10%	-27%	-35%	-38%	-49%	-10%	-16%	-11%						
Sequential % Change																																				
Enterprise	0%	-31%	45%	-9%	7%	2%	2%	-1%	-2%	0%	6%	1%	1%	0%					-	-	-	-	-	-	-	-	-	-	-	-						
Enterprise	-40%	-17%	40%	14%	0%	-50%	-25%	33%	0%	0%	0%	0%	0%	-25%					-	-	-	-	-	-	-	-	-	-	-	-						
NL	21%	-43%	78%	-11%	37%	10%	9%	5%	2%	3%	-1%	4%	3%	2%					20%	12%	2%	21%	28%	-3%	23%	-30%	-26%	-21%	61%	16%						
3.5"ATA	21%	-43%	78%	-11%	37%	10%	9%	5%	2%	3%	-1%	4%	3%	2%					-	-	-	-	-	-	-	-	-	-	-	-						
2.5"	-6%	-22%	30%	-10%	-4%	-10%	-9%	-14%	-8%	6%	19%	0%	5%	0%					-	-	-	-	-	-	-	-	-	-	-	-						

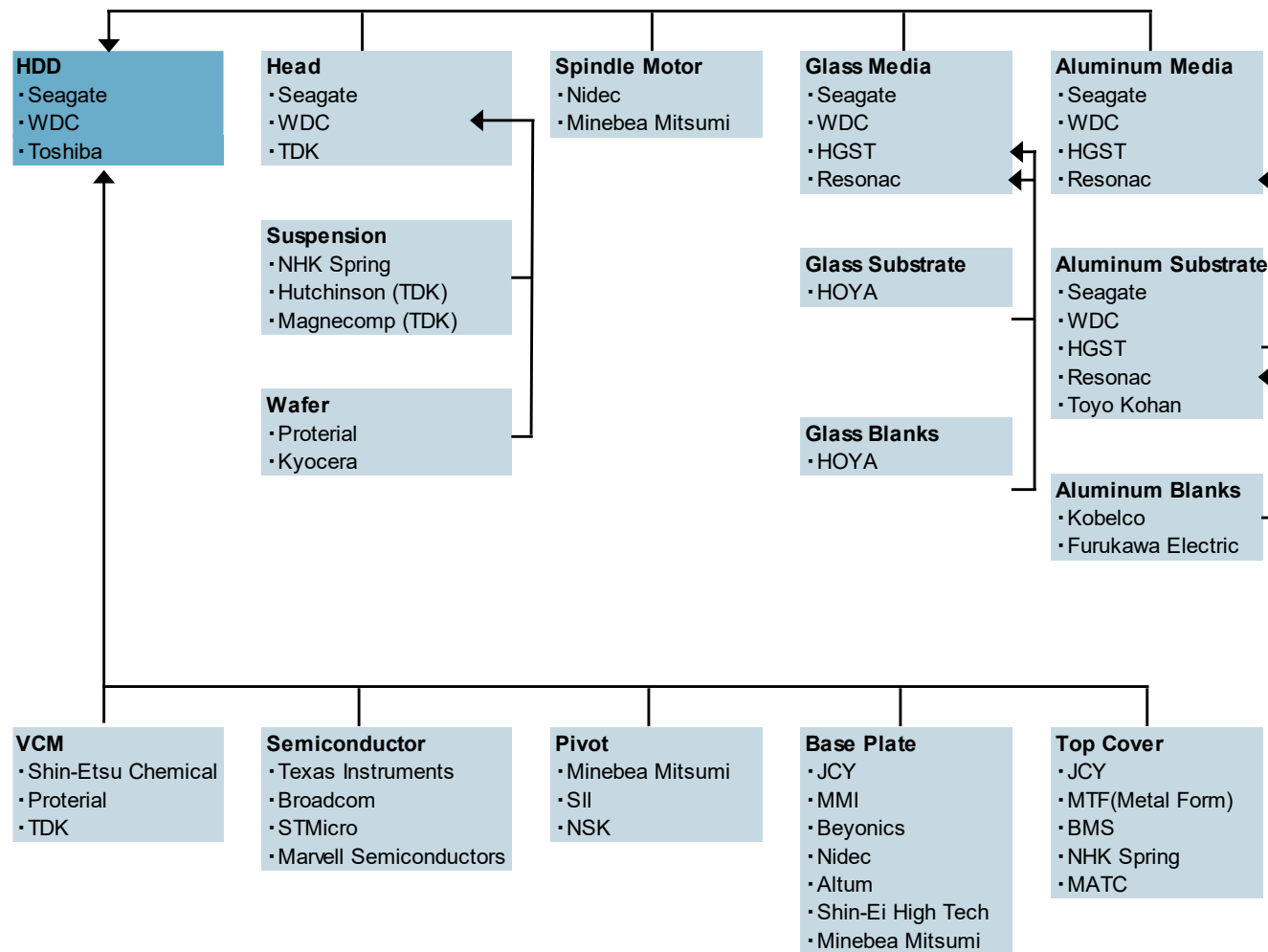
Source: TSR, Morgan Stanley Research

Trends in In-house Production of HDD Heads/Media

	CY23				CY24				CY25				CY26					CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q		F3/15	F3/16	F3/17	F3/18	F3/19	F3/20e	F3/21e	F3/22e	F3/22e	F3/22e				
Seagate																													
Total demand	123.1	95.8	87.9	102.0	103.4	116.2	132.9	140.7	128.3	140.8	155.1	165.2	165.6	173.3		781.6	717.2	664.2	677.2	693.7	650.6	695.3	800.7	607.2	408.8	493.2	589.4		
ex-Seagate	123.1	95.8	87.9	102.0	103.4											651.4	599.2	615.8	677.2	693.7	650.6	695.3	800.7	607.2	408.8	103.4			
ex-Seagate	118.8	92.1	86.9	101.7	103.3	116.1	132.8	140.7	128.3	140.8	155.1	165.2	165.4	166.9		634.0	585.2	590.1	653.7	646.2	600.3	652.2	755.2	579.1	399.5	492.9			
Ratio of in-house production	97%	96%	99%	100%	100%	100%	100%	100%	100%	100%	100%	100%	99.9%	96.3%		81%	86%	93%	97%	93%	92%	94%	94%	95%	98%	100%	100%		
WD																													
Total demand	106.7	86.4	95.4	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	184.4	192.2		882.8	797.4	738.8	749.2	676.2	612.1	611.3	659.3	546.7	384.2	543.0	670.9		
ex-WD	20.7	18.8	17.2	16.5	15.0	15.0	15.2	14.5	13.8	14.2	15.8	15.6	15.2	14.3		526.2	460.8	386.4	395.0	314.4	228.2	216.5	195.5	95.1	73.2	59.7	59.4		
ex-HGST	86.0	67.6	78.2	79.2	100.9	118.6	130.0	133.8	140.0	148.5	157.0	166.0	169.2	177.9		356.6	336.6	352.4	354.2	361.8	383.9	394.8	463.8	451.6	311.0	483.3	611.5		
Volume of in-house production	103.8	84.4	95.2	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	183.7	187.3		826.0	763.9	710.7	717.0	654.2	611.2	611.3	654.5	539.1	379.1	543.0	670.9		
ex-WD	20.7	18.8	17.2	16.5	15.0	15.0	15.2	14.5	13.8	14.2	15.8	15.6	15.2	14.3		482.5	443.3	370.2	385.3	309.4	227.3	216.5	195.5	95.1	73.2	59.7	59.4		
ex-HGST	83.1	65.6	78.0	79.2	100.9	118.6	130.0	133.8	140.0	148.5	157.0	166.0	168.5	173.0		343.5	320.6	340.5	331.7	344.8	383.9	394.8	459.0	444.0	305.9	483.3	611.5		
Shipment volume from TDK	2.9	2.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7	4.4		56.8	33.5	28.1	32.2	22.0	0.9	0.0	4.8	7.6	5.1	0.0	0.0		
Ratio of in-house production	97%	98%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	99.6%	97.5%		94%	96%	96%	96%	97%	100%	100%	99%	99%	99%	100%	100%		
Toshiba																													
Total demand	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9	49.8		267.2	242.5	299.8	294.2	252.2	254.1	211.3	287.9	195.1	126.9	183.1	193.8		
Volume of in-house production	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Shipment volume from TDK	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9	49.8		267.2	242.5	299.8	294.2	252.2	254.1	211.3	287.9	195.1	126.9	183.1	193.8		
Ratio of in-house production	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%		0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%		
Industry Total																													
Total demand	262.2	202.9	217.1	237.7	262.5	297.1	327.3	332.4	330.0	348.7	377.6	397.8	401.9	415.3		1,931.6	1,757.1	1,702.8	1,720.6	1,622.1	1,516.8	1,517.9	1,747.9	1,349.0	1,185.1	1,219.3	1,454.1		
Shipment volume from TDK	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8	60.6		471.6	377.0	375.0	349.9	321.7	305.3	254.4	338.2	230.8	198.1	183.4	193.8		
Market share of TDK	15%	13%	16%	17%	16%	16%	15%	13%	15%	13%	13%	13%	13%	15%		24%	21%	22%	20%	20%	17%	19%	17%	17%	15%	13%			
	CY23				CY24													CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24e	
Seagate																													
Total demand	64.1	48.7	44.8	51.8	53.2	59.8	67.0	71.0	67.7	73.9	77.6	82.6	84.0	86.4		395.5	359.0	330.6	330.3	363.6	327.1	351.8	404.7	308.7	209.4	251.0	301.8		
ex-Seagate																325.6	299.6	306.4	330.3	363.6	327.1	351.8	402.2	100.9	-	0%	0.0		
ex-Seagate																283.1	258.7	280.6	301.6	325.1	281.6	306.1	363.6	92.9	0.0	0.0	0.0		
Shipment volume from Fuji Elec	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		51.0	42.3	33.1	28.7	38.5	45.5	36.2	11.6	0.0	0.0	0.0	0.0		
WD																													
Total demand	55.2	44.0	48.5	48.8	60.1	69.0	73.2	74.8	80.0	84.8	87.4	90.8	93.6	97.0		442.9	398.7	369.9	374.8	365.8	305.8	309.4	333.9	276.3	196.5	277.1	343.0		
ex-WD	11.1	10.2	9.4	9.2	8.4	8.2	8.2	7.9	7.5	7.9	7.9	7.8	7.8	7.4		263.1	230.4	193.2	196.2	171.4	125.0	111.9	102.0	50.5	39.9	32.7	31.1		
ex-HGST	44.1	33.8	39.1	39.6	51.7	60.8	65.0	66.9	72.5	76.9	79.5	83.0	85.8	89.6		179.8	168.3	176.7	178.6	194.4	180.8	197.5	231.9	225.8	156.6	244.4	311.9		
Volume of in-house production	49.7	40.0	45.5	45.8	57.1	64.2	68.2	69.8	75.0	79.8	82.4	85.8	87.4	91.9		399.9	368.0	343.9	330.7	339.7	300.9	309.4	332.3	259.2	181.0	259.3	323.0		
ex-WD	11.1	10.2	9.4	9.2	8.4	8.2	8.2	7.9	7.5	7.9	7.9	7.8	7.8	7.4		233.1	211.9	177.2	182.2	163.4	123.0	111.9	102.0	50.5	39.9	32.7	31.1		
ex-HGST	38.6	29.8	36.1	36.6	48.7	56.0	60.0	61.9	67.5	71.9	74.5	78.0	79.6	84.5		166.8	156.1	166.7	148.5	176.3	177.9	197.5	230.3	208.7	141.1	226.6	291.9		
Shipment volume from Resonac	5.5	4.0	3.0	3.0	3.0	4.8	5.0	5.0	5.0	5.0	5.0	5.0	6.2	5.1		43.0	30.7	26.0	44.1	26.1	3.2	0.0	1.6	17.1	15.5	17.8	20.0		
Shipment volume from Fuji Elec	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Toshiba																													
Total demand	16.5	11.2	19.0	20.0	23.0	22.4	26.0	25.0	29.2	24.0	26.0	27.4	23.8	27.3		135.6	122.0	149.9	152.1	133.0	124.3	107.1	145.8	105.3	66.7	96.4	106.6		
Volume of in-house production	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Shipment volume from Resonac	16.5	11.2	19.0	20.0	23.0	22.4	26.0	25.0	29.2	24.0	26.0	27.4	23.8	27.3		135.6	122.0	149.9	152.1	133.0	124.3	107.1	146.1	105.3	66.7	96.4	106.6		
Shipment volume from Fuji Elec	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Industry Total																													
Total demand	135.8	103.9	112.3	120.6	136.3	151.2	166.2	170.8	176.9	182.7	191.0	200.8	201.4	210.7		974.0	879.7	850.4	857.2	862.4	757.2	768.3	884.4	690.3	472.6	624.5	751.4		
Shipment volume from Resonac	26.5	18.2	24.0	23.0	26.0	27.2	31.0	30.0	34.2	29.0	31.0	32.4	30.0	32.9		240.0	210.7	192.8	196.2	159.1	127.5	116.6	177.0	151.6	91.7	114.2	126.6		
Market share of Resonac	20%	18%	21%	19%	19%	18%	19%	18%	19%	16%	16%	16%	15%	16%		25%	24%	23%	23%	18%	17%	15%	20%	22%	19%	18%	17%		
Shipment volume from Fuji Elec	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		51.0	42.3	33.1	28.7	38.5	45.5	36.2	11.6	0.0	0.0	0.0	0.0		
Market share of Fuji Elec	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%		5%	5%	4%	3%	4%	6%	5%	1%	0%	0%	0%	0%		
HDD Media/HDD	4.06	3.32	3.92	4.17	4.58	4.99	5.18	5.38	6.16	6.01	6.04	6.08	6.10	6.38		1.73	1.88	2.01	2.13	2.30	2.38	2.96	3.41	4.01	3.87	5.04	6.07		

Source: TSR, Morgan Stanley Research

Supply Chain of HDDs



Source: TSR, Morgan Stanley Research

History of Consolidation – HDDs, Media and Heads

CY1990		
HDD 41 Firms	Media 21 Firms	Head 15 Firms
Aiomega	Akashic	AMC
Alps Elec	Asahi Komag	Alps Elec
Areal	Denki Kagaku	Dastek
Censtor	Domain	Fujitsu
Conner Peripheral	Fuji Elec	Hitachi Metals
DEC	Fujitsu	Hitachi
Fuji Elec	Hitachi	Head Way
Fujitsu	HMT	IBM
Greenery	IBM	Kyusyu Matsushita
Hitachi	Komag	Read-Rite
HP	Kowa Mira	Seagate
IBM	Kyusyu Matsushita	Sony
JVC	Mitsubishi Chemical	Sunward
Kalok	Nashua(NCPC)	TDK
Kyocera	Seagate Technology	Yamaha
Magnum	Showa Denko	
Magtron	Sony	
Maxtor	Stolle	
Micropolis	Toso	
Microscience	Trace	
Miniscribe	Xidex	
Mitsubishi Elec		
NEC		
Orca		
PrairieTek		
Priam		
Quantum		
Ricoh		
Rodime		
Samsung		
Seagate Technology		
Seiko Epson		
Siemens		
Sony		
Syquest		
TEAC		
Tokiko		
Toshiba		
Western Digital		
Zentec		

CY1996		
HDD 21 Firms	Media 18 Firms	Head 13 Firms
Aiomega	Akashic	AMC
Areal	Asahi Komag	Alps Elec
Calluna	Fuji Elec	Fujitsu
Castlewood	Fujitsu	Hitachi
Fujitsu	Hitachi	Head Way
Halo Data Device	HMT	IBM
Hitachi	HOYA	Read-Rite
IBM	IBM	Read-Rite SMI
Integral Peripherals	Komag	Quantum-MKE
JTS	Maxtor	Seagate Technology
Maxtor	Mitsubishi Chemical	Sony
MiniStor	NSG	TDK
Myrica	Seagate	Yamaha
NEC	Showa Denko	
Quantum-MKE	StorMedia	
Samsung	Trace	
Seagate Technology	Western Digital	
Syquest		
Toshiba		
Western Digital		

CY2004
HDD 9 Firms
Cornice
Fujitsu
Hitachi
Maxtor
Magicstor
Samsung
Seagate
Toshiba
Western Digital
Media 8 Firms
Fuji Elec
Fujitsu
Hitachi
HOYA
Komag
Maxtor
Seagate Technology
Showa Denko
Head 6 Firms
Alps Elec
Fujitsu
Hitachi
Seagate Technology
TDK
Western Digital

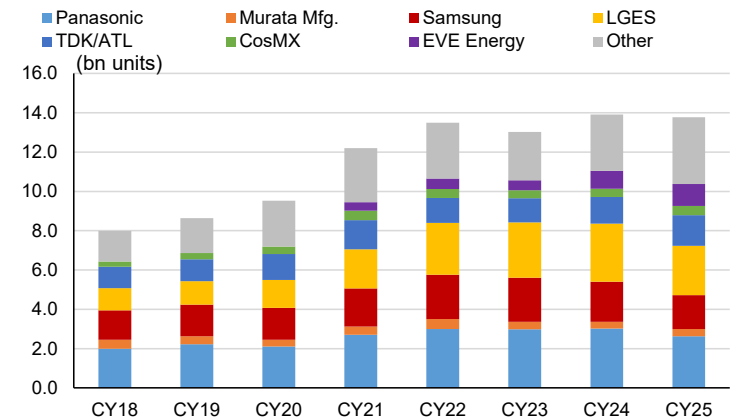
CY2016
HDD 3 Firms
Seagate
Toshiba
Western Digital
Media 4 Firms
Fuji Elec
Seagate Technology
Showa Denko
Western Digital
Head 3 Firms
Seagate Technology
TDK
Western Digital

CY2026
HDD 3 Firms
Seagate Technology
Western Digital
Toshiba
Media 3 Firms
Seagate Technology
Western Digital
Resonac>Showa Denko)
Head 3 Firms
Seagate Technology
Western Digital
TDK

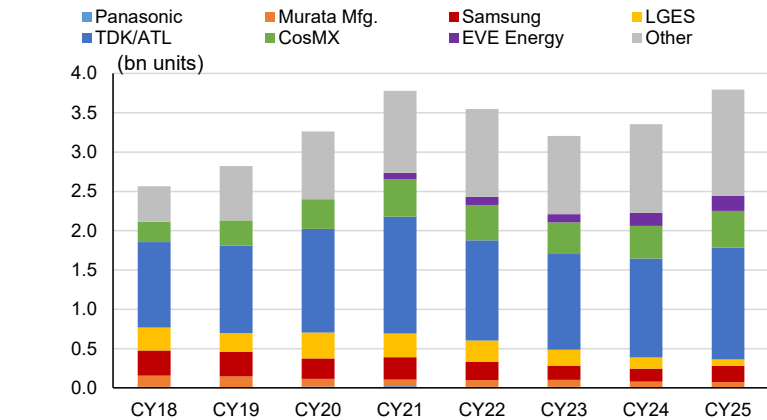
Source: TSR, Morgan Stanley Research

Small Size LiB Market Overview

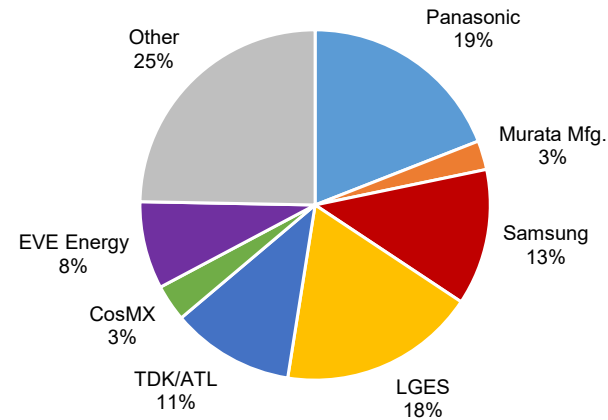
Small size LiB shipment volume by supplier



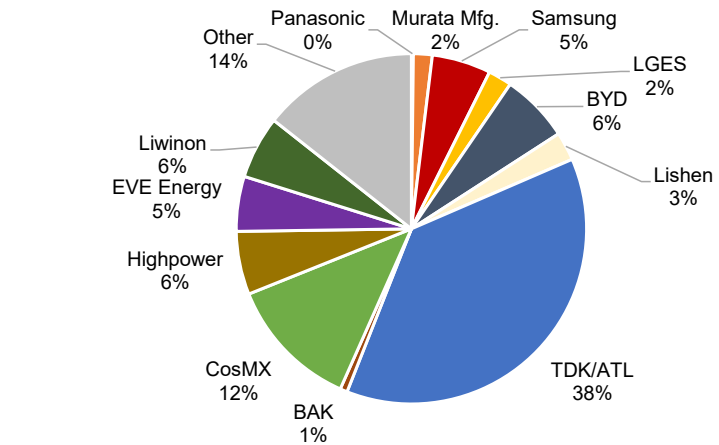
Laminated small size LiB shipment volume by supplier



Small size LiB market share by supplier (CY25)



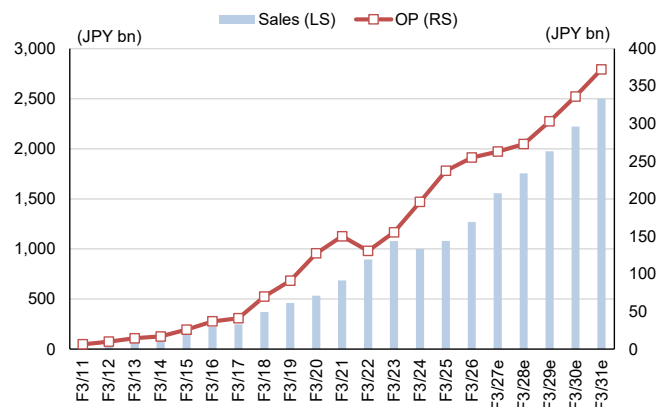
Laminated small size LiB market share by supplier (CY25)



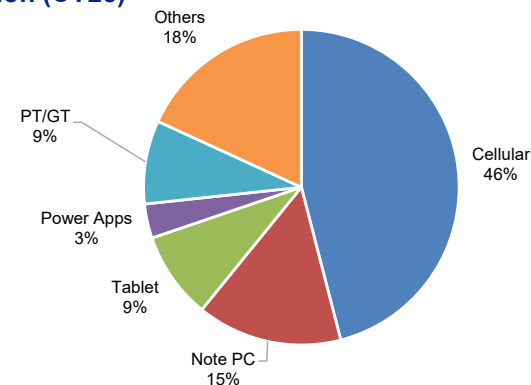
Source: TSR, Morgan Stanley Research

TDK: LiB Earnings Expansion, Major Acquisition and Transfer of Business

TDK: Sales and OP for rechargeable battery business



TDK/ATL: Small size battery shipment volume composition by application (CY25)



TDK: Major acquisition and transfer of business

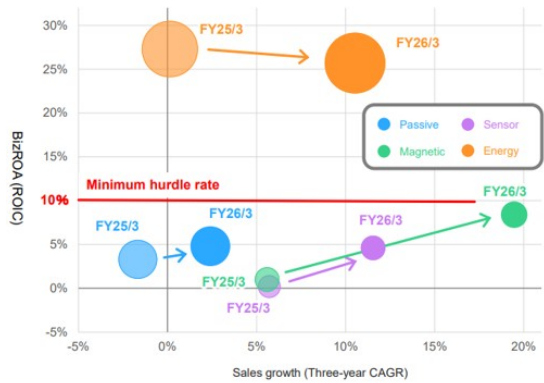
	Acquisition and Transfer of Business	Business	Acquisition/Sales value (JPY mn)
Aug-1986	Acquisition of SAE Magnetics	HDD Heads	unknown
May-1989	Acquisition of Silicon Systems	Mixed-signal Semiconductors	28,000
Jun-1996	Transfer of Silicon Systems to Texas Instruments	Mixed-signal Semiconductors	62,700
Mar-2000	Acquisition of Headway Technologies	HDD Heads	26,410
May-2005	Acquisition of Amperex Technology(ATL)	Lithium Polymer Batteries	8,700
Oct-2005	Acquisition of Lambda Power Group	Power	25,500
Aug-2007	Transfer of Recording Media business to Imation	Recording Media	31,400
Sep-2007	Acquisition of HDD head business from Alps	HDD Heads	34,400
Nov-2007	Acquisition of Magnecomp International	HDD Head Suspensions	20,900
Oct-2008	Acquisition of Epcos	Passive Components	170,600
Mar-2016	Acquisition of Micronas	MEMS Magnetic Sensors	22,500
Oct-2016	Acquisition of Hutchinson Technology	HDD Head Suspensions	14,262
Dec-2016	Acquisition of Tronics	MEMS Inertial Sensors	4,107
Feb-2017	Transfer of Epcos High Frequency Components business to Qualcomm	High Frequency Components	339,000
Mar-2017	Acquisition of ICsense	Gyroscope Sensors	2,396
May-2017	Acquisition of InvenSense	MEMS Inertial Sensors	142,758
Feb-2018	Acquisition of Chirp Microsystems	Ultrasonic 3D Sensors	3,015
May-2018	Acquisition of Faraday Semi LLC	Power Management IC	unknown
Apr-2025	Transfer of Micro Actuator Solution Business to QT Investment	Micro Actuator	unknown
Jun-2025	Acquisition of SoftEye	Smart glasses	unknown
Jun-2025	Acquisition of RF power business from QEI Corporation	RF power business	unknown
Jun-2026	Acquisition of Linergy Power	Mid-sized LiB	38,300
Jun-2026	Acquisition of Fabric8Labs	Thermal Management for DC	64,000

Note: e = Morgan Stanley Research estimates

Source: TSR, Company data, Morgan Stanley Research

TDK: Actions for Business Portfolio Improvement

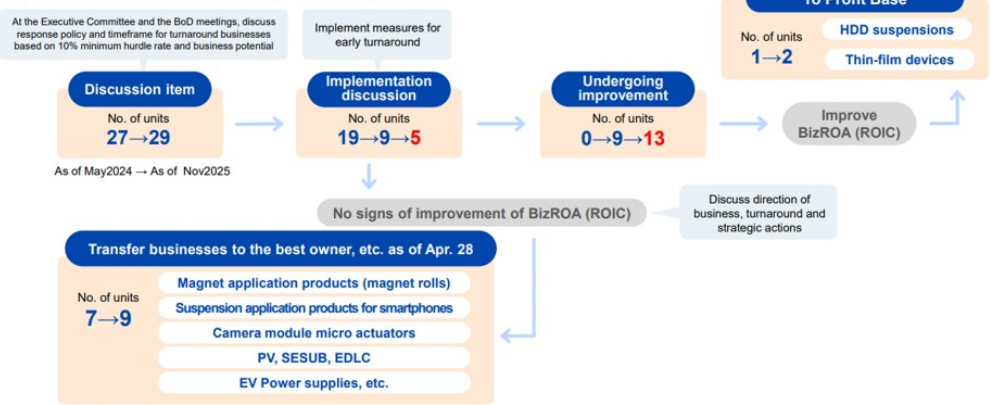
Chart for growth and capital efficiency by segment



Net sales, BizROA by segment

	(¥bn)	FY3/25	FY3/26
Net sales		2,204.8	2,504.8
Passive Components		559.6	593.2
Sensor Application Products		189.5	224.7
Magnetic Application Products		223.6	262.9
Energy Application Products		1,176.5	1,370.3
OP margin		10.2%	10.9%
BizROA (ROIC)		6.7%	7.5%
Passive Components		3.3%	4.8%
Sensor Application Products		0.2%	4.7%
Magnetic Application Products		1.0%	8.4%
Energy Application Products		27.3%	25.7%

Progress of proactive business portfolio management



Source: Company data

TDK: Medium-term Results and Forecasts

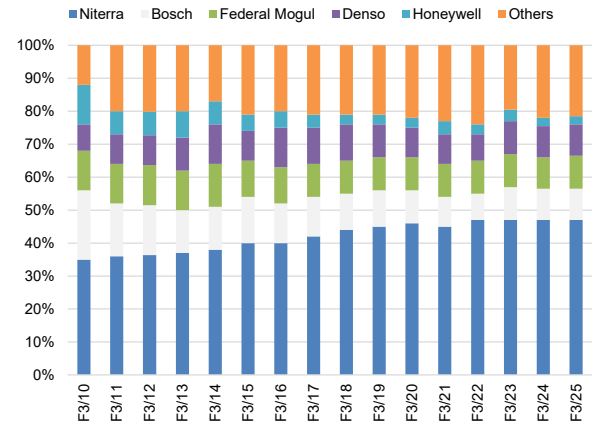
(bn JPY, JPY/share)	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe
Sales	1,271.7	1,381.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	3,071.8	3,360.5	3,667.4	4,007.9	4,384.8
Passive Components	417.6	433.4	395.5	407.1	507.8	578.8	565.6	559.6	593.2	739.7	781.9	815.6	852.0	890.0
Sensor Application Products	77.4	76.5	77.9	81.3	130.8	169.5	180.5	189.5	224.6	254.8	272.0	290.4	310.3	331.6
Magnetic Application Products	277.5	272.8	219.7	199.3	248.4	200.6	184.2	223.6	262.9	359.8	391.8	422.6	456.4	493.2
Energy Application Products	442.8	537.5	597.7	740.2	965.3	1,173.4	1,121.7	1,176.5	1,370.3	1,665.1	1,864.9	2,088.7	2,339.3	2,620.0
Other	56.4	61.6	72.3	51.1	49.7	58.6	51.8	55.6	53.8	52.4	50.0	50.0	50.0	50.0
OP	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	360.4	400.7	454.0	502.4	565.4
Passive Components	46.0	58.4	39.1	40.2	76.8	95.5	53.9	34.1	41.8	72.5	79.0	88.8	99.1	109.9
Sensor Application Products	-19.4	-22.1	-25.0	-24.9	-0.3	10.7	6.0	5.0	20.7	32.6	39.0	44.8	51.8	59.4
Magnetic Application Products	16.1	17.0	0.4	-2.4	4.6	-56.4	-35.6	3.4	27.0	52.6	59.8	67.2	75.2	83.8
Energy Application Products	72.9	91.0	124.1	147.4	123.2	147.4	195.7	234.4	246.7	271.1	280.9	311.2	334.2	370.3
Other	0.6	-6.7	-8.6	-16.1	-1.4	-0.4	-1.8	-4.4	-10.2	-7.8	-6.0	-6.0	-6.0	-6.0
Intersegment Elimination	-30.5	-29.8	-32.2	-32.7	-36.1	-28.0	-45.3	-48.3	-53.6	-60.5	-52.0	-52.0	-52.0	-52.0
OPM	6.7%	7.8%	7.2%	7.5%	8.8%	7.7%	8.2%	10.2%	10.9%	11.7%	11.9%	12.4%	12.5%	12.9%
ROE	7.8%	9.7%	6.7%	8.6%	11.4%	8.3%	7.9%	9.5%	9.7%	12.2%	11.9%	12.5%	12.6%	13.0%
EPS	33.4	43.3	30.4	41.8	69.3	60.2	65.7	88.0	103.0	147.8	157.1	180.7	200.4	226.1
FCF	-154.8	0.1	180.4	-8.7	-102.6	28.4	230.4	201.0	129.9	17.0	202.9	274.7	319.7	375.3

(JPY bn)	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26 MSe	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe	CAGR F3/18-F3/25	CAGR F3/25-F3/31
Total Sales	1,271.7	1,381.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	3,071.8	3,360.5	3,667.4	4,007.9	4,384.8	8.2%	12.1%
Group A Sales	633.2	727.5	752.1	889.5	1,165.4	1,309.7	1,237.6	1,375.7	1,612.9	1,999.5	2,235.8	2,495.4	2,785.1	3,108.4	11.7%	14.6%
HDD Head	175.5	181.4	136.4	116.5	148.5	104.1	100.1	134.5	141.7	201.6	225.8	248.4	273.2	300.5		
HDD Suspension	60.5	56.0	54.3	57.5	66.3	49.7	43.7	56.2	81.0	110.9	118.6	126.9	135.8	145.3		
Magnetic Sensor	26.2	28.1	27.5	30.5	54.6	76.6	91.9	104.3	120.3	129.8	137.6	145.8	154.6	163.8		
LiB	371.0	461.9	533.8	685.0	896.0	1,079.3	1,001.9	1,080.7	1,269.9	1,557.2	1,753.8	1,974.3	2,221.5	2,498.7		
Group B Sales	638.6	654.3	611.0	589.5	736.7	871.1	866.3	829.1	892.0	1,072.3	1,124.7	1,172.0	1,222.9	1,276.5	3.8%	7.5%
Capacitors	157.0	173.3	153.9	158.2	198.1	242.5	245.0	234.3	257.5	337.9	357.5	375.4	394.2	413.9		
Inductive Devices	157.5	150.0	137.6	140.0	180.2	198.5	192.0	204.3	216.2	263.3	281.7	293.0	304.7	316.9		
Other Passive Components	103.1	110.1	104.0	109.0	129.4	137.8	128.7	121.1	119.5	138.5	142.6	147.2	153.1	159.3		
Temperature/Pressure Sensor	24.9	26.4	26.8	27.2	33.0	36.9	40.4	41.6	42.3	52.1	54.2	56.4	58.6	61.0		
MEMS Sensor	25.5	22.0	23.7	23.6	43.2	56.0	48.3	43.5	62.0	72.9	80.2	88.2	97.0	106.8		
Magnet	41.5	35.4	28.9	25.2	33.6	46.8	40.7	32.9	40.2	47.3	47.3	47.3	47.3	47.3		
Power Supply	71.8	75.6	63.9	55.2	69.3	94.0	119.7	95.8	100.4	107.8	111.1	114.4	117.8	121.4		
Others	57.2	61.6	72.3	51.1	49.8	58.6	51.6	55.6	53.8	52.4	50.0	50.0	50.0	50.0		
Total OP	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	360.4	400.7	454.0	502.4	565.4		
Group A OP	86.2	118.1	145.1	154.3	153.4	126.7	193.7	273.0	309.9	346.3	366.8	407.0	451.6	500.0		
Group B OP	30.0	20.0	-15.1	-10.0	53.3	70.1	24.5	-0.6	16.1	74.6	85.9	99.0	102.8	117.4		
Intersegment elimination	-30.5	-30.3	-32.2	-32.7	-39.8	-28.0	-45.3	-48.3	-53.6	-60.5	-52.0	-52.0	-52.0	-52.0		

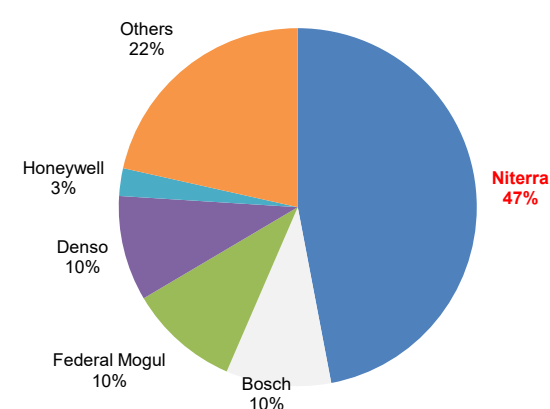
Note: MSe = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

Global Market Share Trend and Split for Spark Plugs / Exhaust Gas Oxygen Sensors

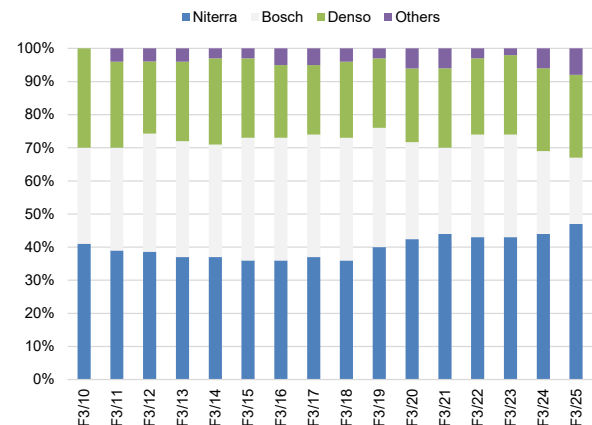
■ Spark plug global market share trend (based on value)



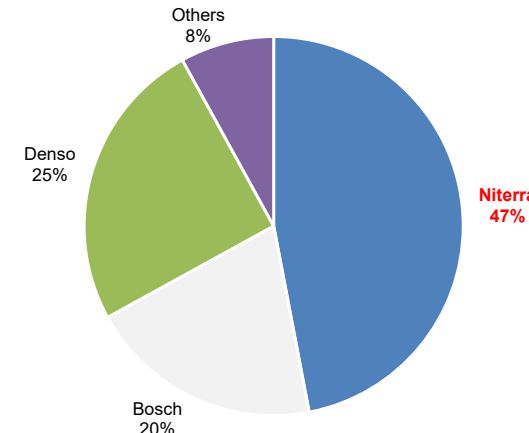
■ Spark plug global market share in F3/25 (based on value)



■ Exhaust gas oxygen sensor global market share (based on value)



■ Exhaust gas oxygen sensor global market share split in F3/25 (based on value)



Source: Company data, Morgan Stanley Research

Niterra: Medium-term Results and Forecasts

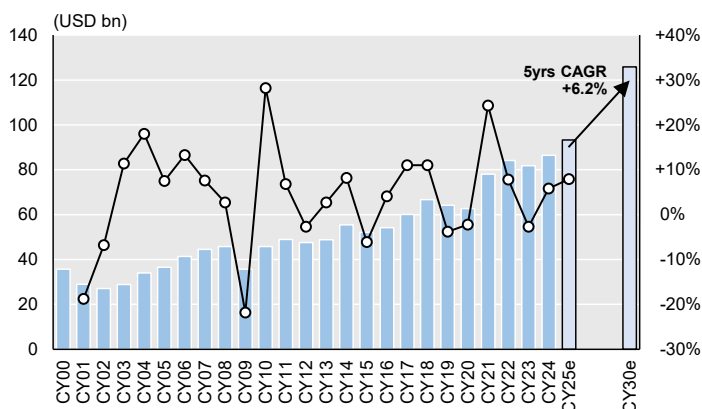
(bn JPY, JPY/share)	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe
Sales	562.6	427.5	491.7	562.6	614.5	653.0	731.2	793.5	816.6	878.3	945.9	1,020.0
Spark & Glow Plugs	203.5	205.3	250.0	285.1	335.3	362.6	390.7	428.5	446.2	490.8	539.9	593.9
Automotive Sensors	140.9	133.4	137.8	158.9	170.0	180.2	196.8	204.2	206.9	213.1	219.5	226.1
Component Solutions	36.3	80.0	95.5	110.8	96.5	103.1	130.7	150.0	157.5	168.4	180.5	194.0
(Electrostatic Chuck)	18.5	22.9	30.0	40.4	33.2	45.0	52.2	60.9	67.0	74.4	82.5	91.6
New Business	0.0	4.5	4.6	4.5	5.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Others/Elimination	4.1	4.4	3.9	3.3	7.4	7.1	13.0	10.9	6.0	6.0	6.0	6.0
OP	48.4	47.4	75.5	89.2	107.6	129.7	138.2	142.9	172.0	198.1	224.6	253.4
Spark & Glow Plugs	36.2	40.4	55.0	67.4	95.8	109.8	109.2	110.8	134.4	150.9	169.0	189.0
Automotive Sensors	16.6	16.2	19.3	25.9	25.4	24.4	26.1	25.3	26.2	28.4	30.7	33.0
Component Solutions	-1.1	-0.2	14.7	11.0	-0.6	-6.3	-4.6	7.2	11.4	18.8	24.9	31.4
(Electrostatic Chuck)	-0.6	0.5	8.6	13.0	7.0	6.4	10.4	12.8	16.0	19.7	23.8	28.3
New Business	0.0	-9.4	-13.6	-17.1	-13.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Others/Elimination	-0.6	0.4	0.1	2.0	0.2	1.8	7.4	-0.4	0.0	0.0	0.0	0.0
OPM	8.6%	11.1%	15.4%	15.9%	17.5%	19.9%	18.9%	18.0%	21.1%	22.6%	23.7%	24.8%
ROE	8.5%	9.1%	12.5%	12.3%	13.8%	14.1%	15.7%	15.3%	15.1%	16.1%	16.7%	17.2%
EPS	82.8	94.3	296.0	163.0	205.8	233.4	287.6	311.8	339.1	399.4	464.1	522.2
FCF	6.9	20.9	82.1	31.9	26.0	98.7	-56.1	97.2	136.4	128.1	145.4	164.8

Note: MSe = Morgan Stanley Research estimates

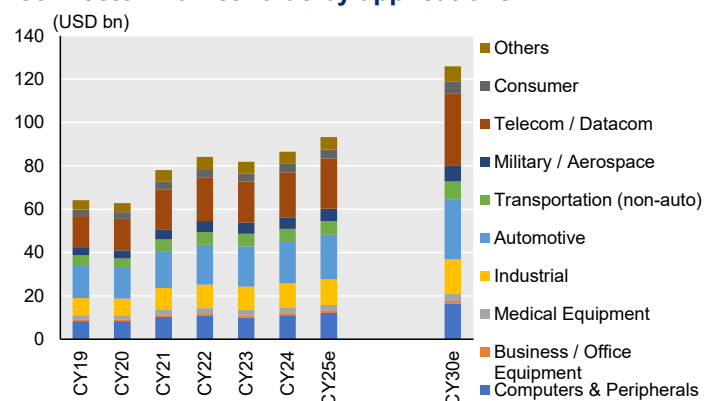
Source: Company data, Morgan Stanley Research

Connector Market

Global connector market value



Global connector market value by applications



Global connector market value CAGR by applications

	CY04-09	CY09-14	CY14-19	CY19-24	CY24-30e
Computers & Peripherals	3.0%	5.9%	-0.3%	5.7%	6.9%
Business / Office Equipment	4.1%	5.1%	-0.7%	3.1%	3.5%
Medical Equipment	8.2%	6.4%	2.6%	6.5%	4.4%
Industrial	-0.6%	9.0%	4.6%	7.0%	6.1%
Automotive	-5.7%	12.0%	4.6%	4.7%	6.4%
Transportation (non-auto)	1.4%	9.4%	7.4%	6.1%	5.4%
Military / Aerospace	4.4%	2.1%	3.4%	6.9%	5.9%
Telecom / Datacom	5.1%	10.2%	5.8%	7.9%	8.0%
Consumer	3.6%	6.4%	3.0%	4.6%	5.1%
Others	2.3%	15.5%	-6.9%	5.3%	4.3%
Market total	1.0%	9.2%	3.0%	6.1%	6.5%

Global connector sales by major suppliers

(USD bn)	CY13		(USD bn)	CY23	
	Sales	Share		Sales	Share
1 TE Connectivity	8.7	17.8%	1 TE Connectivity	12.7	15.6%
2 Amphenol	4.3	8.8%	2 Amphenol	7.0	8.5%
3 Molex	3.6	7.4%	3 Aptiv(Delphi)	4.7	5.8%
4 Aptiv(Delphi)	3.0	6.0%	4 Molex	4.6	5.7%
5 Foxconn	2.7	5.5%	5 Foxconn	2.5	3.1%
6 Yazaki	2.4	4.9%	6 Luxshare Precision	2.3	2.9%
7 JST	1.5	3.1%	7 Yazaki	2.2	2.7%
8 JAE	1.3	2.7%	8 JAE	1.7	2.1%
9 Sumitomo Wiring Systems	1.0	2.1%	9 Rosenberger	1.4	1.7%
10 Hirose Electric	1.0	2.0%	10 Hirose Electric	1.1	1.4%
Top 10	29.5	60.4%	Top 10	40.3	49.3%
Total	48.9	100.0%	Total	81.9	100.0%

Note: e = TTI estimates

Source: TTI, Company data, Morgan Stanley Research

Connector Coverage Companies: Medium-term Results and Forecasts

■ Hirose Electric (6806)

(JPY bn, JPY/share)	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/27 Ce	F3/28e MSe	F3/29e MSe	F3/30e MSe	F3/30 Ce	F3/31e MSe
Sales	133.5	163.7	183.2	165.5	189.4	211.3	256.6	250.0	276.9	298.6	322.5	290.0	348.7
General Industrial	40.4	56.4	69.1	50.5	46.9	62.0	86.2	83.0	92.3	99.2	106.7	84.1	114.7
Smartphone	31.7	33.9	34.0	35.5	46.4	44.3	45.7	44.5	46.9	48.3	49.7	89.9	51.2
Consumer/Mobile equipment	26.3	31.3	30.0	25.3	34.7	34.7	37.7	36.0	39.1	40.6	42.2	89.9	43.9
Automotive/Mobility related	23.8	29.2	36.0	42.9	49.2	54.3	63.0	63.0	70.4	78.8	88.3	89.9	98.9
ns infrastructure equipment	5.2	6.2	6.0	4.5	5.4	6.7	10.5	10.0	12.0	13.9	15.9	11.6	18.3
OA equipment and Others	6.1	6.7	8.1	6.8	6.8	9.3	13.5	13.5	16.2	17.8	19.6	14.5	21.6
OP	27.9	40.8	46.8	34.0	42.7	43.0	55.3	50.0	62.4	67.8	74.0	66.7	81.1
OPM	20.9%	24.9%	25.5%	20.6%	22.5%	20.4%	21.6%	20.0%	22.5%	22.7%	22.9%	23.0%	23.3%
ROE	6.3%	9.4%	10.0%	7.4%	9.0%	8.9%	10.2%	-	12.1%	12.6%	13.1%	12.0%	13.5%
EPS	549.1	885.4	1,002.1	772.4	976.3	991.9	1,185.7	1,069.6	1,470.1	1,619.8	1,791.7	-	1,990.3
FCF	15.1	33.4	52.1	12.9	12.7	36.6	16.6	-	33.7	37.2	42.1	-	44.6

■ JAE (6807)

(JPY bn, JPY/share)	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/27 Ce	F3/28e MSe	F3/29e MSe	F3/29e Ce	F3/30 MSe	F3/31 MSe
Sales	209.7	225.1	235.9	225.8	221.6	227.9	248.9	240.0	256.0	272.2	260.0	285.1	298.8
Connector	195.8	209.0	215.9	204.9	201.8	206.9	224.9	215.0	230.1	244.1	233.0	254.7	265.8
Handhelds	99.7	88.9	86.8	75.5	66.9	65.2	55.8	-	52.7	51.2	-	49.8	48.6
Automotive	70.5	83.1	91.0	105.3	112.6	115.7	131.5	-	136.2	144.4	-	153.0	162.2
Industrial & Infrastructure	20.5	30.1	32.5	19.8	18.0	20.6	31.0	-	32.8	34.7	-	36.8	39.0
Aviation & Space (Defence)	0.0	0.0	0.0	0.0	0.0	0.0	0.9	-	2.6	8.0	-	9.2	10.1
Others	5.1	6.9	5.5	4.4	4.3	5.4	5.8	-	5.8	5.8	-	5.8	5.8
Aerospace	13.3	15.3	19.1	20.1	19.3	20.5	23.4	24.5	25.3	27.5	27.0	29.9	32.5
Automotive	1.3	1.1	1.1	1.0	0.2	0.1	0.1	-	0.1	0.1	-	0.1	0.1
Industrial & Infrastructure	8.0	9.1	13.6	13.7	10.7	9.3	11.2	-	12.0	12.8	-	13.8	14.7
Aviation & Space (Defence)	4.0	5.0	4.4	5.4	8.4	11.1	12.1	-	13.3	14.6	-	16.1	17.7
Others	0.6	0.8	0.9	0.7	0.5	0.5	0.5	0.5	0.5	0.5	0.0	0.5	0.5
OP	8.7	18.0	17.6	14.4	15.6	8.9	8.4	9.5	10.7	12.2	18.0	13.4	14.7
Connector	12.3	20.8	18.1	15.8	18.0	12.0	10.3	-	12.3	13.5	-	14.4	15.3
Aerospace	0.3	1.6	4.1	3.6	2.6	2.2	3.3	-	3.5	3.9	-	4.2	4.6
Others	0.3	0.3	0.3	0.1	0.1	0.1	0.1	-	0.1	0.1	-	0.1	0.1
Adjustment	-4.2	-4.6	-4.8	-5.1	-5.0	-5.4	-5.2	-	-5.3	-5.3	-	-5.3	-5.3
OPM	4.2%	8.0%	7.4%	6.4%	7.0%	3.9%	3.4%	4.0%	4.2%	4.5%	6.9%	4.7%	4.9%
ROE	4.1%	9.6%	8.9%	8.2%	8.9%	5.1%	3.5%	-	5.1%	5.7%	8.0%	6.0%	6.3%
EPS	62.6	157.4	160.7	181.8	172.0	104.9	74.1	89.0	111.2	127.5	-	139.4	152.7
FCF	25.8	-5.8	-2.6	2.7	17.1	-7.4	1.4	-	6.8	6.3	-	7.5	7.5

Note: MSe = Morgan Stanley Research estimates, Ce = Company estimates

Source: Company data, Morgan Stanley Research

iPhone Sell-in Forecasts / iPhone Camera Actuators by Model

	1Q24	2Q24	3Q24	4Q24	2024	1Q25	2Q25	3Q25	4Q25	2025	1Q26	2Q26	3Q26e	4Q26e	2026e
iPhone Builds															
iPhone SE3 / SE 5G / 16e / 17e	1.0	2.0	5.0	3.0	11.0	4.0	9.0	4.0	1.0	18.0	2.0	10.0	8.0	5.0	25.0
iPhone X	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone XR	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 mini (5.4")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 (6.1")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 Pro (6.1")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 Pro Max (6.7")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 mini (5.4")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 (6.1")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 Pro (6.1")	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-	-	-	-
iPhone 13 Pro Max (6.7")	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-	-	-	-
iPhone 14 (6.1")	1.0	0.5	-	-	1.5	-	-	-	-	-	-	-	-	-	-
iPhone 14 Plus (6.7")	1.0	0.5	-	-	1.5	-	-	-	-	-	-	-	-	-	-
iPhone 14 Pro (6.1")	2.0	2.0	1.0	1.0	6.0	1.0	1.0	1.0	-	3.0	-	-	-	-	-
iPhone 14 Pro Max (6.7")	2.0	2.0	1.0	1.0	6.0	1.0	1.0	1.0	-	3.0	-	-	-	-	-
iPhone 15 (6.1")	6.0	5.0	4.0	1.0	16.0	1.0	0.5	0.5	0.5	2.5	-	-	-	-	-
iPhone 15 Plus (6.7")	4.0	3.0	1.0	1.0	9.0	1.0	0.5	0.5	0.5	2.5	-	-	-	-	-
iPhone 15 Pro (6.1")	12.0	10.0	8.5	2.0	32.5	2.0	2.0	2.0	1.0	7.0	1.0	1.0	-	-	2.0
iPhone 15 Pro Max (6.7")	18.0	13.0	10.5	3.0	44.5	3.0	3.0	2.0	1.0	9.0	1.0	1.0	-	-	2.0
iPhone 16 (6.1")	-	-	6.0	12.0	18.0	5.0	4.0	4.0	1.0	14.0	2.0	2.0	1.0	1.0	6.0
iPhone 16 Plus (6.7")	-	-	5.0	6.0	11.0	4.0	3.0	1.0	-	8.0	-	-	-	-	-
iPhone 16 Pro (6.3")	-	-	8.0	17.0	25.0	11.5	9.5	4.0	2.0	27.0	2.0	2.0	2.0	1.0	7.0
iPhone 16 Pro Max (6.9")	-	-	4.0	26.0	30.0	16.5	13.0	10.0	4.0	43.5	5.0	3.0	2.0	1.0	11.0
iPhone 17 (6.1")	-	-	-	-	-	-	-	6.0	16.0	22.0	10.0	8.0	6.0	2.0	26.0
iPhone 17 Air (6.7")	-	-	-	-	-	-	-	4.0	4.0	8.0	-	-	-	-	-
iPhone 17 Pro (6.3")	-	-	-	-	-	-	-	8.0	22.0	30.0	18.0	13.0	9.0	4.0	44.0
iPhone 17 Pro Max (6.9")	-	-	-	-	-	-	-	7.0	23.0	30.0	15.0	12.0	9.0	5.0	41.0
iPhone 18 Pro (6.3")	-	-	-	-	-	-	-	-	-	-	-	-	8.0	24.0	32.0
iPhone 18 Pro Max (6.9")	-	-	-	-	-	-	-	-	-	-	-	-	8.0	24.0	32.0
iPhone Duo	-	-	-	-	-	-	-	-	-	-	-	-	1.0	6.0	7.0
Total Builds	48.0	39.0	54.0	73.0	214.0	50.0	46.5	55.0	76.0	227.5	56.0	52.0	54.0	73.0	235.0
Y/Y Growth	-11%	-5%	8%	-3%	-3%	4%	19%	2%	4%	6%	12%	12%	-2%	-4%	3%
Q/Q Growth	-36%	-19%	38%	35%	-	-32%	-7%	18%	38%	-	-26%	-7%	4%	35%	-

		Rear cam								Front cam	
Type		OIS	SS	SS + FF	SS + AF	SS + OIS + AF	SS + periscope + AF	SS + periscope + AF (var.)		VCM	VCM x 2?
number of cameras		Single	Single	Dual	Dual	Triple	Triple	Triple		Single	Dual
2023	iPhone 15 (6.1")			○						○	
	iPhone 15 Plus (6.7")			○						○	
	iPhone 15 Pro (6.1")					○				○	
	iPhone 15 Pro Max(6.7")						○			○	
2024/25	iPhone 16 (6.1")				○					○	
	iPhone 16 Plus (6.7")				○					○	
	iPhone 16 Pro (6.3")						○			○	
	iPhone 16 Pro Max(6.9")						○			○	
2025/26	iPhone 16e (6.1")	○								○	
	iPhone 17 (6.3")				○					○	
	iPhone Air (6.5")		○							○	
	iPhone 17 Pro (6.3")						○			○	
2026/27	iPhone 17 Pro Max (6.9")						○			○	
	iPhone 17e (6.1")	○								○	
	iPhone 18 Pro (6.3")							○		○	
	iPhone 18 Pro Max (6.9")							○		○	
	iPhone Duo (5.4"/7.6")				○						○

Note: e = Morgan Stanley Research estimates

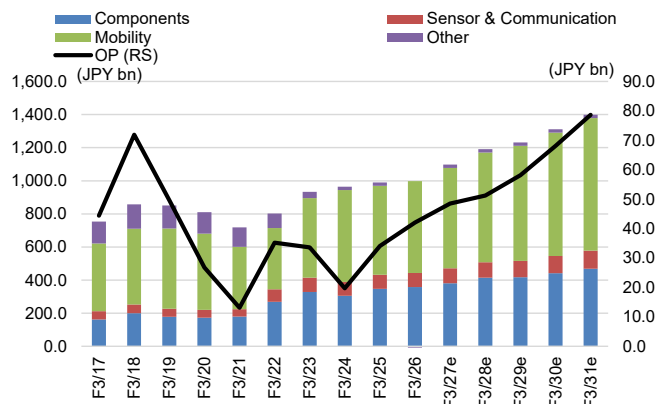
Source: Morgan Stanley Research

Alps Alpine: We Expect New Smartphone Camera Actuators to Contribute to Earnings

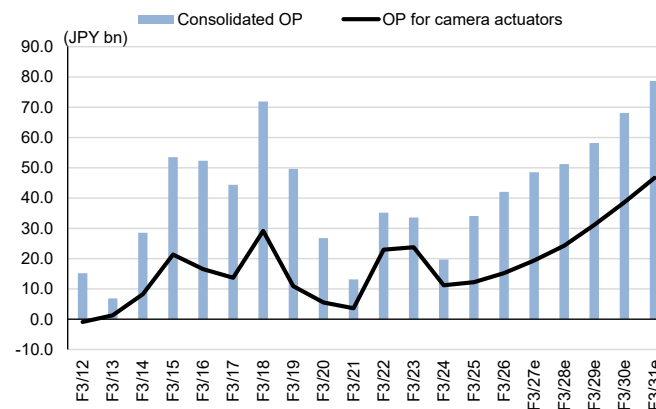
■ Alps Alpine: Medium-term results and forecasts

(bn JPY, JPY/share)	F3/17	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27	F3/28	F3/29	F3/30	F3/31	F3/27	F3/28
											MSe	MSe	MSe	MSe	MSe	Ce	Ce
Sales	753.3	858.3	851.3	810.6	718.0	802.9	933.1	964.1	990.4	990.4	1,097.4	1,191.1	1,231.6	1,311.7	1,399.1	1,045.0	1,085.0
Components	162.8	200.3	178.9	173.6	180.1	268.5	329.0	305.3	348.0	358.4	381.8	414.7	417.7	442.5	469.4	398.0	415.0
Sensor & Communication	50.1	52.1	48.1	46.7	44.3	76.7	85.5	84.1	84.2	85.3	89.1	93.6	98.3	103.2	108.4	87.0	100.0
Mobility	408.3	458.6	484.4	461.1	376.3	369.4	481.4	554.4	537.2	555.1	606.7	662.9	695.6	746.0	801.4	540.0	550.0
Other	132.1	147.4	139.9	129.2	117.3	88.2	37.2	20.3	21.0	-8.3	19.8	20.0	20.0	20.0	20.0	20.0	20.0
OP	44.4	71.9	49.6	26.8	13.1	35.2	33.6	19.7	34.1	42.0	48.5	51.2	58.1	68.1	78.7	48.5	67.5
Components	20.9	41.5	19.5	14.8	10.3	32.6	38.3	20.5	30.4	30.2	31.0	36.2	40.6	48.1	56.2	28.0	35.0
Sensor & Communication	9.6	7.9	6.0	3.0	1.7	3.0	-1.6	-1.5	-3.4	-3.5	-0.7	2.0	2.5	3.0	3.5	0.5	4.0
Mobility	7.9	15.9	18.0	3.7	-4.6	-8.0	-6.6	-1.1	5.6	14.2	17.6	12.0	14.0	16.0	18.0	19.0	27.5
Other/Elimination	6.0	6.6	6.2	5.3	5.8	7.6	3.5	1.9	1.5	1.3	0.7	1.0	1.0	1.0	1.0	1.0	1.0
OPM	5.9%	8.4%	5.8%	3.3%	1.8%	4.4%	3.6%	2.0%	3.4%	4.2%	4.4%	4.3%	4.7%	5.2%	5.6%	4.6%	6.2%
ROE	14.4%	17.0%	6.6%	-1.2%	-1.1%	6.2%	2.9%	-7.6%	9.4%	6.2%	7.2%	7.6%	8.2%	9.1%	9.8%	-	10.0%
EPS	178.3	241.9	110.2	-19.5	-18.7	110.8	55.8	-145.0	184.0	134.8	167.8	187.9	213.0	249.2	287.6	-	-
FCF	3.6	3.7	5.3	44.8	1.5	-11.2	-38.8	34.1	64.1	37.5	-2.7	-0.8	15.3	14.1	20.1	-	-

■ Alps Alpine: Sales by segment and consolidated OP



■ Alps Alpine: Consolidated OP and OP for camera actuators



Note: MSe/e = Morgan Stanley Research estimates, Ce = Company estimates

Source: Company data, Morgan Stanley Research

Ibiden: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	369.4	416.2	562.5	701.0	864.2
COGS	236.1	284.5	322.1	374.8	442.4
Gross profit	113.3	131.7	240.5	326.2	421.9
SG&A	65.7	69.6	105.5	126.0	148.0
Operating Profit	47.6	62.0	135.0	200.2	273.8
OP Incl. depreciation of inactive noncurrent assets	45.9	58.2	134.2	200.2	273.8
Non-operating Income/Loss	0.3	-1.2	1.5	2.3	2.3
Recurring Profit	47.9	60.8	136.5	202.5	276.2
Extraordinary Income/Loss	3.6	30.3	-2.4	-2.4	-2.4
Pre-tax Profit	51.5	91.1	134.1	200.1	273.7
Income Tax	17.4	26.9	40.7	60.0	82.1
(Effective Tax Rate)	34%	30%	30%	30%	30%
Minority Interest	0.3	0.4	0.5	0.2	0.2
Net Profit	33.7	63.7	92.9	139.9	191.4
Capex	157.3	64.3	157.6	158.0	208.0
Depreciation and Amortization	54.2	66.2	81.4	95.5	110.5
R&D Expenses	27.5	29.9	34.8	45.6	56.2
EBITDA	101.8	128.3	216.4	295.7	364.3
ROE	6.8%	12.2%	15.7%	20.1%	22.6%
Average Share Outstanding (Diluted)	558.8	558.7	589.8	589.8	589.8
EPS (Diluted, Yen)	60.3	107.5	157.5	237.2	324.5
BPS (Yen)	890.0	927.7	1,080.1	1,299.5	1,593.3
DPS (Yen)	10.0	30.0	17.5	20.0	45.0
USD/JPY: Average Exchange Rate	152.0	149.0	156.5	155.0	155.0
EUR/JPY: Average Exchange Rate	162.0	172.0	180.8	180.0	180.0
YoY % Change					
Net Sales	0%	13%	35%	25%	23%
Operating Profit	0%	30%	118%	48%	37%
Net Profit	7%	89%	46%	51%	37%
As % of Sales					
Gross profit	30.7%	31.6%	42.7%	46.5%	48.8%
Operating Profit	12.9%	14.9%	24.0%	28.6%	31.7%
Net Profit	9.1%	15.3%	16.5%	20.0%	22.1%
Capex	42.6%	15.4%	28.0%	22.5%	24.1%
Depreciation and Amortization	14.7%	15.9%	14.5%	13.6%	12.8%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	1,061.7	960.4	1,045.6	1,166.7	1,371.9
Total Current Assets	549.6	465.5	477.5	553.0	639.8
Cash and Cash Equivalents	390.7	296.7	288.2	329.4	374.8
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	68.2	82.1	99.3	123.8	152.6
Inventory	67.2	71.0	73.2	83.1	95.7
Other Current Assets	23.5	16.8	16.8	16.8	16.8
Total Long Term Assets	532.1	494.9	571.1	633.6	731.1
Net PP&E	460.1	437.7	514.0	576.5	674.0
Investments in Securities	58.8	32.3	32.3	32.3	32.3
Other Assets	13.3	24.9	24.9	24.9	24.9
Total Current Liabilities	327.7	221.6	230.2	238.8	249.9
ST Debt & Current Portion of LT Debt	90.0	15.0	15.0	15.0	15.0
Accounts Payable	38.8	44.1	52.7	61.3	72.4
Other Current Liabilities	106.9	81.6	81.6	81.6	81.6
Total Long Term Liabilities	256.7	181.4	181.4	181.4	181.4
Long Term Debt	253.0	177.5	177.5	177.5	177.5
Other Long Term Liabilities	3.7	3.9	3.9	3.9	3.9
Net Assets	497.3	557.4	637.0	766.4	939.7
Minority Interests	6.8	7.3	7.3	7.3	7.3
Stock Acquisition Rights	0.0	0.0	0.0	0.0	0.0
Net Debt	-106.5	-135.5	-128.0	-169.2	-214.6

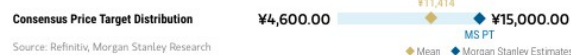
Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	118.9	106.4	163.4	209.6	271.6
Change in Interest-bearing Debt	0.0	-150.0	0.0	0.0	0.0
Depreciation and Amortization	54.2	66.2	81.4	95.5	110.5
Changes in Working Capital	-9.8	-7.1	-10.9	-25.7	-30.3
Other-net	40.7	-16.4	0.0	0.0	0.0
Net Cash from Investing Activities	-164.2	-52.4	-157.6	-158.0	-208.0
Purchase of PP&E	-197.5	-106.1	-157.6	-158.0	-208.0
Sale/Purchase of Investments	34.6	57.1	0.0	0.0	0.0
Other-net	-1.3	-3.4	0.0	0.0	0.0
Free Cash Flow	-45.3	54.0	5.8	51.6	63.6
Net Cash from Financing Activities	-7.1	-157.5	-13.3	-10.5	-18.2
Change in Treasury Stock	0.0	-150.0	0.0	0.0	0.0
Payment of Cash Dividends	-5.6	-7.0	-13.3	-10.5	-18.2
Other-net	-1.0	-0.6	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-0.5	5.8	0.0	0.0	0.0
Net Change in Cash	-52.9	-97.7	-7.5	41.2	45.4
Cash at End of Period	390.7	292.9	285.4	326.6	372.0
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	369.4	416.2	562.5	701.0	864.2
Electronics	197.2	243.3	385.7	525.5	680.4
FC Package	186.9	219.2	376.7	516.5	671.4
Print Circuit Board	10.3	10.0	9.0	9.0	9.0
FC CSP	0.0	0.0	0.0	0.0	0.0
Ceramics	84.1	82.6	86.0	81.5	87.0
DPF	34.6	32.5	36.4	30.0	30.0
Sealing Materials	30.4	29.3	28.9	30.0	30.0
Graphite Specialty Products	16.2	13.2	16.8	16.5	16.7
Others	3.0	5.7	3.9	5.0	10.3
Others	88.1	90.3	90.8	94.0	96.8
Total Operating Profit	47.6	62.0	135.0	200.2	273.8
Electronics	28.9	45.2	115.3	179.7	252.2
Ceramics	12.2	7.6	8.7	9.0	9.5
Others	8.6	9.1	11.0	11.6	12.1
Operating Profit Margin	12.9%	14.9%	24.0%	28.6%	31.7%
Electronics	13.6%	18.6%	29.9%	34.2%	37.1%
Ceramics	14.5%	9.3%	10.1%	11.0%	10.9%
Others	9.7%	10.1%	12.1%	12.3%	12.5%

Risk Reward – Ibiden (4062.T)

Probability of upside to F3/31 guidance has increased.

PRICE TARGET ¥15,000

Derived from the base case. DCF assumptions: 3.0% risk-free rate, 1.52 beta, 3.2% risk premium, for 7.7% WACC. Terminal growth of 0% from F3/36.



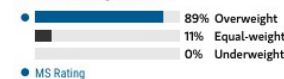
RISK REWARD CHART



OVERWEIGHT THESIS

- We expect sales of ABF packages for AI servers to increase and earnings growth to continue.
- We see Ibiden sustaining a competitive advantage in ABF package technology and keeping dominant market share in GPU ABF packages for AI servers.
- We expect earnings growth to continue due to customer and product mix improvement, even though costs increase due mainly to new lines being activated.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥18,000

Derived from DCF

Earnings ahead of the base case on stronger sales of value-added ABF packages for AI servers: OP of ¥142.1bn in F3/27, ¥235.8bn in F3/28 and ¥504.2bn in F3/31. Same FX and DCF assumptions as the base case. Fair value equates to 108.5x F3/27 diluted EPS of ¥165.8, 64.4x F3/28 diluted EPS of ¥279.4, and 30.1x F3/31 diluted EPS of ¥597.9.

BASE CASE

¥15,000

Derived from DCF

Higher sales of advanced FC packages should drive FY3/31 earnings above the company's guidance: OP ¥135.0bn in F3/27, ¥200.2bn in F3/28, ¥407.9bn in F3/31. FX assumptions ¥155/\$, ¥180/€. Fair value is based on DCF as explained in the valuation methodology section. Base case fair value equates to 95.2x our F3/27e diluted EPS of ¥157.5, 63.2x our F3/28e diluted EPS of ¥237.2 and 31.0x our F3/31e diluted EPS of ¥483.7.

BEAR CASE

¥5,000

F3/27 BPS ¥1,012 x P/B 4.9x

Limited earnings improvement from F3/27: OP of ¥77.0bn in F3/27, ¥94.2bn in F3/28 and ¥163.7bn in F3/31. Same FX assumptions as the base case. ROE declines from 12.2% in F3/26 to 9.5% in F3/27, but starts to improve from F3/28 and 11.2% in F3/28, higher than 7.7% cost of capital, but below the market consensus.

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Niterra: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	653.0	731.2	793.5	816.6	878.3
COGS	394.9	450.3	506.7	498.7	532.2
Gross profit	258.1	280.9	286.9	317.9	346.1
SG&A	128.4	142.7	144.0	145.8	148.1
Operating Profit	129.7	138.2	142.9	172.0	198.1
Non-operating Income/Loss	3.7	27.3	56.7	6.7	6.7
Recurring Profit	133.3	165.5	199.6	178.7	204.8
Extraordinary Income/Loss	0.0	0.0	0.0	0.0	0.0
Pre-tax Profit	133.3	165.5	199.6	178.7	204.8
Income Tax	40.5	49.0	77.1	50.0	57.3
(Effective Tax Rate)	30%	30%	39%	28%	28%
Minority Interest	0.2	3.6	0.9	0.9	0.9
Net Profit	92.6	112.9	121.6	127.8	146.5
Capex	37.5	49.0	47.2	45.0	45.0
Depreciation and Amortization	39.7	45.0	42.1	43.0	43.0
R&D	0.0	0.0	30.0	31.0	31.0
EBITDA	169.4	183.2	185.0	215.0	241.1
ROE	14.1%	15.7%	15.3%	15.1%	16.1%
Average Share Outstanding (Diluted)	397.2	395.8	386.9	376.9	366.9
EPS (Diluted, Yen)	233.4	287.6	311.8	339.1	399.4
BPS (Yen)	1,698.7	1,957.8	2,111.0	2,312.6	2,512.4
DPS (Yen)	89.0	102.5	105.0	110.0	125.0
USD/JPY: Average Exchange Rate	153.0	151.0	155.0	155.0	155.0
EUR/JPY: Average Exchange Rate	164.0	175.0	180.0	180.0	180.0
YoY % Change					
Sales	6%	12%	9%	3%	8%
Operating Profit	21%	7%	3%	20%	15%
Net Profit	12%	22%	8%	5%	15%
As % of Sales					
Gross profit	39.5%	38.4%	36.2%	38.9%	39.4%
Operating Profit	19.9%	18.9%	18.0%	21.1%	22.6%
Net Profit	14.2%	15.4%	15.3%	15.6%	16.7%
Capex	5.7%	6.7%	5.9%	5.5%	5.1%
Depreciation and Amortization	6.1%	6.2%	5.3%	5.3%	4.9%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	991.0	1,221.1	1,249.1	1,274.6	1,322.7
Total Current Assets	588.5	658.2	681.1	704.7	750.7
Cash and Cash Equivalents	208.2	187.7	182.6	218.0	242.5
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	147.6	172.2	186.8	192.3	206.8
Inventory	183.9	217.7	231.1	213.8	220.9
Other Current Assets	48.8	80.6	80.6	80.6	80.6
Total Long Term Assets	402.4	562.9	568.0	570.0	572.0
Net PP&E	220.0	265.7	270.7	272.7	274.7
Investments in Securities	93.1	81.6	81.6	81.6	81.6
Other Assets	89.4	215.7	215.7	215.7	215.7
Total Current Liabilities	153.6	195.8	204.5	203.2	208.4
ST Debt & Current Portion of LT Debt	16.7	31.4	31.4	31.4	31.4
Accounts Payable	57.7	69.7	78.5	77.2	82.4
Other Current Liabilities	79.3	94.6	94.6	94.6	94.6
Total Long Term Liabilities	162.6	254.1	224.1	194.1	164.1
Long Term Debt	140.0	211.0	181.0	151.0	121.0
Other Long Term Liabilities	22.6	43.1	43.1	43.1	43.1
Net Assets	674.7	771.3	820.5	877.3	950.2
Minority Interests	0.1	3.8	3.8	3.8	3.8
Net Debt	-144.6	-26.8	-51.7	-117.1	-171.6

Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	132.9	109.4	144.4	151.4	173.1
Net Income	92.6	112.9	121.6	127.8	146.5
Depreciation and Amortization	39.7	45.0	42.1	43.0	43.0
Changes in Working Capital	-6.2	-27.3	-19.3	10.6	-16.4
Other-net	6.8	-21.2	0.0	0.0	0.0
Net Cash from Investing Activities	-34.2	-165.5	-47.2	-45.0	-45.0
Purchase of PP&E	-27.8	-32.3	-47.2	-45.0	-45.0
Sale/Purchase of Investments	-3.4	7.7	0.0	0.0	0.0
Other-net	-3.0	-140.9	0.0	0.0	0.0
Free Cash Flow	98.7	-56.1	97.2	136.4	128.1
Net Cash from Financing Activities	-71.0	28.6	-102.3	-101.0	-103.7
Change in Interest-bearing Debt	-25.9	86.4	-30.0	-30.0	-30.0
Change in Treasury Stock	-10.0	-16.8	-30.0	-30.0	-30.0
Payment of Cash Dividends	-34.4	-36.4	-42.3	-41.0	-43.7
Other-net	-0.7	-4.6	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-0.2	7.1	0.0	0.0	0.0
Net Change in Cash	27.5	-20.4	-5.1	35.4	24.5
Cash at End of Period	208.2	187.7	182.6	218.0	242.5
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net sales	653.0	731.2	793.5	816.6	878.3
Automotive Components	542.8	587.5	632.6	653.1	704.0
Spark & Glow Plugs	362.6	390.7	428.5	446.2	490.8
Automotive Sensors	180.2	196.8	204.2	206.9	213.1
Ceramics	103.1	100.7	150.0	157.5	168.4
New Business	0.0	0.0	0.0	0.0	0.0
Others/Elimination	7.1	13.0	10.9	6.0	6.0
Total Operating Profit	129.7	138.2	142.9	172.0	198.1
Automotive Components	134.2	135.3	136.9	160.6	179.3
Ceramics	-4.3	-4.6	7.2	11.4	18.8
New Business	0.0	0.0	0.0	0.0	0.0
Others/Elimination	1.8	7.4	-0.4	0.0	0.0
Operating Profit Margin	19.9%	18.9%	18.0%	21.1%	22.6%
Automotive Components	24.7%	23.0%	21.5%	24.6%	25.9%
Ceramics	-6.1%	-3.9%	4.8%	7.3%	11.2%

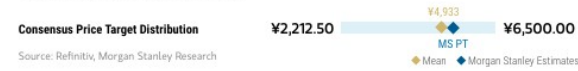
Risk Reward – Niterra (5334.T)

Expect earnings growth in auto components with volume & unit price going up

PRICE TARGET ¥5,250

Calculated from our DCF model derived from our base case. Our DCF model assumes a 3.0% risk-free rate, 1.11 equity beta, and 3.2% risk premium, yielding a 5.8% WACC. We assume a 0% terminal growth rate from F3/36.

Consensus Price Target Distribution



RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect the firm to maintain competitiveness in automotive plugs and exhaust gas sensors, and market share to keep gradually expanding.
- Although demand for plugs is unlikely to grow substantially (Niterra boasts the top global share at 47%), we expect Niterra's market share to increase gradually and the firm to book large profits on aftermarket products.
- In automotive exhaust gas oxygen sensors, we expect Niterra to gradually increase its 47% global market share, and earnings to keep growing alongside the tightening of environmental regulations.
- In non-auto business, we believe earnings from electrostatic chucks for SPE will continue growing, having bottomed in F3/24.

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: Positive

Pricing Power: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥6,500

Derived from DCF

Automotive plug & exhaust gas sensor sales exceed the base case by 4% and earnings continue to grow further: This scenario assumes OP of ¥169.8bn in F3/27, ¥200.1bn in F3/28, and ¥291.1bn in F3/31, while using the same FX and DCF assumptions as in the base case. Bull-case fair value equates to 18.2x F3/27e diluted EPS of ¥356.9, 16.6x F3/28e diluted EPS of ¥392.7, and 10.9x F3/31e diluted EPS of ¥598.2.

BASE CASE

¥5,250

Derived from DCF

Expect profit to continue to reach new heights, driven by increased sales of automobile plugs and SPE electrostatic chucks: We forecast OP of ¥142.9bn in F3/27, ¥172.0bn in F3/28, and ¥253.4bn in F3/31, assuming FX rates of ¥155/US\$. Our PT is calculated using a DCF valuation based on the assumptions outlined in our valuation methodology. Our base-case PT equates to 16.8x F3/27e diluted EPS of ¥311.8, 15.5x F3/28e diluted EPS of ¥339.1, and 10.0x F3/31e diluted EPS of ¥522.2.

BEAR CASE

¥1,850

F3/27e diluted BPS ¥2,066 x P/B 0.9

Lack of growth in autos-related volumes, limited improvement in SPE-related earnings: This scenario assumes OP of ¥113.2bn in F3/27, ¥101.5bn in F3/28, and ¥73.7bn in F3/31, while using the same FX and DCF assumptions as in the base case. Under this scenario, market consensus is revised downward and the share price falls to 0.9x P/B.

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Minebea Mitsumi: Key Financial Data and Risk Reward

Income Statement					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,522.7	1,664.4	1,733.9	1,780.7	1,855.0
Cost of Goods Sold	1,252.3	1,371.4	1,357.1	1,375.7	1,429.4
Gross Profit	270.4	293.0	376.8	405.0	425.6
SG&A Expenses	175.9	189.0	252.2	266.7	270.5
Operating Profit	94.5	104.0	124.6	138.2	155.1
Interest/Dividend Income	6.9	40.9	9.7	1.8	1.8
Interest Expense	18.8	11.1	12.2	2.5	2.5
Other Non-Operating Income/Loss	0.0	0.0	0.0	0.0	0.0
Recurring Profit	82.6	133.8	122.1	137.5	154.4
Extraordinary Income/Loss	0.0	0.0	0.0	0.0	0.0
Pretax profit	82.6	133.8	122.1	137.5	154.4
Income Tax	22.8	34.5	32.6	35.8	40.2
(Effective Tax Rate)	28%	26%	27%	26%	26%
Minority Interest	0.4	0.2	-0.1	0.0	0.0
Net Profit	59.5	99.0	89.6	101.8	114.3

Capex	94.8	96.5	111.9	108.8	111.3
Depreciation and Amortization	66.2	70.4	75.2	77.3	80.5
R&D	45.5	49.8	51.5	53.4	55.6
EBITDA	160.7	174.4	199.9	215.5	235.6
ROE	8.2%	12.1%	9.8%	10.1%	10.5%

Average Share Outstanding (mil, Diluted)	406.3	401.6	401.6	401.6	401.6
EPS (Diluted, Yen)	147.7	246.6	223.1	253.4	284.6
BPS (Yen)	1,830.0	2,238.6	2,408.6	2,608.0	2,831.6
DPS (Yen)	45.0	50.0	53.0	56.0	59.0

USD/JPY: Average Exchange Rate	152.6	150.0	155.0	155.0	155.0
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YoY % Change					
Net Sales	0%	0%	0%	0%	0%
Operating Profit	28%	10%	20%	11%	12%
Net Profit	10%	67%	-10%	14%	12%
As % of Sales					
Gross Profit	17.8%	17.6%	21.7%	22.7%	22.9%
Operating Profit	6.2%	6.2%	7.2%	7.8%	8.4%
Net Profit	3.9%	6.0%	5.2%	5.7%	6.2%

Balance Sheet					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	1,584.8	1,814.8	1,876.2	1,960.8	2,064.0
Total Current Assets	904.1	1,028.7	1,053.4	1,108.4	1,180.8
Cash and Cash Equivalents	214.3	227.5	227.1	266.5	307.1
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	293.3	356.5	385.6	398.0	412.5
Inventory	350.9	391.3	387.3	392.6	407.9
Other Current Assets	170.3	220.9	220.9	220.9	220.9
Total Long Term Assets	680.7	786.1	822.8	852.3	883.2
Net PP&E	529.0	585.3	622.0	651.5	682.3
Investments in Securities	27.1	33.3	33.3	33.3	33.3
Other assets	0.0	0.0	0.0	0.0	0.0
Total Current Liabilities	477.2	692.2	595.1	606.4	613.1
ST Debt & Current Portion of LT Debt	175.1	264.1	264.1	264.1	264.1
Accounts Payable	213.6	235.6	217.8	220.8	229.4
Other Current Liabilities	88.5	102.5	113.1	115.5	119.5
Total Long Term Liabilities	353.0	301.6	301.6	301.6	301.6
Long Term Debt	288.5	218.4	218.4	218.4	218.4
Other Long Term Liabilities	64.4	83.2	83.2	83.2	83.2
Accrued pension cost	0.0	0.0	0.0	0.0	0.0
Minority Interests	11.2	12.1	12.2	12.2	12.2
Total Equity	754.6	911.9	979.5	1,056.7	1,148.3
Net Debt	249.3	255.0	255.4	216.0	175.4

Cash Flow Statement					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	133.7	94.9	122.2	166.3	171.6
Net Income After Tax	59.5	99.0	89.6	101.8	114.3
Depreciation and Amortization	66.2	70.4	75.2	77.3	80.5
Changes in Working Capital:	9.7	-53.9	-42.8	-12.7	-23.2
Other-net	-1.7	-20.7	0.1	0.0	0.0
Net Cash from Investing Activities	-125.8	-82.8	-101.3	-104.5	-107.3
Purchase of PP&E	-77.4	-77.1	-111.9	-106.8	-111.3
Sale/Purchase of Investments	-0.4	-0.2	0.0	0.0	0.0
Other-net	-48.0	-5.5	10.6	2.3	4.0
Free cash flow	7.9	12.1	20.9	61.8	64.3
Net Cash from Financing Activities	64.0	-16.1	-21.3	-22.5	-23.7
Change in Interest-bearing Debt	96.4	13.0	0.0	0.0	0.0
Change in Treasury Stock	-8.7	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-16.1	-20.1	-21.3	-22.5	-23.7
Other-net	-7.6	-9.1	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-4.3	17.3	0.0	0.0	0.0
Cash at End of Period	214.3	227.5	227.1	266.5	307.1

Segment Information					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales by Segment					
Total Sales	1,522.7	1,664.4	1,733.9	1,780.7	1,855.0
Precision Technologies	255.7	281.2	332.5	338.6	359.0
Ball bearings	171.8	186.4	214.7	215.0	229.4
Rod-ends/Fasteners	59.1	66.0	94.2	89.3	94.7
Pivot assemblies	24.8	28.7	33.6	34.3	35.0
Motor, Lighting & Sensing	407.7	456.5	509.4	530.3	552.1
Motors	322.4	340.3	393.9	413.6	434.2
Electronic devices	41.1	68.0	76.5	76.5	76.5
Measuring components	36.9	38.5	39.0	40.2	41.4
Other	7.4	9.8	0.0	0.0	0.0
Semiconductors & Electronics	527.6	590.3	535.2	541.8	555.4
Access Solutions	328.1	332.2	351.5	369.0	387.5
Others	3.5	4.2	5.3	1.0	1.0

Operating Profit by Segment					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Operating Profit	94.5	104.0	124.6	138.2	155.1
Machined Components Business	55.7	62.2	76.6	77.7	85.9
Electronic Devices & Components Business	23.0	26.9	32.7	36.9	41.5
Semiconductors & Electronics	22.0	26.7	27.7	30.2	33.3
Access Solutions	15.9	17.1	15.6	20.3	21.3
Others	-1.2	-2.7	-2.1	-1.0	-1.0
Adjustments	-20.9	-26.3	-25.8	-25.8	-25.8

Operating Profit Margin					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Machined Components Business	6.2%	6.2%	7.2%	7.8%	8.4%
Electronic Devices & Components Business	21.9%	22.1%	23.0%	22.9%	23.9%
Semiconductors & Electronics	5.6%	5.9%	6.4%	7.0%	7.5%
Access Solutions	4.2%	4.5%	5.2%	5.6%	6.0%
Others	4.9%	5.1%	4.4%	5.5%	5.5%
Others	-33.8%	-63.2%	-40.3%	-100.0%	-100.0%

Risk Reward – Minebea Mitsumi (6479.T)

Earnings should recover gradually, but this is mostly priced into the stock

PRICE TARGET ¥4,300

Derived from the base case. DCF assumptions: 3.0% risk-free rate, 1.27 equity beta, 3.2% risk premium, yielding 5.7% WACC; zero growth from F3/36.



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE

¥5,100

Derived from DCF

Semiconductor & Electronics and Precision Technology sales higher than in our base case: OP ¥144.2bn in F3/27, ¥162.6bn in F3/28, and ¥226.4bn in F3/31. Same FX and DCF model assumptions as the base case. Fair value for the shares equates to 19.7x F3/27e diluted EPS of ¥258.8, 17.1x F3/28e diluted EPS of ¥298.3, and 12.3x F3/28e diluted EPS of ¥415.8.

BASE CASE

¥4,300

Derived from DCF

Earnings continue to grow, driven by rising sales of core products, as the market expects: OP ¥124.6bn in F3/27, ¥138.2bn in F3/28, and ¥187.9bn in F3/31. FX assumptions of ¥155/¥, ¥180/¥. DCF model assumptions as shown in the valuation methodology section. Base case fair value equates to 19.3x F3/27e diluted EPS of ¥223.1, 17.0x F3/28e diluted EPS of ¥253.4, and 12.5x F3/31e diluted EPS of ¥344.9.

BEAR CASE

¥2,100

F3/27e BPS ¥2,350 x P/B 0.9

Earnings especially in Semiconductor & Electronics and Motor, Lighting & Sensing weaker than base case: OP ¥92.5bn in F3/27, ¥97.6bn in F3/28, and ¥117.6bn in F3/31. 6.9% ROE in F3/27 is lower than cost of capital of 7.1%, and earnings are lower than market consensus and the share price declines to 0.9 x P/B.

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Mabuchi Motor: Key Financial Data and Risk Reward

Income Statement (Y bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Sales	196.2	200.4	224.5	242.1	255.2
COGS	143.3	140.6	157.7	173.1	183.5
Gross Profit	52.9	59.8	66.8	69.0	71.4
SG&A Expenses	31.2	34.3	39.2	39.6	39.8
Operating Profit	21.6	25.5	27.5	29.4	31.6
Non-operating Income/Loss	10.8	9.6	7.9	7.9	4.4
Recurring Profit	32.4	35.1	35.4	37.4	36.0
Extraordinary Income/Loss	-8.7	-0.1	-0.2	-0.2	-0.2
Pretax Profit	23.7	35.0	35.2	37.1	35.7
Income Tax (Effective Tax Rate)	10.9	8.7	10.8	10.8	10.4
Minority Interest	40%	25%	31%	29%	29%
Net Profit	12.8	26.3	24.4	26.4	25.4
Capex	14.1	10.3	13.6	15.0	15.0
Depreciation and Amortization	14.2	13.3	14.9	14.0	14.0
R&D Expenses	7.0	7.8	8.6	9.0	9.0
EBITDA	35.8	38.8	42.4	43.4	45.6
ROE	4.1%	8.0%	7.4%	8.3%	8.2%
Average Share Outstanding (Diluted)	254.1	248.1	234.1	234.1	234.1
EPS (Diluted, Yen)	50.5	105.9	99.4	112.6	108.4
BPS (Yen)	1,257.7	1,347.0	1,378.4	1,338.5	1,295.5
DPS (Yen)	38.0	53.0	70.0	64.0	68.0
USD/JPY: Average Exchange Rate	151.6	149.7	156.6	155.0	155.0
YoY % Change					
Net Sales	10%	2%	12%	8%	5%
Operating Profit	39%	18%	8%	7%	7%
Net Profit	-34%	105%	-7%	8%	-4%
As % of Sales					
Gross Profit	26.9%	29.8%	29.7%	28.5%	28.0%
Operating Profit	11.0%	12.7%	12.2%	12.1%	12.4%
Net Profit	6.5%	13.1%	10.9%	10.9%	9.9%
Capex	7.2%	5.1%	6.1%	6.2%	5.9%
Depreciation and Amortization	7.2%	6.6%	6.6%	5.8%	5.5%
Balance Sheet (Y bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Total Assets	355.0	370.2	358.8	350.5	341.2
Total Current Assets	238.5	247.7	237.7	228.4	218.0
Cash and Cash Equivalents	134.9	143.4	126.9	113.8	102.9
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	37.3	38.7	39.0	41.4	42.9
Inventory	59.8	57.8	63.9	65.4	64.4
Other Current Assets	6.4	7.8	7.8	7.8	7.8
Total Long Term Assets	116.5	122.4	121.1	122.1	123.1
Net PP&E	95.5	95.9	94.6	95.6	96.6
Investments in Securities	12.4	16.4	16.4	16.4	16.4
Other Assets	8.6	10.2	10.2	10.2	10.2
Total Current Liabilities	28.8	28.3	28.5	29.5	30.3
ST Debt and Current Portion of LT Debt	0.0	0.0	0.0	0.0	0.0
Accounts Payable	7.7	10.4	10.6	11.6	12.3
Other Current Liabilities	21.1	17.9	17.9	17.9	17.9
Total Long Term Liabilities	6.5	7.6	7.6	7.6	7.6
Long Term Debt	0.5	1.3	1.3	1.3	1.3
Other Long Term Liabilities	6.0	6.3	6.3	6.3	6.3
Net Assets	319.6	334.2	322.6	313.3	303.2
Minority Interests	0.0	0.0	0.0	0.0	0.0
Net Debt	-146.8	-158.5	-142.0	-128.8	-117.9

Cash Flow Statement (Y bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Cash from Operations	40.1	35.4	33.1	37.5	39.6
Net Income	12.8	26.3	24.4	26.4	25.4
Depreciation and Amortization	14.2	13.5	14.9	14.0	14.0
Changes in Working Capital	2.2	8.1	-6.2	-2.9	0.2
Other-net	10.9	-12.5	0.0	0.0	0.0
Net Cash from Investing Activities	-15.8	-10.5	-13.6	-15.0	-15.0
Purchase of PP&E	-14.0	-10.2	-13.6	-15.0	-15.0
Other-net	-0.5	0.0	0.0	0.0	0.0
Net Cash from Financing Activities	-16.2	-17.4	-35.9	-36.7	-35.4
Change in Interest-bearing Debt	0.0	0.0	0.0	0.0	0.0
Change in Treasury Stock	-5.7	-6.7	-19.0	-20.0	-20.0
Payment of Cash Dividends	-10.2	-9.7	-16.9	-15.7	-15.4
Other-net	-0.3	-1.0	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	10.1	1.9	0.0	0.0	0.0
Net Change in Cash	18.3	9.4	-16.5	-13.2	-10.9
Cash at End of Period	130.6	139.9	123.5	110.3	99.4

Sales by Application (Y bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Sales	196.2	200.4	224.5	242.1	255.2
Automotive Products	152.5	154.5	165.0	173.4	183.9
Life & Industrial Products	43.7	45.9	59.5	68.7	71.3

YoY % Change	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Sales	10%	2%	12%	8%	5%
Automotive Products	9%	1%	7%	5%	6%
Life & Industrial Products	13%	5%	30%	15%	4%

As % of Sales	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Automotive Products	78%	77%	73%	72%	72%
Life & Industrial Products	22%	23%	27%	28%	28%

Shipment Volume by Application (mn units)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Total Sales Quantity	1,348	1,340	1,311	1,369	1,438
Automotive Products	1,055	1,071	1,060	1,108	1,163
Life & Industrial Products	293	269	249	261	275

Precision & Office Equipment	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Total Sales Quantity	2%	-1%	-2%	4%	5%
Automotive Products	2%	2%	-1%	4%	5%
Life & Industrial Products	-1%	-8%	-7%	5%	5%

As % of Sales	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Automotive Products	78%	80%	81%	81%	81%
Life & Industrial Products	22%	20%	19%	19%	19%

Risk Reward – Mabuchi Motor (6592.T)

Gradual profit growth on auto output recovery, headcount increases

PRICE TARGET ¥1,750

Our PT is calculated from our DCF model derived from our base case, and assumes a risk-free rate of 3.0%, equity beta of 0.83, and risk premium of 3.2%, yielding WACC of 5.7%. We assume a 0% perpetual growth rate for F12/36 onwards.



RISK REWARD CHART



EQUAL-WEIGHT THESIS

- ROE improves gradually due to moderate recovery in profit, driven by global auto production recovery and headcount increases.
- Higher automotive motor volumes drive earnings growth over the longer term.

Consensus Rating Distribution



Risk Reward Themes

Secular Growth: **Positive**

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥2,500	BASE CASE	¥1,750	BEAR CASE	¥1,200
Derived from DCF		Derived from DCF		End-F12/26e diluted BPS ¥1,369 x P/B 0.9	
Automotive motor output exceeds our base case by 8% thanks to demand growth and market share gains. This scenario assumes OP of ¥36.7bn in F12/26, ¥40.1bn in F12/27 and ¥52.8bn in F12/30 using the same FX and DCF assumptions as the base case. Our bull case fair value estimate equates to P/E of 14.8x (12.4x excluding net financial assets) on F12/26e diluted EPS of ¥168.7.		Earnings recover gradually due mainly to global auto production recovery. We forecast OP of ¥27.5bn in F12/26, ¥29.4bn in F12/27 and ¥42.3bn in F12/30, assuming an FX rate of ¥155/\$. Our PT is derived from our DCF model, the conditions for which are outlined in our Valuation Methodology. Our PT equates to P/E of 17.6x (11.5x excluding net financial assets) on F12/26e diluted EPS of ¥99.4, 15.5x on F12/27e diluted EPS of ¥112.6 and 12.4x on F12/30e diluted EPS of ¥140.9.		Profit drops in F12/26 due to rising costs and automobile production decline. This scenario is for OP of ¥23.2bn in F12/26, ¥23.8bn in F12/27 and ¥27.1bn in F12/30, assuming ROE declines to 6.6% in F12/26, remaining above the cost of capital of 5.6%; however, the market consensus has been revised down sharply, and the share price is expected to fall to a P/B of 0.9x, but end-F12/25 net financial assets/share of ¥64.5 and shareholder returns underpin the share price.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

TDK: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,204.8	2,504.8	3,071.8	3,360.5	3,667.4
Cost of Goods Sold	1,516.8	1,721.4	2,107.7	2,319.7	2,547.9
Gross Profit	688.0	783.4	964.1	1,040.8	1,119.5
SG&A Expenses	494.0	544.0	616.2	640.1	665.4
Others*	30.2	33.0	12.5	0.0	0.0
Operating Profit	224.2	272.4	360.4	400.7	454.0
Non-operating Income/Loss	13.6	4.4	11.2	7.6	8.7
Pretax Profit	237.8	276.8	371.6	408.3	462.7
Income Tax	67.4	77.6	87.3	106.2	115.7
(Effective Tax Rate)	28%	28%	24%	26%	25%
Minority Interest	3.2	3.5	3.7	4.0	4.0
Net Profit	167.2	195.7	280.6	298.1	343.0
Capex	225.3	296.6	364.9	350.0	350.0
Depreciation and Amortization	196.2	204.2	244.3	260.0	280.0
R&D Expenses	253.6	289.7	310.5	320.0	330.0
EBITDA	420.4	476.6	604.7	660.7	724.0
ROE	9.5%	9.7%	12.2%	11.9%	12.5%
Average Share Outstanding (Diluted)	1,900.0	1,900.2	1,896.7	1,898.2	1,898.2
EPS (Diluted, Yen)	88.0	103.0	147.8	157.1	180.7
BPS (Yen)	953.3	1,159.6	1,263.1	1,377.2	1,511.9
DPS (Yen)	30.0	36.0	40.0	44.0	48.0
USD/JPY: Average Exchange Rate	152.7	150.8	155.0	155.0	155.0
EUR/JPY: Average Exchange Rate	163.9	174.8	180.0	180.0	180.0
YoY % Change					
Net Sales	5%	14%	23%	9%	9%
Operating Profit	30%	22%	32%	11%	13%
Net Profit	34%	17%	43%	6%	15%
As % of Sales					
Gross Profit	31.2%	31.3%	31.4%	31.0%	30.5%
Operating Profit	10.2%	10.9%	11.7%	11.9%	12.4%
Net Profit	7.6%	7.8%	9.1%	8.9%	9.4%
Capex	10.2%	11.9%	11.9%	10.4%	9.5%
Depreciation and Amortization	8.9%	8.2%	8.0%	7.7%	7.6%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	3,541.4	4,415.2	4,384.6	4,653.7	4,963.7
Total Current Assets	1,836.9	2,455.2	2,224.1	2,403.2	2,643.2
Cash and Cash Equivalents	697.3	842.8	787.7	910.9	1,098.3
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	583.1	780.6	668.6	703.8	738.0
Inventory	410.0	585.4	521.4	542.1	560.5
Other Current Assets	146.4	246.4	246.4	246.4	246.4
Total Long Term Assets	1,704.6	1,959.9	2,160.5	2,250.5	2,320.5
Net PP&E	1,030.1	1,225.8	1,386.3	1,476.3	1,546.3
Investments in Securities	183.8	226.8	226.8	226.8	226.8
Other Assets	276.6	269.2	269.2	269.2	269.2
Total Current Liabilities	1,098.5	1,574.5	1,331.9	1,378.5	1,428.8
ST Debt and Current Portion of LT Debt	187.1	211.0	211.0	211.0	211.0
Accounts Payable	392.5	706.7	464.0	510.7	560.9
Other Current Liabilities	518.8	656.9	656.9	656.9	656.9
Total Long Term Liabilities	631.7	637.1	637.1	637.1	637.1
Long Term Debt	346.0	332.7	332.7	332.7	332.7
Other Long Term Liabilities	285.7	304.4	304.4	304.4	304.4
Minority Interests	11.2	16.3	20.0	24.0	28.0
Total Equity	1,811.3	2,203.5	2,395.7	2,614.2	2,869.9
Net Debt	-348.0	-525.9	-470.9	-594.1	-781.5

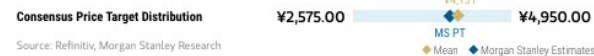
Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	445.8	507.7	461.9	552.9	624.7
Net Income	167.2	195.7	280.6	298.1	343.0
Depreciation and Amortization	196.2	204.2	244.3	260.0	280.0
Changes in Working Capital	2.5	-8.9	-66.7	-9.2	-2.3
Other-net	80.0	116.7	3.7	4.0	4.0
Net Cash from Investing Activities	-244.8	-377.8	-444.9	-350.0	-350.0
Purchase of PP&E	-225.3	-296.6	-364.9	-350.0	-350.0
Sale/Purchase of Investments	9.9	-4.3	0.0	0.0	0.0
Other-net	-29.5	-74.9	-80.0	0.0	0.0
Free Cash Flow	201.0	129.9	17.0	202.9	274.7
Net Cash from Financing Activities	-143.3	-64.7	-72.1	-79.7	-87.3
Change in Interest-bearing Debt	-81.3	9.9	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-48.5	-60.7	-72.1	-79.7	-87.3
Other-net	-13.5	-13.9	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-10.4	80.3	0.0	0.0	0.0
Net Change in Cash	47.3	145.5	-55.1	123.2	187.4
Cash at End of Period	697.3	842.8	787.7	910.9	1,098.3
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,204.8	2,504.8	3,071.8	3,360.5	3,667.4
Passive Components	559.6	593.2	739.7	781.9	815.6
Capacitors	234.3	257.5	337.9	357.5	375.4
Inductive Devices	204.3	216.2	263.3	281.7	293.0
Other Passive Components	121.1	119.5	138.5	142.6	147.2
Sensor Application Products	189.5	224.6	254.8	272.0	290.4
Magnetic Application Products	223.6	262.9	359.8	391.8	422.6
Energy Application Products	1,176.5	1,370.3	1,665.1	1,864.9	2,088.7
Other	55.6	53.8	52.4	50.0	50.0
Total Operating Profit	224.2	272.4	360.4	400.7	454.0
Passive Components	34.1	41.8	72.5	79.0	88.8
Sensor Application Products	5.0	20.7	32.6	39.0	44.8
Magnetic Application Products	3.4	27.0	52.6	59.8	67.2
Energy Application Products	234.4	246.7	271.1	280.9	311.2
Other	-4.4	-10.2	-7.8	-6.0	-6.0
Corporate and Eliminations	-48.3	-53.6	-60.5	-52.0	-52.0
Profit Margin	10.2%	10.9%	11.7%	11.9%	12.4%
Passive Components	6.1%	7.1%	9.8%	10.1%	10.9%
Sensor Application Products	2.6%	9.2%	12.8%	14.4%	15.4%
Magnetic Application Products	1.5%	10.3%	14.6%	15.3%	15.9%
Energy Application Products	19.9%	18.0%	16.3%	15.1%	14.9%
Other	-8.0%	-19.0%	-14.9%	-12.0%	-12.0%

Risk Reward – TDK (6762.T)

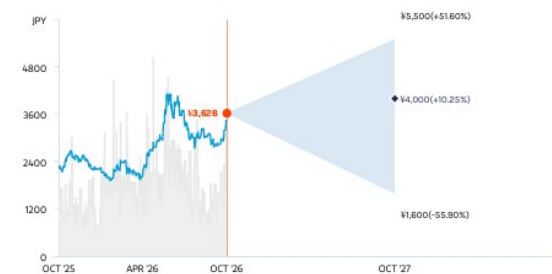
We expect LIB profit growth and continued improvement beyond LIB

PRICE TARGET ¥4,000

Derived from the base case. DCF assumptions: risk-free rate 3.0%, equity beta 1.31, risk premium 3.2%, yielding WACC 6.6%. 0% perpetual growth rate from F3/36.



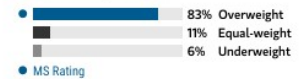
RISK REWARD CHART



OVERWEIGHT THESIS

- We expect Passive Components and Sensor Application earnings to continue to improve after the bottom in F3/25.
- We envision only modest LIB profit growth in F3/27 due to higher cobalt prices, etc., but sustained LIB profit growth in the mid-long term as sales expand in high value-added silicon cathode products and DC BBU batteries.
- HDD head and suspension earnings continue to expand due to increased demand for NL HDD for DC.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥5,500	BASE CASE	¥4,000	BEAR CASE	¥1,600
Derived from DCF		Derived from DCF		End-F3/27e diluted BPS ¥1,213 x P/B 1.3x	
Earnings from rechargeable batteries, passive components, sensors, and HDD heads ahead of base case. OP ¥401.2bn in F3/27, ¥474.4bn in F3/28, ¥800.4bn in F3/31. Same currency and DCF model assumptions as base case. Fair value equates to P/E 33.5x on F3/27e diluted EPS ¥164.0, P/E 29.7x on F3/28e diluted EPS ¥185.5, P/E 17.3x on F3/31e diluted EPS ¥318.6.		Modest rechargeable battery profit gains in F3/27, but mid-long term profit growth not only for rechargeable batteries, but also HDDs, sensors & passive components. OP ¥360.4bn in F3/27, ¥400.7bn in F3/28, ¥565.4bn in F3/31. Assumes ¥155/\$; DCF derived from assumptions listed in valuation methodology. PT equates to P/E 27.1x F3/27e diluted EPS ¥147.8, 25.5x F3/28e diluted EPS ¥157.1, 17.7x F3/31e diluted EPS ¥226.1.		Limited profit growth due to rechargeable battery slowdown. OP ¥238.8bn in F3/27, ¥270.7bn in F3/28, ¥355.7bn in F3/31. ROE 8.3% in F3/27 and 8.4% in F3/28, above the cost of capital of 7.2%. P/B sinks to 1.3x due to earnings missing market consensus.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Alps Alpine: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Net Sales	990.4	1,019.5	1,097.4	1,191.1	1,231.6
Cost of Goods Sold	815.2	837.6	897.4	989.5	997.3
Gross Profit	175.2	181.9	200.0	221.6	234.3
SG&A Expenses	141.1	139.8	151.5	170.3	176.1
Operating Profit	34.1	42.0	48.5	51.2	58.1
Non-Operating Income/Loss	-3.6	7.1	-1.3	2.4	2.4
Pretax profit	57.9	43.9	51.4	51.6	58.5
Income Tax	19.8	16.7	18.3	15.0	17.0
(Effective Tax Rate)	34%	38%	36%	29%	29%
Minority Interest	0.2	0.4	0.4	0.0	0.0
Net Profit	37.8	26.9	32.8	36.7	41.6
Capex	51.8	61.6	79.2	60.0	60.0
Depreciation and Amortization	35.1	34.0	43.0	42.0	42.0
R&D	24.3	21.9	27.8	33.0	35.0
EBITDA	69.2	76.0	91.5	93.2	100.1
ROE	9.4%	6.2%	7.2%	7.6%	8.2%
Average Share Outstanding (Diluted)	205.6	199.4	195.1	195.1	195.1
BPS (Diluted, Yen)	184.0	134.8	167.8	187.9	213.0
BPS (Yen)	2,013.1	2,293.7	2,397.5	2,515.4	2,653.4
DPS (Yen)	60.0	62.0	64.0	70.0	75.0
USD/JPY: Average Exchange Rate	151.3	150.8	155.0	155.0	155.0
YoY % Change					
Net Sales	3%	3%	8%	9%	3%
Operating Profit	73%	23%	15%	6%	13%
Net Profit	-	-29%	22%	12%	13%
As % of Sales					
Gross Profit	17.7%	17.8%	18.2%	18.6%	19.0%
Operating Profit	3.4%	4.1%	4.4%	4.3%	4.7%
Net Profit	3.8%	2.6%	3.0%	3.1%	3.4%
Capex	5.2%	6.0%	7.2%	5.0%	4.9%
Depreciation and Amortization	3.5%	3.3%	3.9%	3.5%	3.4%
Balance Sheet					
(Y bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Total Assets	740.7	783.2	808.6	839.5	869.4
Total Current Assets	494.9	500.7	489.9	502.8	514.8
Cash and Cash Equivalents	147.9	153.6	138.4	123.9	124.6
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	166.8	169.4	180.9	196.4	203.1
Inventory	138.6	138.3	148.4	160.3	164.9
Other Current Assets	41.5	39.4	22.2	22.2	22.2
Total Long Term Assets	245.8	282.4	318.6	336.6	354.6
Net PP&E	136.7	158.0	194.2	212.2	230.2
Investments in Securities	60.0	67.7	67.7	67.7	67.7
Other assets	49.1	56.8	56.8	56.8	56.8
Total Current Liabilities	226.9	240.7	245.9	253.8	256.8
ST Debt & Current Portion of LT Debt	44.3	37.1	37.1	37.1	37.1
Accounts Payable	90.4	93.2	98.3	106.2	109.3
Other Current Liabilities	92.2	110.5	110.5	110.5	110.5
Total Long Term Liabilities	98.3	93.0	93.0	93.0	93.0
Long Term Debt	56.5	55.2	55.2	55.2	55.2
Other Long Term Liabilities	41.8	37.8	37.8	37.8	37.8
Minority Interests	1.5	1.9	1.9	1.9	1.9
Total Equity	415.5	448.4	469.7	492.7	519.6
Net Debt	-107.2	-129.0	-113.8	-99.3	0.0

Cash Flow Statement					
(Y bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Net Cash from Operations	65.8	95.9	59.3	59.2	75.3
Net Income	37.8	26.9	32.8	36.7	41.6
Depreciation and Amortization	35.1	34.0	43.0	42.0	42.0
Changes in Working Capital:	9.0	16.0	-16.5	-19.5	-8.2
Other-net	-16.2	19.1	0.0	0.0	0.0
Net Cash from Investing Activities	-1.7	-58.4	-62.0	-60.0	-60.0
Purchase of PP&E	-43.0	-43.6	-79.2	-60.0	-60.0
Sale/Purchase of Investments	0.0	0.0	0.0	0.0	0.0
Other-net	41.4	-14.8	17.2	0.0	0.0
Free cash flow	64.1	37.5	-2.7	-0.8	15.3
Net Cash from Financing Activities	-37.3	-41.1	-12.5	-13.7	-14.6
Change in Interest-bearing Debt	-28.5	-9.5	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-8.2	-12.2	-12.5	-13.7	-14.6
Other-net	-0.5	-19.4	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-1.7	9.5	0.0	0.0	0.0
Net Change in Cash	25.2	5.9	-15.2	-14.5	0.7
Segment Information					
(Y bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Net Sales	990.4	1,019.5	1,097.4	1,191.1	1,231.6
Components	348.0	358.4	381.8	414.7	417.7
Sensor & Communication	84.2	85.3	89.1	93.6	98.3
Mobility	537.2	555.1	606.7	662.9	695.6
Logistics/Others	21.0	20.8	19.8	20.0	20.0
Total Operating Profit	34.1	42.0	48.5	51.2	58.1
Components	30.4	30.2	31.0	36.2	40.6
Sensor & Communication	-3.4	-3.5	-0.7	2.0	2.5
Mobility	5.6	14.2	17.6	12.0	14.0
Logistics/Others	1.5	1.3	0.7	1.0	1.0
Operating Profit Margin	3.4%	4.1%	4.4%	4.3%	4.7%
Components	8.7%	8.4%	8.1%	8.7%	9.7%
Sensor & Communication	-4.0%	-4.1%	-0.8%	2.1%	2.5%
Mobility	1.0%	2.6%	2.9%	1.8%	2.0%
Logistics/Others	7.0%	6.0%	3.5%	5.0%	5.0%

Risk Reward – ALPS ALPINE (6770.T)

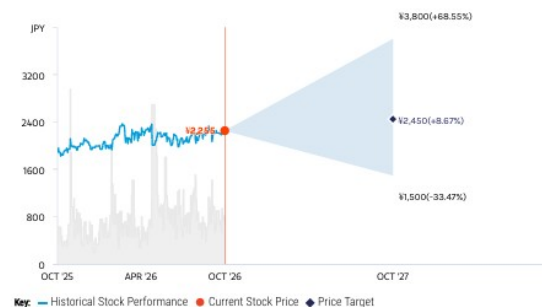
Earnings growth in FY/28 to be limited, difficult to achieve medium-term plan.

PRICE TARGET ¥2,450

Derived from the base case. DCF model assumptions: risk-free rate 3.0%, equity beta 0.91, risk premium 3.2%, yielding WACC 5.3%. 0% perpetual growth rate from F3/36.



RISK REWARD CHART



EQUAL-WEIGHT THESIS

- We expect a significant expansion in profits from smartphone camera actuators and components starting in F3/27.
- We expect only limited improvement in earnings for the Module & System and Mobility segments.
- We estimate that the company received compensation from a mobility customer in 4Q FY3/27 related to the cancellation of a project. Assuming this compensation will not recur in FY3/28, earnings growth is likely to be limited, making the company's FY3/28 medium-term management plan target of ¥67.5bn in OP a challenging hurdle to achieve.

Consensus Rating Distribution



Risk Reward Themes

Contrarian: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥3,800	BASE CASE	¥2,450	BEAR CASE	¥1,500
Derived from DCF					
Earnings improve at a pace exceeding the mid-term management plan. OP ¥67.0bn in F3/27, ¥74.7bn in F3/28, ¥102.2bn in F3/31. Same currency and DCF model assumptions as base case. Fair value equates to P/E 17.5x F3/27e diluted EPS ¥217.3, 14.5x F3/28e diluted EPS ¥261.8, 10.2x F3/31e diluted EPS ¥373.1.					
Derived from DCF					
We expect earnings growth in FY3/28 to be limited, difficult to achieve medium-term plan. We forecast OP ¥48.5bn in F3/27, ¥51.2bn in F3/28, ¥78.7bn in F3/31. Base case assumes ¥155/¥ PT equates to P/E 14.6x F3/27e diluted EPS ¥167.8, 13.0x F3/28e diluted EPS ¥187.9 and 8.5x F3/31e EPS ¥287.6.					
F3/27e diluted BPS ¥2,233 x P/B 0.7x					
Sluggish output of high-end smartphones and declining market share cause smartphone camera actuator sales to fall 10% short of our base case. OP ¥36.4bn in F3/27, ¥35.2bn in F3/28, ¥43.2bn in F3/31. ROE 5.5% in F3/27, lower than the 5.9% cost of capital we assume, and the share drops to 0.7x F3/27e diluted BPS from earnings missing market consensus.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Nihon Dempa Kogyo: Key Financial Data and Risk Reward

Income Statement (JPY bn)						Cashflow Statement (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	53.1	54.6	64.5	72.7	78.2	Cash flow from operating activities	6.1	4.2	8.7	8.2	10.7
Costs of Sales	37.0	38.9	44.5	49.2	52.3	EBIT	3.0	2.6	4.6	6.5	8.0
Gross Profit	16.1	15.7	20.0	23.5	25.9	Depreciation	3.4	3.9	4.4	4.9	5.6
SG&A (incl. other income/losses)	11.4	12.4	14.5	16.0	16.8	Changes in Receivables	0.2	-0.5	-1.2	-1.9	-1.3
Operating Income	4.6	3.4	5.5	7.5	9.1	Change in inventory	0.6	-1.5	-0.3	-1.2	-0.6
Financial income/losses	-1.7	-0.8	-0.9	-1.0	-1.1	Changes in Payables	0.0	1.2	1.9	1.5	1.0
Profit before income taxes	3.0	2.6	4.6	6.5	8.0	Other	-1.1	-1.5	-0.6	-1.6	-2.0
Income taxes	1.2	0.5	1.2	2.0	2.4	Cash flow from investing activities	-4.5	-7.4	-10.0	-9.0	-9.8
(Effective tax rate)	39%	19%	26%	31%	30%	Purchase of PP&E	-3.7	-5.7	-9.5	-8.6	-9.3
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Sales of PP&E	0.1	0.0	0.0	0.0	0.0
Net Profit	1.8	2.1	3.4	4.5	5.6	Other	-0.9	-1.7	-0.5	-0.4	-0.5
Capex	4.6	6.5	10.0	9.0	9.8	Cash flow from financing activities	1.9	-2.4	1.7	1.3	1.4
Depreciation & amortization	3.4	3.9	4.4	4.9	5.6	Net Change In Borrowings	3.3	-0.7	2.4	2.3	2.5
R&D	2.1	2.9	3.7	4.3	4.6	Purchase of treasury shares	0.0	-0.2	0.0	0.0	0.0
EBITDA	8.0	7.3	9.9	12.4	14.7	Cash dividend paid	-0.7	-0.7	-0.7	-0.9	-1.1
ROE	6.3%	6.8%	10.3%	12.4%	13.8%	Other	-0.7	-0.9	0.0	0.0	0.0
No of average shares (mn)	23.0	23.0	22.9	22.9	22.9	Effect of exchange rate change	0.0	0.5	0.0	0.0	0.0
EPS (JPY)	77.8	89.7	150.1	196.3	244.2	Changes in cash and cash equivalents	3.6	-5.1	0.4	0.6	2.4
BPS (JPY)	1,265.0	1,382.1	1,510.2	1,666.5	1,863.7	Cash and cash equivalents at end of the period	15.9	10.8	11.2	11.8	14.2
DPS (JPY)	30.0	30.0	35.0	42.0	50.0	FCF	1.7	-3.2	-1.3	-0.8	0.9
USD/JPY (Average)	152.5	151.0	156.3	155.0	155.0						
Balance Sheet (JPY bn)						Sales by Application (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	71.5	76.3	83.3	90.8	98.9	Automotive	29.2	30.1	33.1	32.9	34.3
Current assets	43.5	44.0	45.9	49.6	53.8	Mobile communications / IoT	13.5	13.0	12.8	12.9	13.2
Cash and cash equivalents	15.9	10.8	11.2	11.8	14.2	Industrial equipment	3.7	4.4	10.0	18.0	21.6
Accounts receivable	12.7	13.8	15.0	16.9	18.2	Special products	1.4	1.7	2.0	2.2	2.4
Inventories	10.5	12.4	12.7	13.9	14.6	Optical products	1.4	1.2	1.6	1.7	1.8
Other financial assets	4.4	6.9	6.9	6.9	6.9	Others	3.9	4.3	5.0	5.0	5.0
Fixed assets	28.0	32.3	37.5	41.2	45.1	Automotive	4%	3%	10%	0%	4%
Tangible fixed assets	19.7	23.2	28.1	31.8	35.5	Mobile communications / IoT	22%	-4%	-1%	1%	2%
Intangible fixed assets	2.3	3.2	3.4	3.5	3.7	Industrial equipment	26%	19%	128%	79%	20%
Other non-current assets	6.0	5.9	5.9	5.9	5.9	Special products	-	21%	16%	10%	10%
Current liabilities	12.2	14.0	15.9	17.6	18.8	Optical products	-	-18%	38%	5%	6%
Short term borrowings	0.8	1.3	1.3	1.4	1.5	Others	-	11%	18%	0%	0%
Accounts payable	9.5	10.3	12.2	13.7	14.8						
Other current liabilities	1.9	2.3	2.4	2.5	2.5						
Fixed liabilities	30.2	30.4	32.8	35.0	37.4						
Long term borrowings	25.8	24.8	27.2	29.3	31.8						
Other fixed liabilities	4.4	5.6	5.6	5.6	5.6						
Net assets	29.2	31.9	34.6	38.2	42.7						
Total liabilities and equity	71.5	76.3	83.3	90.8	98.9						

Risk Reward – Nihon Dempa Kogyo (6779.T)

Ensuring profitability of DC, server products is key

PRICE TARGET ¥4,500

Our price target is ¥4,500, based on a DCF valuation (risk-free rate 3.0%; beta 1.2; risk premium 3.2%; WACC 4.6%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥330, our price target equates to P/E 22.9x our F3/28e EPS of ¥196.3, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

RISK REWARD CHART



BULL CASE

¥7,200

Accelerated sales growth for industrials/defence

OP and EPS forecasts of ¥5.5bn/¥150.1 in F3/27, ¥8.1bn/¥213.7 in F3/28, ¥11.6bn/¥314.0 in F3/29, ¥14.7bn/¥405.6 in F3/30, and ¥18.8bn/¥523.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥500. The price equates to 33.7x our F3/28e EPS of ¥213.7 and 4.3x BPS of ¥1,683.9.

BASE CASE

¥4,500

A/DC oscillator demand grows, mainly in optics

OP and EPS forecasts of ¥5.5bn/¥150.1 in F3/27, ¥7.5bn/¥196.3 in F3/28, ¥9.1bn/¥244.2 in F3/29, ¥10.5bn/¥283.5 in F3/30, and ¥12.3bn/¥331.4 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥330. The price equates to 22.9x our F3/28e EPS of ¥196.3 and 2.7x BPS of ¥1,666.5.

BEAR CASE

¥600

Weak demand for automotive, N America smartphones

OP and EPS forecasts of ¥5.5bn/¥150.1 in F3/27, ¥5.0bn/¥122.1 in F3/28, ¥4.8bn/¥113.4 in F3/29, ¥4.3bn/¥100.3 in F3/30, and ¥4.2bn/¥95.9 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥100. The price equates to 4.9x our F3/28e EPS of ¥122.1 and 0.4x BPS of ¥1,592.3.

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Meiko Electronics: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	206.8	240.6	361.5	475.4	555.4	Cash flow from operating activities	21.7	27.5	23.4	41.2	71.6
Costs of Sales	167.0	190.1	285.2	368.1	425.4	EBIT	18.5	27.4	36.3	53.8	68.7
Gross Profit	39.8	50.4	76.4	107.3	130.0	Depreciation	12.5	13.7	19.0	29.0	39.0
SG&A	20.7	25.9	37.1	47.9	55.4	Changes in Receivables	-3.3	-8.3	-15.8	-16.7	-9.2
Operating Income	19.1	24.6	39.3	59.4	74.6	Changes in Inventory	2.9	6.5	19.6	18.5	13.0
Non-operating income/expenses	-0.3	1.9	-3.0	-5.6	-5.9	Changes in Payables	-3.4	-3.3	-8.3	-16.5	-21.0
Recurring Profit	18.8	26.5	36.3	53.8	68.7	Other	-5.5	-8.5	-27.4	-26.9	-18.9
Extraordinary income	-0.2	0.9	0.0	0.0	0.0	Cash flow from investing activities	-24.3	-55.5	-70.0	-70.0	-70.0
Profit before income taxes	18.5	27.4	36.3	53.8	68.7	Purchase of PP&E	-24.4	-49.8	-69.8	-69.8	-69.8
Income taxes	3.4	7.5	8.3	16.5	21.0	Sales of PP&E	0.2	0.0	0.0	0.0	0.0
(Effective tax rate)	19%	27%	23%	31%	31%	Other	-0.1	-5.7	-0.2	-0.2	-0.2
Profit attributable to non-controlling interests	0.2	0.2	0.2	0.2	0.2	Cash flow from financing activities	4.1	30.1	49.6	28.6	3.2
Net Profit	14.9	19.8	27.8	37.1	47.5	Net Change In Borrowings	7.4	33.1	53.6	33.6	9.8
Capex	29.8	49.7	70.0	70.0	70.0	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	12.5	13.7	19.0	29.0	39.0	Cash dividend paid	-2.4	-2.7	-4.0	-5.0	-4.5
R&D	6.3	6.0	9.0	11.0	12.5	Other	-0.8	-0.3	0.0	0.0	0.0
EBITDA	31.6	38.3	58.3	88.4	113.6	Effect of exchange rate changes	0.1	1.4	-7.5	0.0	0.0
ROE	14.5%	16.2%	19.2%	22.0%	23.1%	Changes in cash and cash equivalents	1.6	3.6	-4.5	-0.1	4.8
No. of average shares (mn)	26.2	26.1	25.7	25.7	25.7	Cash and cash equivalents at end of the period	22.9	26.5	21.9	21.8	26.6
EPS (JPY)	569.5	758.6	1,081.9	1,445.5	1,850.6	FCF	-2.7	-27.9	-46.6	-28.8	1.6
BPS (JPY)	4,218.2	5,309.2	5,946.9	7,197.3	8,793.0	Sales by application					
DPS (JPY)	88.0	115.0	170.0	225.0	285.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	152.6	151.1	156.4	155.0	155.0	Automotive	90.4	97.5	130.8	132.2	135.5
Balance Sheet						Smartphone/Tablet	25.5	29.4	38.2	57.8	67.8
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Satellite	16.6	28.2	57.9	96.2	116.2
Total assets	256.4	335.3	427.2	513.7	579.0	AI server	4.2	7.3	23.5	62.5	96.5
Current assets	114.1	142.5	183.8	229.5	264.1	Moludes	9.2	9.2	13.7	16.0	17.4
Cash and cash equivalents	23.2	27.5	21.9	21.8	26.6	Amusement/Smart appliance/Others	23.7	24.4	24.9	24.7	24.8
Accounts receivable	46.2	61.6	92.6	121.8	142.3	Electronic equipment	36.2	42.6	70.0	82.1	92.1
Inventories	38.5	49.5	65.4	82.1	91.3	Semi Package	1.0	2.0	2.5	4.0	5.0
Other financial assets	6.3	3.9	3.9	3.9	3.9	(YoY)					
Fixed assets	142.3	192.8	243.4	284.2	314.9	Automotive	-5%	8%	34%	1%	3%
Tangible fixed assets	129.6	174.9	225.6	266.4	297.1	Smartphone/Tablet	8%	15%	30%	51%	17%
Intangible fixed assets	5.7	5.0	5.0	5.0	5.0	Satellite	118%	70%	105%	66%	21%
Other non-current assets	7.0	12.8	12.8	12.8	12.8	AI server	-	74%	222%	166%	54%
Current liabilities	100.3	128.3	156.2	180.9	198.0	Moludes	-	0%	49%	17%	9%
Short term borrowings	52.1	63.7	69.5	73.5	76.0	Amusement/Smart appliance/Others	69%	3%	2%	-1%	1%
Accounts payable	29.6	39.0	58.7	77.1	90.1	Electronic equipment	19%	18%	64%	17%	12%
Other current liabilities	18.6	25.6	28.0	30.3	31.8	Semi Package	43%	100%	25%	60%	25%
Fixed liabilities	40.5	63.2	110.9	140.5	147.8						
Long term borrowings	34.7	57.6	105.3	135.0	142.2						
Other fixed liabilities	5.8	5.5	5.5	5.5	5.5						
Net assets	115.6	143.8	160.2	192.3	233.2						
Total liabilities and equity	256.4	335.3	427.2	513.7	579.0						

Risk Reward – Meiko Electronics (6787.T)

Geopolitical risks drive continued growth in high-value, high-ASP PCB products

PRICE TARGET ¥39,600

Our price target is ¥39,600, based on a DCF valuation (risk-free rate 3.0%; beta 1.4; risk premium 3.2%; WACC 4.7%); beyond our 3-year base earnings forecasts, we also build forecasts for F3/30–F3/31 into our DCF-based valuation, and assume the company maintains FCF/share of ¥2,950 from F3/32 onward. Our price target equates to 27x our F3/28e EPS of ¥1,445.5, and our forex rate assumption for the next quarter and thereafter is ¥155/US\$.

OVERWEIGHT THESIS

Our scenario has the firm benefiting from production capacity enhancements undertaken to meet growing demand. We evaluate positively its proactive stance on expanding production capacity for build-up substrates, including those for satellite communications and high-end smartphones. In our view, Meiko's expertise in producing build-up substrates in Vietnam will enable it to benefit from growing demand for high layer/density buildup PCBs, as well as from efforts to move supply chains out of Greater China.

RISK REWARD CHART



Risk Reward Themes

New Data Era: Positive
Secular Growth: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥56,700	BASE CASE	¥39,600	BEAR CASE	¥9,300
Faster demand growth in buildup/semi PKG substrate		Continued growth in high layer/density buildup PCB		AI server/satellite demand weakens	
OP and EPS forecasts of ¥39.3bn/¥1,081.9 in F3/27, ¥69.3bn/¥1,695.6 in F3/28, ¥103.3bn/¥2,497.0 in F3/29, ¥136.1bn/¥3,318.2 in F3/30, and ¥172.9bn/¥4,257.8 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥4,250. Our fair value equates to 33.4x our F3/28e EPS of ¥1,695.6 and 6.9x our F3/28e BPS of ¥8,192.5.		OP and EPS forecasts of ¥39.3bn/¥1,081.9 in F3/27, ¥59.4bn/¥1,445.5 in F3/28, ¥74.6bn/¥1,850.6 in F3/29, ¥93.5bn/¥2,364.9 in F3/30, and ¥115.2bn/¥2,968.8 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥2,950. Our PT equates to 27.4x our F3/28e EPS of ¥1,445.5 and 5.5x our F3/28e BPS of ¥7,197.3.		OP and EPS forecasts of ¥39.3bn/¥1,081.9 in F3/27, ¥42.4bn/¥993.5 in F3/28, ¥44.9bn/¥1,055.8 in F3/29, ¥45.4bn/¥1,075.3 in F3/30, and ¥50.5bn/¥1,223.4 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥1,200. Our fair value equates to 9.4x our F3/28e EPS of ¥993.5 and 1.4x our F3/28e BPS of ¥6,745.4.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Hirose Electric: Key Financial Data and Risk Reward

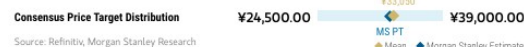
Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	189.4	211.3	256.6	276.9	298.6	Cash flow from operating activities	55.7	45.9	44.8	64.2	70.0
Costs of Sales	104.0	122.3	146.0	155.3	167.4	EBIT	46.2	46.6	58.3	67.2	72.6
Gross Profit	85.4	89.0	110.5	121.6	131.2	Depreciation	17.9	19.5	21.0	23.6	26.1
SG&A (incl. other income/losses)	42.7	46.0	55.2	59.2	63.4	Changes in Receivables	-1.9	-7.2	-10.7	-4.8	-5.1
Operating Income	42.7	43.0	55.3	62.4	67.8	Change in inventory	-1.4	-3.5	-5.0	-2.8	-3.0
Financial income/losses	3.5	3.6	3.0	4.8	4.8	Changes in Payables	2.0	3.5	3.6	1.6	1.7
Profit before income taxes	46.2	46.6	58.3	67.2	72.6	Other	-7.2	-13.0	-22.4	-20.6	-22.3
Income taxes	13.2	13.5	19.8	20.6	22.2	Cash flow from investing activities	-42.9	-9.2	-28.2	-30.5	-32.8
(Effective tax rate)	29%	29%	34%	31%	31%	Purchase of PP&E	-26.7	-20.0	-28.2	-30.5	-32.8
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
Net Profit	33.0	33.1	38.5	46.6	50.4	Other	-16.3	10.8	0.0	0.0	0.0
Capex	26.7	20.0	28.2	30.5	32.8	Cash flow from financing activities	-16.7	-37.9	-39.1	-31.9	-33.4
Depreciation & amortization	17.9	19.5	21.0	23.6	26.1	Net Change In Borrowings	0.0	0.0	0.0	0.0	0.0
R&D	13.5	13.5	14.0	15.1	16.3	Purchase of treasury shares	0.0	-20.4	-22.5	-15.0	-15.0
EBITDA	60.6	62.5	75.3	86.0	93.9	Cash dividend paid	-15.7	-16.5	-16.6	-16.9	-18.4
ROE	8.9%	8.9%	10.2%	12.1%	12.6%	Other	-0.9	-1.1	0.0	0.0	0.0
No of average shares (mn)	33.8	33.4	32.5	31.7	31.1	Effect of exchange rate change	-0.7	3.0	0.0	0.0	0.0
EPS (JPY)	976.3	991.9	1,185.7	1,470.1	1,619.8	Changes in cash and cash equivalents	-4.7	1.7	-22.5	1.8	3.8
BPS (JPY)	10,940.1	11,554.4	11,854.9	12,547.4	13,340.4	Cash and cash equivalents at end of the period	85.7	87.3	64.9	66.7	70.5
DPS (JPY)	490.0	520.0	520.0	570.0	630.0	FCF	12.7	36.6	16.6	33.7	37.2
USD/JPY (Average)	152.6	150.8	156.1	155.0	155.0	Sales by Application					
Balance Sheet						(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	416.9	431.2	431.6	447.9	466.5	General Industrial	46.9	62.0	86.2	92.3	99.2
Current assets	248.8	233.8	227.0	236.4	248.3	Smartphone	46.4	44.3	45.7	46.9	48.3
Cash and cash equivalents	85.7	87.3	64.9	66.7	70.5	Consumer/Mobile equipment	34.7	34.7	37.7	39.1	40.6
Marketable securities	87.6	57.6	57.6	57.6	57.6	Automotive/Mobility related	49.2	54.3	63.0	70.4	78.8
Accounts receivable	40.4	49.9	60.6	65.4	70.5	ns infrastructure equipment	5.4	6.7	10.5	12.0	13.9
Inventories	25.1	30.2	35.1	37.9	40.9	OA equipment and Others	6.8	9.3	13.5	16.2	17.8
Other financial assets	97.6	66.4	66.4	66.4	66.4	(YoY)					
Fixed assets	168.1	197.4	204.6	211.5	218.2	General Industrial	-7%	32%	39%	7%	7%
Tangible fixed assets	91.3	96.3	103.5	110.4	117.1	Smartphone	31%	-5%	3%	3%	3%
Intangible fixed assets	6.6	7.7	7.7	7.7	7.7	Consumer/Mobile equipment	37%	0%	9%	4%	4%
Other non-current assets	70.2	93.4	93.4	93.4	93.4	Automotive/Mobility related	15%	10%	16%	12%	12%
Current liabilities	29.1	32.8	33.8	35.3	37.0	ns infrastructure equipment	20%	24%	57%	15%	15%
Short term borrowings	0.0	0.0	0.0	0.0	0.0	OA equipment and Others	0%	36%	46%	20%	10%
Accounts payable	12.9	16.7	20.3	21.9	23.7						
Other current liabilities	16.2	16.1	13.4	13.4	13.3						
Fixed liabilities	17.7	20.3	20.3	20.3	20.3						
Long term borrowings	0.0	0.0	0.0	0.0	0.0						
Other fixed liabilities	17.7	20.3	20.3	20.3	20.3						
Net assets	370.1	378.1	377.5	392.2	409.2						
Total liabilities and equity	416.9	431.2	431.6	447.9	466.5						

Risk Reward – Hirose Electric (6806.T)

Sales growth to sustain historically high levels while maintaining high margins

PRICE TARGET ¥32,600

Our PT is ¥32,600, based on a DCF valuation (risk-free rate 3.0%; beta 0.9; risk premium 3.2%; WACC 5.5%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥2,000; our price target equates to P/E 22.2x our F3/28e EPS of ¥1,470.1, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.



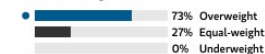
RISK REWARD CHART



OVERWEIGHT THESIS

The equity story is the continued top-line growth across applications through the acquisition of high-margin projects. We like the continued high level of sales growth relative to historical levels, while maintaining a high-profitability structure enabled by its semi-fabrics production model. We believe that if the equity market comes to view profit growth as a mid-term trend, there is further upside to valuations.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive
Secular Growth: Positive
Technology Diffusion: Positive

View descriptions of Risk Rewards Themes [here](#)

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Japan Aviation Electronics: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	221.6	227.9	248.9	256.0	272.2	Cash flow from operating activities	36.3	17.0	20.9	26.3	25.8
Costs of Sales	179.6	191.4	207.4	210.5	222.1	EBIT	15.2	10.2	7.2	10.8	12.4
Gross Profit	42.1	36.4	41.4	45.5	50.1	Depreciation	20.9	19.7	20.5	20.7	20.8
SG&A	26.4	27.5	33.0	34.8	37.9	Changes in Receivables	3.8	-0.1	-3.6	-1.2	-2.8
Operating Income	15.6	8.9	8.4	10.7	12.2	Change in inventory	-1.0	-2.6	-2.3	-0.3	-1.5
Non-operating income/expenses	-0.8	-0.7	-1.2	0.1	0.2	Changes in Payables	1.6	-2.9	2.3	0.8	1.8
Recurring Profit	14.8	8.2	7.2	10.8	12.4	Other	-4.0	-7.2	-3.2	-4.5	-4.9
Extraordinary income	0.3	1.9	0.0	0.0	0.0	Cash flow from investing activities	-19.2	-24.4	-19.5	-19.5	-19.5
Profit before income taxes	15.2	10.2	7.2	10.8	12.4	Purchase of PP&E	-18.0	-23.7	-21.0	-21.0	-21.0
Income taxes	3.6	3.1	2.2	3.3	3.8	Sales of PP&E	0.4	0.0	0.0	0.0	0.0
(Effective tax rate)	24%	30%	31%	31%	31%	Other	-1.5	-0.7	1.5	1.5	1.5
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Cash flow from financing activities	-31.6	0.5	-7.6	-5.8	-4.6
Net Profit	11.6	7.1	5.0	7.5	8.6	Net Change In Borrowings	-27.5	4.8	-3.9	-2.4	-1.2
Capex	18.0	23.7	21.0	21.0	21.0	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	20.9	19.7	20.5	20.7	20.8	Cash dividend paid	-4.0	-4.0	-3.7	-3.4	-3.4
R&D	11.3	12.1	13.2	13.3	13.9	Other	0.0	-0.3	0.0	0.0	0.0
EBITDA	36.5	28.7	28.9	31.4	33.0	Effect of exchange rate changes	-1.0	2.6	-0.9	-0.3	0.0
ROE	8.9%	5.1%	3.5%	5.1%	5.7%	Changes in cash and cash equivalents	-15.4	-4.4	-7.1	0.7	1.7
No of average shares (mn)	67.4	67.4	67.4	67.4	67.4	Cash and cash equivalents at end of the period	52.9	48.5	41.4	42.1	43.8
EPS (JPY)	172.0	104.9	74.1	111.2	127.5	FCF	17.1	-7.4	1.4	6.8	6.3
BPS (JPY)	1,983.6	2,141.0	2,147.1	2,204.1	2,281.6	Sales by Product/Application					
DPS (JPY)	60.0	60.0	50.0	50.0	50.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	152.6	150.7	156.2	155.0	155.0	Connector	201.8	206.9	224.9	230.1	244.1
Balance Sheet						Handhelds	66.9	65.2	55.8	52.7	51.2
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Automotive	112.6	115.7	131.5	136.2	144.4
Total assets	215.4	232.1	231.2	233.4	239.3	Industrial & Infrastructure	18.0	20.6	31.0	32.8	34.7
Current assets	125.3	130.7	129.5	131.7	137.7	Aviation & Space (Defence)	0.0	0.0	0.9	2.6	8.0
Cash and cash equivalents	52.9	48.5	41.4	42.1	43.8	Others	4.3	5.4	5.8	5.8	5.8
Marketable securities	0.0	0.0	0.0	0.0	0.0	Aerospace	19.3	20.5	23.4	25.3	27.5
Accounts receivable	37.7	36.9	42.5	43.7	46.5	Automotive	0.2	0.1	0.1	0.1	0.1
Inventories	29.2	32.8	35.2	35.5	37.0	Industrial & Infrastructure	10.7	9.3	11.2	12.0	12.8
Other financial assets	5.6	10.4	10.4	10.4	10.4	Aviation & Space (Defence)	8.4	11.1	12.1	13.3	14.5
Fixed assets	90.2	101.5	101.7	101.6	101.6	Others	0.5	0.5	0.5	0.5	0.5
Tangible fixed assets	70.3	77.6	78.1	78.4	78.6	Connector	-2%	3%	9%	2%	6%
Intangible fixed assets	2.7	3.1	2.8	2.5	2.3	Handhelds	-11%	-2%	-14%	-5%	-3%
Other non-current assets	17.2	20.8	20.8	20.8	20.8	Automotive	7%	3%	14%	4%	6%
Current liabilities	48.9	52.9	56.2	57.2	59.7	Industrial & Infrastructure	-9%	14%	50%	6%	6%
Short term borrowings	6.0	11.0	11.9	12.0	12.6	Aviation & Space (Defence)	-	-	-	199%	207%
Accounts payable	28.4	25.5	27.8	28.6	30.4	Others	-1%	24%	8%	0%	0%
Other current liabilities	14.6	16.4	16.5	16.6	16.7	Aerospace	-4%	6%	14%	8%	9%
Fixed liabilities	32.8	34.9	30.2	27.6	25.8	Automotive	-79%	-56%	6%	-1%	0%
Long term borrowings	31.0	30.8	28.0	23.4	21.6	Industrial & Infrastructure	-22%	-13%	20%	7%	7%
Other fixed liabilities	1.8	4.1	4.1	4.1	4.1	Aviation & Space (Defence)	55%	31%	9%	10%	10%
Net assets	133.7	144.4	144.8	148.6	153.9	Others	-29%	-2%	7%	0%	0%
Total liabilities and equity	215.4	232.1	231.2	233.4	239.3	OP by Segment					
						(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
						Connector	16.0	12.0	10.3	12.3	13.1
						Aerospace	2.6	2.2	3.3	3.5	3.9
						Others	0.1	0.1	0.1	0.1	0.1
						(OPM)					
						Connector	9%	6%	5%	5%	6%
						Aerospace	13%	11%	14%	14%	14%
						Others	18%	23%	22%	21%	21%

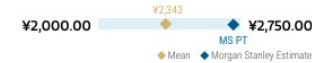
Risk Reward – Japan Aviation Electronics Industry (6807.T)

Key to profitability recovery is regaining North American smartphone share

PRICE TARGET ¥2,700

Our price target is ¥2,700, based on a DCF valuation (risk-free rate 3.0%; beta 1.0; risk premium 3.2%; WACC 4.9%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥155; our price target equates to P/E 24.3x our F3/28e EPS of ¥111.2, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

Consensus Price Target Distribution



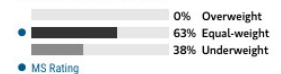
RISK REWARD CHART



EQUAL-WEIGHT THESIS

The equity story centers on margin improvement driven by a recovery in North American smartphone connector share. We expect top-line growth to continue, driven not only by increased sales of wire harnesses for ADAS cameras but also by connectors for BEV chargers and expanded connector sales in India. However, for the stock to rise, we believe profitability must improve through a rebound in North American B-to-B connector market share and expansion of the relatively high-margin aerospace business.

Consensus Rating Distribution



Risk Reward Themes

Secular Growth: *Negative*
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥3,800	BASE CASE	¥2,700	BEAR CASE	¥1,500
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Growth accelerates led by auto/phone connectors

OP and EPS forecasts of ¥8.4bn/¥74.1 in F3/27, ¥12.2bn/¥126.1 in F3/28, ¥15.8bn/¥163.1 in F3/29, ¥19.5bn/¥200.2 in F3/30, and ¥23.2bn/¥237.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥240. The price equates to 30.1x our F3/28e EPS of ¥126.1 and 1.7x BPS of ¥2,218.9.

Sales grow, but midterm margin flat

OP and EPS forecasts of ¥8.4bn/¥74.1 in F3/27, ¥10.7bn/¥111.2 in F3/28, ¥12.2bn/¥127.5 in F3/29, ¥13.4bn/¥139.4 in F3/30, and ¥14.7bn/¥152.7 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥155. Our PT equates to 24.3x our F3/28e EPS of ¥111.2 and 1.2x BPS of ¥2,204.1.

Auto growth slows; smartphone decline continues

OP and EPS forecasts of ¥8.4bn/¥74.1 in F3/27, ¥8.6bn/¥90.5 in F3/28, ¥8.5bn/¥89.0 in F3/29, ¥8.3bn/¥87.5 in F3/30, and ¥7.2bn/¥77.1 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥80. The price equates to 16.6x our F3/28e EPS of ¥90.5 and 0.7x BPS of ¥2,183.3.

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

CMK: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	95.5	100.2	112.3	116.2	120.7	Cash flow from operating activities	9.1	7.8	10.3	12.6	13.6
Costs of Sales	79.9	85.3	94.9	97.6	100.8	EBIT	5.3	5.3	4.7	4.7	5.6
Gross Profit	15.6	14.9	17.4	18.6	19.9	Depreciation	5.9	6.5	9.0	9.5	10.0
SG&A	11.8	12.1	12.3	12.8	13.2	Changes in Receivables	-0.2	0.5	-1.9	-0.4	-0.5
Operating Income	3.8	2.8	5.1	5.8	6.7	Other	0.7	0.3	1.5	0.5	0.6
Non-operating income/expenses	1.7	1.3	-0.4	-1.1	-1.1	Changes in Payables	-0.8	-1.0	-1.5	-1.4	-1.7
Recurring Profit	5.5	4.1	4.7	4.7	5.6	Other	-2.0	-3.8	-1.5	-0.3	-0.4
Extraordinary income	-0.2	1.2	0.0	0.0	0.0	Cash flow from investing activities	-18.8	-6.9	-6.3	-10.0	-10.7
Profit before income taxes	5.3	5.3	4.7	4.7	5.6	Purchase of PP&E	-18.3	-8.6	-5.9	-9.4	-10.0
Income taxes	1.4	1.1	1.5	1.4	1.7	Sales of PP&E	0.3	0.0	0.0	0.0	0.0
(Effective tax rate)	27%	20%	32%	30%	30%	Other	-0.8	1.8	-0.4	-0.6	-0.7
Profit attributable to non-controlling interests	0.1	0.2	0.2	0.2	0.2	Cash flow from financing activities	4.7	-4.1	-4.2	10.7	-3.1
Net Profit	3.8	4.0	3.0	3.1	3.7	Net Change In Borrowings	8.2	-1.8	-2.6	12.9	-0.7
						Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Capex	18.9	9.2	6.3	10.0	10.7	Cash dividend paid	-1.4	-1.4	-1.4	-2.0	-2.1
Depreciation & amortization	5.9	6.5	9.0	9.5	10.0	Other	-2.2	-0.9	-0.2	-0.2	-0.2
R&D	0.6	0.5	0.6	0.6	0.6	Effect of exchange rate changes	1.8	0.4	-1.8	-0.1	0.0
EBITDA	9.7	9.3	14.1	15.3	16.7	Changes in cash and cash equivalents	-3.1	-2.8	-1.8	13.2	-0.2
RDE	5.1%	5.0%	3.6%	3.7%	4.3%	Cash and cash equivalents at end of the period	22.2	19.4	17.6	30.8	30.6
						FCF	-9.7	0.9	4.0	2.6	2.9
No. of average shares (mn)	71.2	71.3	71.3	71.3	71.3						
EPS (JPY)	53.2	56.5	41.9	43.5	51.9	Sales by Product					
BPS (JPY)	1,110.3	1,171.1	1,170.7	1,184.7	1,206.6	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
DPS (JPY)	20.0	20.0	28.0	30.0	32.0	Double-sided/Multi-layer	61.1	63.4	71.6	71.2	70.8
						Double-sided	10.0	9.5	9.4	9.0	8.6
USD/JPY (Average)	151.8	149.8	155.4	155.0	155.0	Multi-layer	51.0	53.9	62.2	62.2	62.2
						HDI	30.2	33.0	37.0	41.3	46.2
						Others	4.2	3.8	3.7	3.7	3.7
						(YoY)					
Balance Sheet						Double-sided/Multi-layer	2%	4%	13%	-1%	-1%
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Double-sided	0%	-5%	-2%	-4%	-4%
Total assets	148.5	151.5	161.2	165.9	167.7	Multi-layer	2%	6%	15%	0%	0%
Current assets	62.4	61.7	64.1	78.5	79.6	HDI	16%	9%	12%	12%	12%
Cash and cash equivalents	22.2	19.5	17.6	30.8	30.6	Others	-8%	-10%	-2%	0%	0%
Accounts receivable	17.8	19.9	22.3	23.1	24.0						
Inventories	18.6	18.4	20.4	20.8	21.2						
Other financial assets	3.8	3.8	3.8	3.8	3.8						
Fixed assets	86.1	89.8	87.1	87.5	88.1						
Tangible fixed assets	76.5	80.3	77.3	77.2	77.4						
Intangible fixed assets	1.0	1.5	1.7	2.2	2.6						
Other non-current assets	8.6	8.1	8.1	8.1	8.1						
Current liabilities	33.7	38.0	39.8	39.8	39.9						
Short term borrowings	12.8	10.3	9.9	9.3	8.6						
Accounts payable	14.0	12.5	14.0	14.5	15.1						
Other current liabilities	6.9	15.3	15.8	16.0	16.2						
Fixed liabilities	33.4	27.4	25.2	38.7	38.6						
Long term borrowings	31.3	25.3	23.0	36.6	36.5						
Other fixed liabilities	2.1	2.2	2.2	2.2	2.2						
Net assets	81.4	86.0	86.2	87.4	89.2						
Total liabilities and equity	148.5	151.5	161.2	165.9	167.7						

Risk Reward – CMK (6958.T)

Further vehicle electrification and growth in new business areas awaited

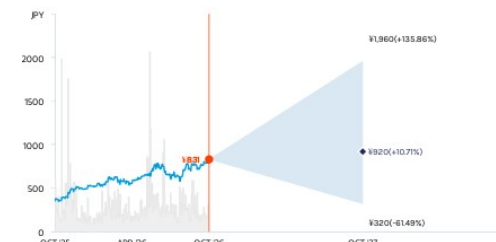
PRICE TARGET ¥920

Our price target is ¥920, based on a DCF valuation (risk-free rate 3.0%; beta 1.3; risk premium 3.2%; WACC 5.4%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥70; our price target equates to P/E 21.2x our F3/28e EPS of ¥43.5, and our forex rate assumption for the next quarter and thereafter is ¥155/¥.

EQUAL-WEIGHT THESIS

Our scenario calls for earnings growth driven by automotive PCBs. We maintain the view that earnings will resume growing after bottoming in F3/26 1H, driven by recovery in production mainly at J-ODM automakers. However, we assign an EW rating as the ramp-up in demand for Japanese Tier-1 suppliers' integrated ECUs appears to be delayed, and the negative impact from sluggish car demand in Europe due to the economic slowdown is likely to persist. That said, the high quality and reliability of CMK's buildup PCBs honed across many years, and its supply system which allows the same boards to be produced globally (including in new facilities in Thailand), continue to be well regarded.

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE	¥1,960	BASE CASE	¥920	BEAR CASE	¥320
Integrated ECU demand growth accelerates		BU sales for Integrated ECU rise fully from F3/27		Weak automotive production, slow uptake for ADAS	
OP and EPS forecasts of ¥6.3bn/¥53.1 in F3/27, ¥8.1bn/¥65.9 in F3/28, ¥9.8bn/¥81.4 in F3/29, ¥12.0bn/¥103.8 in F3/30, and ¥15.5bn/¥138.9 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥140. The price equates to 29.7x our F3/28e EPS of ¥65.9 and 1.6x BPS of ¥1,218.4.		OP and EPS forecasts of ¥5.1bn/¥41.9 in F3/27, ¥5.8bn/¥43.5 in F3/28, ¥6.7bn/¥51.9 in F3/29, ¥7.1bn/¥56.1 in F3/30, and ¥8.4bn/¥68.7 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥70. Our PT equates to 21.2x our F3/28e EPS of ¥43.5 and 0.8x BPS of ¥1,184.7.		OP and EPS forecasts of ¥4.5bn/¥36.3 in F3/27, ¥4.5bn/¥30.9 in F3/28, ¥4.5bn/¥30.9 in F3/29, ¥4.2bn/¥28.1 in F3/30, and ¥4.5bn/¥32.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥30. The price equates to 10.4x our F3/28e EPS of ¥30.9 and 0.3x BPS of ¥1,166.5.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Daishinku: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	38.6	39.6	43.8	46.2	49.5	Cash flow from operating activities	2.3	-1.8	3.0	5.2	5.7
Costs of Sales	29.3	30.4	33.5	35.0	36.9	EBIT	1.6	1.1	0.3	1.1	2.2
Gross Profit	9.3	9.2	10.3	11.2	12.6	Depreciation	4.0	4.2	5.0	5.2	5.5
SG&A	8.4	8.1	9.0	9.2	9.6	Changes in Receivables	0.5	-0.2	-0.6	-0.6	-0.8
Operating Income	0.9	1.1	1.3	2.0	3.0	Change in inventory	-1.9	-5.6	-2.4	-0.9	-1.3
Non-operating income/expenses	-0.5	-0.4	-0.6	-0.9	-0.8	Changes in Payables	1.0	0.1	0.4	0.3	0.3
Recurring Profit	0.4	0.7	0.7	1.1	2.2	Other	-2.8	-1.5	0.3	0.1	-0.2
Extraordinary income	1.2	0.3	-0.4	0.0	0.0	Cash flow from investing activities	-6.3	-3.1	-4.0	-4.6	-5.0
Profit before income taxes	1.6	1.1	0.3	1.1	2.2	Purchase of PP&E	-6.9	-3.0	-4.0	-4.6	-5.0
Income taxes	0.6	0.5	0.1	0.3	0.7	Sales of PP&E	0.0	0.1	0.0	0.0	0.0
(Effective tax rate)	39%	42%	35%	27%	32%	Other	0.5	-0.2	0.0	0.0	0.0
Profit attributable to non-controlling interests	0.7	0.2	0.2	0.2	0.2	Cash flow from financing activities	-1.7	1.2	1.8	-2.8	-2.7
Net Profit	0.3	0.4	0.0	0.6	1.3	Net Change in Borrowings	-0.1	2.7	3.2	-1.5	-1.4
Capex	1.0	4.8	4.0	4.6	5.0	Purchase of treasury shares	-0.3	-0.2	0.0	0.0	0.0
Depreciation & amortization	4.0	4.2	5.0	5.2	5.5	Cash dividend paid	-0.9	-0.9	-0.9	-0.9	-0.9
R&D	0.6	1.9	2.4	2.5	2.7	Other	-0.5	-0.4	-0.5	-0.5	-0.5
EBITDA	4.9	5.4	6.3	7.2	8.5	Effect of exchange rate changes	-0.1	1.2	-2.1	-0.1	0.0
ROE	0.8%	1.1%	0.0%	1.7%	3.6%	Changes in cash and cash equivalents	-5.9	-2.5	-1.2	-2.3	-2.0
No. of average shares (mn)	32.1	31.8	31.8	31.8	31.8	Cash and cash equivalents at end of the period	18.5	16.0	14.8	12.4	10.4
EPS (JPY)	8.9	13.2	-0.4	18.9	40.9	FCF	-4.0	-4.9	-1.0	0.6	0.7
BPS (JPY)	1,166.1	1,226.9	1,133.5	1,121.0	1,133.9	Sales by application					
DPS (JPY)	28.0	28.0	28.0	28.0	28.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	152.2	150.7	156.1	155.0	155.0	Communication	10.2	10.2	11.2	12.3	13.8
Balance Sheet						Automotive	13.8	14.8	15.9	16.9	18.2
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Consumer	10.7	10.4	10.9	11.2	11.5
Total assets	89.9	98.8	98.4	97.0	96.5	Industrial	4.0	4.2	5.7	5.9	6.2
Current assets	48.9	53.4	54.8	54.0	54.1	(YoY)					
Cash and cash equivalents	18.7	16.3	14.8	12.4	10.4	Communication	-13%	0%	10%	10%	11%
Accounts receivable	9.1	10.2	10.8	11.4	12.2	Automotive	3%	8%	8%	6%	8%
Inventories	18.0	24.6	27.0	27.8	29.2	Consumer	15%	-3%	6%	2%	3%
Other financial assets	3.1	2.3	2.3	2.3	2.3	Industrial	-19%	5%	35%	3%	7%
Fixed assets	41.0	45.4	43.6	43.0	42.5						
Tangible fixed assets	34.9	38.4	36.6	36.0	35.5						
Intangible fixed assets	1.0	0.8	0.8	0.8	0.8						
Other non-current assets	5.1	6.2	6.2	6.2	6.2						
Current liabilities	24.9	32.9	31.4	30.8	30.7						
Short term borrowings	16.9	22.6	21.6	20.7	19.9						
Accounts payable	3.4	4.1	4.5	4.8	5.1						
Other current liabilities	4.6	6.3	5.2	5.4	5.6						
Fixed liabilities	19.8	17.9	22.1	21.5	20.9						
Long term borrowings	16.3	14.1	18.3	17.8	17.1						
Other fixed liabilities	3.5	3.8	3.8	3.8	3.8						
Net assets	45.2	48.0	45.0	44.6	45.0						
Total liabilities and equity	89.9	98.8	98.4	97.0	96.5						

Risk Reward – Daishinku (6962.T)

Awaiting demand acceleration for new devices and production system upgrades

PRICE TARGET ¥650

Our price target is ¥650, based on a DCF valuation (risk-free rate 3.0%; beta 1.1; risk premium 3.2%; WACC 4.7%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥80, our price target equates to P/E 34.5x our F3/28e EPS of ¥18.9, and our forex rate assumption for the next quarter and thereafter is ¥155/¥.

RISK REWARD CHART



UNDERWEIGHT THESIS

Sales weighting for photolithographic products has risen as demand for high-frequency products from clients has increased. We give the firm high marks for progressively ramping up capacity of TCXO, demand for which has been rising for use in communication modules or GPS applications. However, we believe Daishinku still needs time to build a network to expand sales of the Arkh Series, which has technological advantages in terms of both cost and productivity. We also think the firm will continue facing downward pressure on prices, esp. for consumer applications. Ongoing production issues could also become a bottleneck in the demand upcycle, and countermeasures should be closely monitored.

Risk Reward Themes

New Data Era: **Negative**

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥1,800	BASE CASE	¥650	BEAR CASE	¥220
Telecom/auto growth speeds up					
OP and EPS forecasts of ¥1.3bn/¥(0.4) in F3/27, ¥2.7bn/¥31.4 in F3/28, ¥4.6bn/¥75.5 in F3/29, ¥6.3bn/¥113.2 in F3/30, and ¥8.3bn/¥154.1 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥150. The price equates to 57.2x our F3/28e EPS of ¥31.4 and 1.6x BPS of ¥1,133.5.					
Gold surge and higher costs limit recovery					
OP and EPS forecasts of ¥1.3bn/¥(0.4) in F3/27, ¥2.0bn/¥18.9 in F3/28, ¥3.0bn/¥40.9 in F3/29, ¥3.9bn/¥62.9 in F3/30, and ¥4.8bn/¥81.8 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥80. Our PT equates to 34.5x our F3/28e EPS of ¥18.9 and 0.6x BPS of ¥1,133.5.					
Investment burden, slowing topline growth					
OP and EPS forecasts of ¥1.3bn/¥(0.4) in F3/27, ¥1.8bn/¥15.7 in F3/28, ¥2.2bn/¥25.2 in F3/29, ¥2.7bn/¥40.9 in F3/30, and ¥2.8bn/¥40.9 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥40. The price equates to 14.0x our F3/28e EPS of ¥15.7 and 0.2x BPS of ¥1,133.5.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Hamamatsu Photonics: Key Financial Data and Risk Reward

Income Statement						Cash Flow Statement					
(Y bn)	F9/24	F9/25	F9/26e	F9/27e	F9/28e	(Y bn)	F9/24	F9/25	F9/26e	F9/27e	F9/28e
Net Sales	204.0	212.1	236.9	245.5	265.2	Net Cash from Operations	38.1	37.8	34.0	52.4	49.7
COGS	100.1	110.7	123.6	128.1	140.4	Net Income	25.1	14.2	17.5	20.5	24.5
Gross Profit	103.9	101.4	113.3	117.4	124.8	Depreciation and Amortization	16.7	18.9	23.4	27.7	26.7
SG&A Expenses	71.8	85.2	89.9	91.2	93.0	Changes in Working Capital	-3.9	3.7	-7.9	3.2	-2.5
Operating Profit	32.1	16.2	23.4	26.2	31.8	Other-net	0.0	1.0	1.0	1.0	1.0
Non-operating Income/Loss	2.4	2.6	3.0	3.0	3.0	Net Cash from Investing Activities	-73.7	-42.2	-36.4	-16.0	-16.0
Recurring Profit	34.5	18.8	26.4	29.2	34.8	Purchase of PP&E	-28.1	-35.8	-33.7	-15.0	-15.0
Extraordinary Income/Loss	0.9	1.9	0.2	0.2	0.2	Sale/Purchase of Investments	-0.5	-4.1	0.0	0.0	0.0
Pretax Profit	35.4	20.7	26.6	29.4	35.0	Other-net	-45.1	-2.2	-2.8	-1.0	-1.0
Income Tax	10.0	6.2	8.9	8.8	10.5	Free Cash Flow	-35.6	-4.4	-2.5	36.4	33.7
(Effective Tax Rate)	35%	35%	35%	35%	35%	Net Cash from Financing Activities	12.6	-2.8	-31.7	-27.7	-28.4
Minority Interest	0.3	0.4	0.2	0.0	0.0	Change in Interest-bearing Debt	24.5	29.0	0.0	0.0	0.0
Net Profit	25.1	14.2	17.5	20.5	24.5	Change in Treasury Stock	0.0	-20.0	-20.0	-15.0	-15.0
Capex	32.2	39.1	33.7	15.0	15.0	Payment of Cash Dividends	-11.8	-11.6	-11.7	-12.7	-13.4
Depreciation and Amortization	17.7	22.3	23.4	27.7	26.7	Other-net	-0.2	-0.3	0.0	0.0	0.0
R&D Expenses	13.6	18.4	19.2	20.3	21.7	Net Effect of Exchange Rate Changes	1.2	0.7	0.0	0.0	0.0
EBITDA	49.8	38.5	46.8	53.9	58.5	Net Change in Cash	-21.8	-6.5	-34.1	8.7	5.3
ROE	7.7%	4.4%	5.6%	6.8%	8.2%	Cash at End of Period	92.6	86.0	51.9	60.7	65.9
Average Share Outstanding (Diluted)	309.7	298.7	298.8	283.2	277.9	Segment Information					
EPS (Diluted, Yen)	81.2	47.4	60.2	72.4	88.0	(Y bn)	F9/24	F9/25	F9/26e	F9/27e	F9/28e
BPS (Yen)	1,075.3	1,082.9	1,065.6	1,069.7	1,067.8	Net Sales	204.0	212.1	236.9	245.5	265.2
DPS (Yen)	38.0	38.0	40.0	45.0	45.0	Electron Tube	78.5	71.9	81.5	84.8	89.9
USD/JPY: Average Exchange Rate	149.4	149.2	155.2	155.0	155.0	Opto-semiconductor	78.9	79.5	92.6	100.9	108.0
YoY % Change						Imaging and Measurement Instruments	32.8	32.7	34.3	36.3	39.2
Net Sales	-8%	4%	12%	4%	8%	Others	3.0	5.7	4.7	-2.3	-0.5
Operating Profit	-43%	-50%	45%	12%	21%	YoY % Change					
Net Profit	-41%	-44%	23%	17%	19%	Electron Tube	-8%	4%	12%	4%	8%
As % of Sales						Opto-semiconductor	-6%	-8%	13%	4%	6%
Gross Profit	50.9%	47.8%	47.8%	47.8%	47.1%	Imaging and Measurement Instruments	-21%	1%	16%	9%	7%
Operating Profit	15.7%	7.6%	9.9%	10.7%	12.0%	Others	3%	0%	5%	6%	8%
Net Profit	12.3%	6.7%	7.4%	8.4%	9.2%	Operating Profit by Segment	180%	89%	-18%	-150%	-80%
Capex	15.8%	18.4%	14.2%	6.1%	5.7%	Electron Tube	32.1	16.2	23.4	26.2	31.8
Depreciation and Amortization	8.7%	10.5%	9.9%	11.3%	10.1%	Total Operating Profit	23.8	19.0	25.4	27.0	29.2
Balance Sheet						Opto-semiconductor	17.9	12.6	17.3	15.9	19.1
(Y bn)	F9/24	F9/25	F9/26e	F9/27e	F9/28e	Imaging and Measurement Instruments	10.4	9.7	8.7	10.9	11.9
Total Assets	434.6	455.0	439.7	432.8	429.6	Others	0.7	0.9	0.0	-0.2	0.0
Total Current Assets	239.0	236.4	210.9	216.7	225.2	Adjustments	-23.3	-21.6	-21.7	-22.0	-23.8
Cash and Cash Equivalents	97.0	90.6	56.5	65.2	70.4	Operating Margin	16%	8%	10%	11%	12%
Marketable Securities	2.9	6.5	6.5	6.5	6.5	Electron Tube	30%	26%	31%	32%	33%
Account Receivables	45.7	46.6	50.9	52.1	55.6	Opto-semiconductor	23%	16%	19%	16%	18%
Inventory	80.6	77.0	81.2	77.1	76.8	Imaging and Measurement Instruments	32%	30%	26%	30%	30%
Other Current Assets	12.8	15.8	15.8	15.8	15.8	Others	24%	15%	0%	10%	7%
Total Long Term Assets	195.6	218.6	228.8	216.1	204.4	Total Operating Profit					
Net PP&E	132.9	150.7	160.9	148.2	136.5	Electron Tube	23.8	19.0	25.4	27.0	29.2
Investments in Securities	35.7	34.6	34.6	34.6	34.6	Opto-semiconductor	17.9	12.6	17.3	15.9	19.1
Other Assets	27.0	33.3	33.3	33.3	33.3	Imaging and Measurement Instruments	10.4	9.7	8.7	10.9	11.9
Total Current Liabilities	80.7	108.5	109.2	109.4	110.2	Others	0.7	0.9	0.0	-0.2	0.0
ST Debt and Current Portion of LT	28.4	55.5	55.5	55.5	55.5	Adjustments	-23.3	-21.6	-21.7	-22.0	-23.8
Accounts Payable	6.6	6.8	7.5	7.8	8.5	Operating Margin	16%	8%	10%	11%	12%
Other Current Liabilities	45.8	46.2	46.2	46.2	46.2	Electron Tube	30%	26%	31%	32%	33%
Total Long Term Liabilities	20.9	23.0	23.0	23.0	23.0	Opto-semiconductor	23%	16%	19%	16%	18%
Long Term Debt	8.5	10.6	10.6	10.6	10.6	Imaging and Measurement Instruments	32%	30%	26%	30%	30%
Other Long Term Liabilities	12.4	12.5	12.5	12.5	12.5	Others	24%	15%	0%	10%	7%
Net Assets	333.0	323.5	307.5	300.2	296.7	Total Operating Profit					
Net Debt	-63.0	-31.0	3.1	-5.6	-10.9	Electron Tube	23.8	19.0	25.4	27.0	29.2

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Risk Reward – Hamamatsu Photonics (6965.T)

Stands out for high potential of its applied photonics products.

PRICE TARGET ¥2,500

DCF model assumptions: 3.0% risk-free rate, 0.91 equity beta, 3.2% risk premium, yielding 5.5% WACC; zero growth from F9/36.



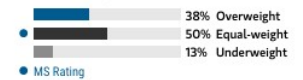
RISK REWARD CHART



EQUAL-WEIGHT THESIS

- Following the completion of customer inventory adjustments, we expect sales to bottom out not only in semiconductor manufacturing and inspection equipment, but also in medical and biotech applications, leading to a recovery in earnings.
- While we expect a recovery from F9/26 following a trough in F9/25, it may take time to exceed the peak OP of ¥57.0bn in F9/22 given higher fixed costs, including depreciation and labor expenses.
- Photonics applications, such as optoelectronic conversion and photonic integration, hold significant potential, including the prospect of yet-to-be-anticipated new products contributing to earnings.

Consensus Rating Distribution



BULL CASE	¥3,700	BASE CASE	¥2,500	BEAR CASE	¥1,000
Derived from DCF		Derived from DCF		0.9x diluted book value ¥1,112 for F9/26 end	
Earnings from SPE, FA, and automobile markets ahead of base case. OP ¥29.7bn in F9/26, ¥38.5bn in F9/27, and ¥77.8bn in F9/30. Same currency and DCF model assumptions as base case. Fair value equates to P/E 50.3 x F9/26e diluted EPS ¥73.6, 37.3 x F9/27e diluted EPS ¥99.3 and 18.5 x F9/27e diluted EPS ¥200.3.		Earnings to bottom out in F9/25 and recover from F9/26, although it is likely to take time to reach a new record high. OP ¥23.4bn in F9/26, ¥26.2bn in F9/27, and ¥43.7bn in F9/30. Assumes ¥155/USD; DCF model assumptions is the same as explained above. Zero growth from F9/36. Base case fair value equates to P/E 41.6x F9/26e diluted EPS ¥60.2, 34.5x F9/27e diluted EPS ¥72.4 and 20.4x F9/30e diluted EPS ¥122.9.		Expect continued earnings decline in F9/26, with growth remaining modest from F9/27 onward. OP ¥16.9bn in F9/26, ¥17.8bn in F9/27, and ¥34.4bn in F9/30. Same currency assumptions as base case. ROE 3.8% in F9/26 and 3.9% in F9/27, 5.7% in F9/29 lower than the 5.9% cost of capital; trades at 0.9x diluted book value for F9/26-end.	

Kyocera: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	2,014.5	2,070.2	2,078.9	2,174.9	2,273.1
COGS	1,455.3	1,462.6	1,435.6	1,514.6	1,963.6
Gross Profit	559.2	607.6	643.3	660.3	309.5
SG&A	531.9	489.5	478.0	488.8	125.0
Operating Profit	27.3	118.1	165.3	171.5	184.5
Non-operating Income/Loss	36.3	50.9	44.8	29.0	19.0
Pretax Profit	63.6	169.0	210.1	200.5	203.5
Income Tax	36.2	24.1	40.5	40.1	40.7
(Effective Tax Rate)	1%	0%	0%	0%	0%
Minority Interest	3.4	4.0	2.4	2.0	2.0
Net Profit	24.1	141.0	167.2	158.4	160.8
Capex	161.7	149.1	222.2	200.0	200.0
Depreciation and Amortization	111.7	110.9	120.0	130.0	135.0
R&D	104.3	115.7	119.7	125.0	125.0
EBITDA	139.0	229.1	285.3	301.5	319.5
ROE	0.7%	4.3%	5.1%	5.1%	5.4%
Average Share Outstanding (Diluted)	1,412.4	1,372.6	1,291.5	1,194.8	1,127.3
EPS (Diluted, Yen)	17.1	102.7	129.5	132.6	146.9
BPS (Yen)	2,195.0	2,203.5	2,279.7	2,328.5	2,385.6
DPS (Yen)	50.0	52.0	65.0	70.0	70.0
USD/JPY: Average Exchange Rate	151.8	150.8	155.0	155.0	155.0
EUR/JPY: Average Exchange Rate	163.8	174.8	180.0	180.0	180.0
YoY % Change					
Sales	1%	3%	0%	5%	5%
Operating Profit	-71%	333%	40%	4%	8%
Net Profit	-76%	485%	19%	-5%	2%
As % of Sales					
Gross Profit	27.8%	29.4%	30.9%	30.4%	13.6%
Operating Profit	1.4%	5.7%	8.0%	7.9%	8.1%
Net Profit	1.2%	6.8%	8.0%	7.3%	7.1%
Capex	8.0%	7.2%	10.7%	9.2%	8.9%
Depreciation and Amortization	5.5%	5.4%	5.8%	6.0%	5.9%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	4,511.3	4,646.3	4,490.8	4,477.7	4,728.3
Total Current Assets	1,435.7	1,501.2	1,243.5	1,160.4	1,337.9
Cash and Cash Equivalents	444.7	455.9	187.9	68.5	73.0
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	411.2	416.6	391.0	409.1	427.6
Inventory	521.8	522.0	557.8	576.1	730.7
Other Current Assets	58.0	106.7	106.7	106.7	106.7
Total Long Term Assets	3,075.6	3,145.2	3,247.3	3,317.3	3,382.3
Net PP&E	651.9	685.5	787.6	857.6	922.6
Investments and Advances	1,770.3	1,804.0	1,804.0	1,804.0	1,804.0
Goodwill	282.2	274.0	274.0	274.0	274.0
Intangible Assets	142.1	121.2	121.2	121.2	121.2
Other Assets	229.1	260.5	260.5	260.5	260.5
Total Current Liabilities	491.7	525.0	530.6	541.7	604.4
ST Debt and Current Portion of LT Debt	45.8	56.1	56.1	56.1	56.1
Accounts Payable	207.0	194.8	200.4	211.4	274.1
Other Current Liabilities	238.8	274.2	274.2	274.2	274.2
Total Long Term Liabilities	776.4	753.9	753.9	903.9	1,103.9
Long Term Debt	202.6	189.0	189.0	339.0	539.0
Other Long Term Liabilities	573.8	565.0	565.0	565.0	565.0
Minority Interests	25.4	27.9	25.6	23.6	21.6
Total Equity	3,243.2	3,367.4	3,180.6	3,008.5	2,990.4
Net Debt (excl. KDDI Shares)	-385.3	-483.8	-285.3	-265.9	-320.4

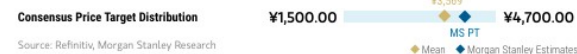
Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	237.9	226.2	280.2	261.1	183.4
Net Income	24.1	141.0	167.2	158.4	160.8
Depreciation and Amortization	160.9	158.2	120.0	130.0	135.0
Changes in Working Capital:	26.0	-41.4	-4.6	-25.3	-110.4
Other-net	26.8	-31.5	-2.4	-2.0	-2.0
Net Cash from Financing Activities	-150.5	74.5	-222.2	-200.0	-200.0
Purchase of PP&E	-154.7	-159.2	-222.2	-200.0	-200.0
Sale/Purchase of Investments	0.0	0.0	0.0	0.0	0.0
Other-net	4.2	233.7	0.0	0.0	0.0
Free cash flow	87.4	300.8	58.1	61.1	-16.6
Net Cash from Financing Activities	-64.9	-312.0	-326.0	-180.5	21.1
Change in Interest-bearing Debt	38.0	-6.2	0.0	150.0	200.0
Change in Treasury Stock	0.0	-200.0	-250.0	-100.0	-100.0
Payment of Cash Dividends	-73.3	-72.2	-76.0	-80.5	-78.9
Other-net	-29.6	-33.5	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-2.5	22.3	0.0	0.0	0.0
Net Change in Cash	20.0	11.1	-268.0	-119.4	4.5
Cash at End of Period	444.7	455.9	187.9	68.5	73.0
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,014.5	2,070.2	2,078.9	2,174.9	2,273.1
Core Components Business	567.1	653.4	709.1	748.3	798.3
Industrial & Automotive Components Unit	233.1	250.1	260.4	273.5	287.1
Semiconductor Components Unit	300.8	379.4	421.1	454.8	491.2
Others	33.3	23.9	27.5	20.0	20.0
Electronic Components Business	354.6	363.5	396.1	419.9	445.1
Solutions Business	1,111.0	1,070.9	985.1	1,016.8	1,039.7
Industrial Tools Unit	305.9	285.9	184.3	189.0	193.7
Document Solutions Unit(Information Eq.	480.0	478.5	507.1	519.8	532.8
Communications Unit	225.5	219.2	203.0	208.1	213.3
Others	99.7	87.3	90.7	100.0	100.0
Others	17.7	17.7	17.7	17.7	17.7
Elimination	-35.4	-31.8	-27.3	-26.0	-26.0
Net OP	24.0	133.2	196.8	201.5	214.5
Core Components Business	-1.1	63.1	98.5	105.5	112.5
Industrial & Automotive Components Unit	25.0	18.7	37.0	40.0	42.0
Semiconductor Components Unit	-27.8	46.9	61.9	65.0	70.0
Others	1.7	-2.8	-0.4	0.5	0.5
Electronic Components Business	-0.8	7.3	25.8	28.0	29.0
Solutions Business	72.9	103.9	98.3	95.0	100.0
Others	0.0	0.0	0.0	0.0	0.0
Segment Margin	1.2%	6.4%	9.5%	9.3%	9.4%
Core Components Business	-0.2%	9.7%	13.9%	14.1%	14.1%
Industrial & Automotive Components Unit	10.7%	7.5%	14.2%	14.6%	14.6%
Semiconductor Components Unit	-9.3%	12.4%	14.7%	14.3%	14.3%
Electronic Components Business	-0.2%	2.0%	6.5%	6.7%	6.5%
Solutions Business	6.6%	9.7%	10.0%	9.3%	9.6%

Risk Reward – Kyocera (6971.T)

After balance sheet reform, sustainability of profit growth will be the key.

PRICE TARGET ¥4,000

Derived from the base case. DCF model assumptions: 3.0% risk-free rate, 0.91 equity beta, 3.2% risk premium, for WACC of 5.7%; zero growth from F3/36.



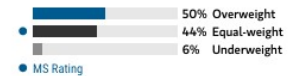
RISK REWARD CHART



EQUAL-WEIGHT THESIS

- Needs to further strengthen core components business and restructure solution business.
- While the divestment of KDDI shares and enhanced shareholder returns should support ROE improvement, we also want to assess how much of the numerator is being driven by profit growth.
- We expect ROE to improve to 5.1% in F3/27, 5.4% in F3/29, and 6.2% in F3/31, and we forecast that ROE in F3/31 will be higher than cost of capital of 5.9%.

Consensus Rating Distribution



Risk Reward Themes

Self-help: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥5,000

Derived from DCF

Sales top the base case by 5% as parts demand recovers: OP ¥204.3bn in F3/27, ¥230.5bn in F3/28, ¥308.5bn in F3/31. Same FX & DCF model assumptions as the base case. Bull case fair value equates to P/E of 34.8x on F3/27 diluted EPS of ¥143.5, 30.8x F3/28e diluted EPS of ¥162.3, 25.2x F3/31e diluted EPS of ¥198.1.

BASE CASE

¥4,000

Derived from DCF

We aim to assess the trajectory of ROE improvement, driven by balance sheet enhancement through the sale of KDDI shares and shareholder returns, as well as by profit growth. OP ¥165.3bn in F3/27, ¥171.5bn in F3/28, ¥249.6bn in F3/31. FX assumptions: ¥155/US\$. Base case fair value equates to 30.9x F3/27e diluted EPS of ¥129.5, 30.2x F3/28e diluted EPS of ¥132.6, 22.8x F3/31e diluted EPS of ¥175.5.

BEAR CASE

¥1,500

F3/27e diluted BPS ¥2,235 x P/B 0.7

Limited improvement in components business earnings. Improvement of capital efficiency limited, although shareholder return improvement continues. OP ¥101.1bn in F3/27, OP ¥131.5bn in F3/28, OP ¥113.6bn in F3/31. Same FX assumptions as the base case. ROE of 3.4% in F3/27 and 3.5% in F3/28 remains below capital costs of 5.9%, prompting the stock to trade on P/B of 0.7x.

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Taiyo Yuden: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	341.4	355.3	428.7	486.0	558.4
COGS	259.9	273.4	315.5	350.2	402.8
Gross Profit	71.6	81.9	113.0	135.7	155.6
SG&A Expenses	61.1	61.9	67.9	69.6	71.8
Operating Profit	10.5	20.0	45.1	66.1	83.8
Non-operating Income/Loss	0.1	4.1	-1.5	-1.5	-1.5
Recurring Profit	10.5	24.1	43.6	64.6	82.4
Extraordinary Income/Loss	-2.2	-4.1	-0.5	-2.0	-2.0
Pretax Profit	8.4	20.1	43.1	62.6	80.4
Income Tax	6.0	5.3	13.1	18.8	24.1
(Effective Tax Rate)	72%	26%	30%	30%	30%
Minority Interest	0.0	0.0	0.0	0.0	0.0
Net Profit	2.3	14.8	30.1	43.9	56.3
Capex	64.2	40.4	42.0	60.0	60.0
Depreciation and Amortization	46.3	49.1	50.0	52.0	52.0
R&D Expenses	15.0	14.5	15.0	16.0	17.0
EBITDA	56.7	69.1	95.1	118.1	135.8
ROE	0.7%	4.5%	8.5%	11.6%	13.6%
Average Share Outstanding (Diluted)	144.3	137.7	137.7	137.7	137.7
EPS (Diluted, Yen)	16.1	107.6	218.3	318.5	408.6
BPS (Yen)	2,206.4	2,501.8	2,633.8	2,857.0	3,161.2
DPS (Yen)	90.0	90.0	100.0	110.0	120.0
USD/JPY: Average Exchange Rate	152.6	150.0	156.0	155.0	155.0
YoY % Change					
Net Sales	6%	4%	21%	13%	15%
Operating Profit	15%	91%	125%	47%	27%
Net Profit	-72%	536%	103%	46%	28%
As % of Sales					
Gross Profit	21.0%	23.1%	26.4%	27.9%	27.9%
Operating Profit	3.1%	5.6%	10.5%	13.6%	15.0%
Net Profit	0.7%	4.2%	7.0%	9.0%	10.1%
Capex	18.8%	11.4%	9.8%	12.3%	10.7%
Depreciation and Amortization	13.5%	13.8%	11.7%	10.7%	9.3%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	573.2	615.5	639.0	673.2	720.4
Total Current Assets	279.3	318.3	349.8	376.0	415.1
Cash and Cash Equivalents	78.2	100.1	124.9	130.9	141.4
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	80.5	86.4	96.8	110.6	125.6
Inventory	110.0	125.8	120.1	128.4	142.1
Other Current Assets	10.5	6.0	6.0	6.0	6.0
Total Long Term Assets	293.9	297.2	289.2	297.2	305.2
Net PP&E	286.8	290.0	282.0	290.0	298.0
Investments in Securities	0.0	0.0	0.0	0.0	0.0
Other Assets	7.0	7.3	7.3	7.3	7.3
Total Current Liabilities	79.6	93.8	99.0	102.5	107.8
ST Debt and Current Portion of LT Debt	19.7	35.2	35.2	35.2	35.2
Accounts Payable	27.2	26.5	31.8	35.2	40.5
Other Current Liabilities	32.8	32.1	32.1	32.1	32.1
Total Long Term Liabilities	174.4	177.4	177.4	177.4	177.4
Long Term Debt	144.7	136.0	136.0	136.0	136.0
Other Long Term Liabilities	29.7	41.4	41.4	41.4	41.4
Net Assets	319.2	344.4	362.6	393.3	435.2
Minority Interests	0.0	0.0	0.0	0.0	0.0
Stock Acquisition Rights	0.7	0.0	0.0	0.0	0.0
Net Debt	86.2	71.1	46.3	40.3	29.8

Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	33.9	58.1	78.7	79.1	84.8
Net Income	2.3	14.8	30.1	43.9	56.3
Depreciation and Amortization	46.3	49.1	50.0	52.0	52.0
Changes in Working Capital	-13.5	-9.0	-1.4	-16.7	-23.4
Other-net	-1.2	3.2	0.0	0.0	0.0
Net Cash from Investing Activities	-63.5	-25.7	-42.0	-60.0	-60.0
Purchase of PP&E	-62.6	-40.4	-42.0	-60.0	-60.0
Sale/Purchase of Investments	1.3	0.0	0.0	0.0	0.0
Other-net	-2.2	14.7	0.0	0.0	0.0
Free Cash Flow	-29.6	32.4	36.7	19.1	24.8
Net Cash from Financing Activities	3.0	-6.8	-11.9	-13.1	-14.4
Change in Interest-bearing Debt	15.7	6.5	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-11.2	-11.2	-11.9	-13.1	-14.4
Other-net	-1.5	-2.1	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-0.9	4.9	0.0	0.0	0.0
Net Change in Cash	-27.4	30.5	24.8	6.0	10.5
Cash at End of Period	67.5	98.1	122.9	128.9	139.4
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	341.4	355.3	428.7	486.0	558.4
Capacitors	232.1	251.8	320.8	374.3	441.6
Elina	0.0	0.0	0.0	0.0	0.0
Ferrite and Applied Products	61.5	64.3	70.2	72.2	77.3
Complex Devices	23.0	14.8	12.1	14.5	14.5
Others	24.8	24.5	25.6	25.0	25.0
Total Operating Profit	10.5	20.0	45.1	66.1	83.8
Operating Profit Margin	3.1%	5.6%	10.5%	13.6%	15.0%

Risk Reward – Taiyo Yuden (6976.T)

Share price has fallen, but upside potential likely to remain limited

PRICE TARGET ¥10,500

Derived from the base case. DCF assumptions: 3.0% risk-free rate, 1.54 equity beta, 3.2% risk premium, for 7.4% WACC; zero growth from F3/36.



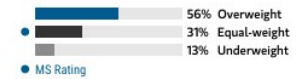
RISK REWARD CHART



EQUAL-WEIGHT THESIS

- While we expect earnings growth to continue, we believe the upside potential for the stock is likely to be limited.
- We expect continued earnings growth in the core MLCC business.
- We expect the contribution from premium MLCC products for AI/data center applications to remain limited.
- We expect sales growth for power inductors used in 5G smartphones and base stations to continue.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥14,000	BASE CASE	¥10,500	BEAR CASE	¥4,500
Derived from DCF					
MLCC sales exceed the base case. OP ¥56.5bn in F3/27, ¥85.5bn in F3/28, ¥176.3bn in F3/31. Same FX & DCF model assumptions as the base case. Fair value equates to P/E 50.7x on F3/27e diluted EPS of ¥276.1, 33.6x on F3/28e diluted EPS of ¥417.0, and 15.9x on F3/31e diluted EPS of ¥879.1.					
Earnings are expanding driven by growth in MLCC sales, but market expectations appear overly high: OP ¥45.1bn in F3/27, ¥66.1bn in F3/28, ¥107.7bn in F3/31. FX assumption: ¥155/US\$. Base case fair value equates to P/E 48.1x on F3/27e diluted EPS of ¥218.3, 33.0x on F3/28e diluted EPS of ¥318.5, and 19.8x on F3/31e diluted EPS of ¥530.1.					
F3/27e diluted BPS ¥2,552 x P/B 1.8					
Stays competitive in core products, but slowing end demand drags profits down. OP ¥39.2bn in F3/27, OP ¥56.1bn in F3/28, ¥65.8bn in F3/31. Same FX and DCF assumptions as in the base case. ROE 7.3% in F3/27, 10.1% in F3/28, improving from 4.5% in F3/26 and higher in F3/28 than cost of capital (7.9%), but with earnings lower than market expectation, P/B falls to 1.8x.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Murata Manufacturing: Key Financial Data and Risk Reward

Income Statement (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,743.4	1,830.9	2,194.6	2,628.3	3,104.1
Cost of Goods Sold	1,025.7	1,056.0	1,183.8	1,380.9	1,639.4
Gross Profit	717.7	774.8	1,010.8	1,247.5	1,464.7
SG&A Expenses	438.0	493.0	520.1	544.9	573.7
Operating Profit	279.7	281.8	490.8	702.6	891.0
Non-Operating Income/Loss	24.7	26.8	18.3	12.0	12.0
Pretax profit	304.4	308.6	509.0	714.6	903.0
Income Tax	71.4	74.9	125.8	175.1	221.2
(Effective Tax Rate)	23%	24%	25%	25%	25%
Minority Interest	-0.8	-0.1	0.0	0.0	0.0
Net Profit	233.8	233.8	383.2	539.5	681.8

Capex	180.5	247.8	253.0	250.0	250.0
Depreciation and Amortization	173.3	178.2	180.6	190.0	200.0
R&D	149.3	158.9	169.9	173.0	178.0
EBITDA	453.0	460.0	671.4	802.6	1,091.0
ROE	9.1%	8.8%	13.8%	18.2%	20.8%
Average Share Outstanding (Diluted)	1,869.3	1,831.4	1,811.9	1,786.3	1,769.6
EPS (Diluted, Yen)	125.08	127.65	211.52	302.05	385.28
BPS (Yen)	1,380.1	1,484.0	1,559.2	1,729.3	1,964.7
DPS (Yen)	57.0	65.0	70.0	75.0	82.0

USD/JPY: Average Exchange Rate	152.6	150.8	156.1	155.0	155.0
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YoY % Change					
Net Sales	6%	5%	20%	20%	18%
Operating Profit	30%	1%	74%	43%	27%
Net Profit	29%	0%	64%	41%	26%

As % of Sales					
Gross Profit	41.2%	42.3%	46.1%	47.5%	47.2%
Operating Profit	16.0%	15.4%	22.4%	26.7%	28.7%
Net Profit	13.4%	12.8%	17.5%	20.5%	22.0%
Capex	10.4%	13.5%	11.5%	9.5%	8.1%
Depreciation and Amortization	9.9%	9.7%	8.2%	7.2%	6.4%

Balance Sheet (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	3,028.2	3,199.1	3,315.1	3,593.8	4,001.2

Total Current Assets	1,497.9	1,581.4	1,625.0	1,843.7	2,201.0
Cash and Cash Equivalents	625.1	653.7	585.3	650.8	824.9
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	294.4	328.2	393.4	471.1	556.4
Inventory	482.8	520.5	567.2	642.7	740.6
Other Current Assets	95.5	79.1	79.1	79.1	79.1
Total Long Term Assets	1,530.3	1,617.7	1,690.1	1,750.1	1,800.1
Net PP&E	1,183.7	1,300.9	1,373.3	1,433.3	1,483.3
Investments in Securities	40.3	50.4	50.4	50.4	50.4
Other assets	306.3	266.4	266.4	266.4	266.4

Total Current Liabilities	276.5	316.5	326.1	341.0	366.6
ST Debt & Current Portion of LT Debt	0.8	1.7	1.7	1.7	1.7
Accounts Payable	69.6	79.8	89.5	104.3	123.9
Other Current Liabilities	206.1	234.9	234.9	234.9	234.9
Total Long Term Liabilities	171.7	164.8	164.8	164.8	164.8
Long Term Debt	1.7	1.5	1.5	1.5	1.5
Other Long Term Liabilities	170.0	163.3	163.3	163.3	163.3
Minority Interests	-0.8	-0.9	-0.9	-0.9	-0.9
Total Equity	2,580.0	2,717.8	2,825.1	3,088.9	3,476.7

Net Debt	-663.0	-700.8	-632.4	-697.9	-872.1
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Cash Flow Statement (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	451.9	425.2	461.5	591.2	718.1
Net Income	233.8	233.8	383.2	539.5	681.8
Depreciation and Amortization	173.3	178.2	180.6	190.0	200.0
Changes in Working Capital:					
Other-net	24.4	-18.0	-102.3	-138.3	-163.7
Net Cash from Investing Activities	-208.1	-193.8	-253.0	-250.0	-250.0
Purchase of PP&E	-179.4	-243.0	-253.0	-250.0	-250.0
Sale/Purchase of Investments	3.4	2.2	0.0	0.0	0.0
Other-net	-32.1	46.9	0.0	0.0	0.0
Free cash flow	243.8	231.4	208.5	341.2	468.1
Net Cash from Financing Activities	-242.7	-221.8	-276.9	-275.7	-294.0
Change in Interest-bearing Debt	-50.4	0.8	0.0	0.0	0.0
Change in Treasury Stock	-80.0	-100.0	-150.0	-150.0	-150.0
Payment of Cash Dividends	-101.6	-110.7	-126.9	-125.7	-144.0
Other-net	-10.7	-11.9	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	2.0	19.0	0.0	0.0	0.0
Net Change in Cash	3.1	28.6	-68.4	65.5	174.1

Segment Information (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,793.3	1,830.9	2,194.6	2,628.3	3,104.1
Components Total	1,033.1	1,159.7	1,526.1	1,838.7	2,156.5
Capacitors	831.8	936.4	1,274.5	1,577.0	1,884.3
Inductors and EMI Filters	201.3	223.3	251.6	261.7	272.2
Devices/Modules	697.2	656.0	653.2	776.7	934.7
High Frequency Devices/Communications Module	443.6	394.8	377.9	453.5	544.2
Battery/Power supply	155.7	154.1	162.3	203.4	263.5
Functional Devices	97.8	107.1	113.0	119.8	127.0
Others	13.1	15.2	15.3	13.0	13.0

Total Operating Profit	279.7	281.8	490.8	702.6	891.0
Components Total	275.2	315.1	501.4	671.4	834.5
Devices/Modules	10.0	-26.5	-2.4	37.2	62.5
Others	-5.4	-6.8	-6.2	-6.0	-6.0
Operating Profit Margin	15.6%	15.4%	22.4%	26.7%	28.7%
Components Total	26.6%	27.2%	32.9%	36.5%	38.7%
Devices/Modules	2.3%	-6.7%	-0.6%	8.2%	11.5%

Sales by Application (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales (excl. Non-product Sales)	1,793.3	1,830.9	2,194.6	2,628.3	3,104.1
AV	674.2	653.0	-	-	-
Communication	453.1	474.5	-	-	-
Computer and Peripheral	281.9	310.4	-	-	-
Automobile Electronics	150.4	142.7	-	-	-
Home Appliances and Others	183.7	250.3	-	-	-

As % of Total					
AV	38.7%	36%	-	-	-
Communication	26.0%	26%	-	-	-
Computer and Peripheral	16.2%	17%	-	-	-
Automobile Electronics	8.6%	8%	-	-	-
Home Appliances and Others	10.5%	14%	-	-	-

Risk Reward – Murata Manufacturing (6981.T) Top Pick

We expect AI/DC use value-added MLCC sales and earnings to continue to expand

PRICE TARGET ¥11,000

Derived from our base case and a DCF model, with the following assumptions: 3.0% risk-free rate, 1.39 equity beta, 3.2% risk premium, yielding 7.5% WACC; zero growth from F3/36.



RISK REWARD CHART



BULL CASE	¥14,000	BASE CASE	¥11,000	BEAR CASE	¥3,500
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Derived from DCF

Demand for AI/DC use MLCC is stronger than in the base case. OP assumptions: ¥560.3bn in F3/27, ¥794.0bn in F3/28, ¥1,515bn in F3/31. Same FX & DCF model assumptions as the base case. Bull case fair value equates to P/E of 58.2x on F3/27 diluted EPS of ¥240.4, 41.1x on F3/28 diluted EPS of ¥340.7, and 21.1x on F3/31 diluted EPS of ¥664.0.

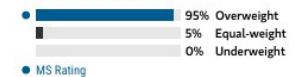
Derived from DCF

Earnings continue to grow and beat consensus over the medium/long term due to technological advantages & competitiveness, especially in MLCCs. Assumptions: OP ¥490.8bn in F3/27, ¥702.6bn in F3/28, ¥1,350.6bn in F3/31; FX ¥155/US\$, DCF model (see Valuation Methodology); zero growth from F3/36. Base case fair value equates to P/E 52.0x on F3/27e diluted EPS of ¥211.5, 36.4x on F3/28e diluted EPS of ¥302.0 and 18.6x on F3/31e diluted EPS of ¥592.5.

OVERWEIGHT THESIS

- We expect AI/DC use value-added MLCC sales and earnings expansion to continue.
- Expect a significant increase in the total capacitance required for MLCCs in AI/data center applications, alongside continued miniaturization and further increases in capacitance per unit.
- We expect Murata to remain competitive in MLCCs and other products over the long term.
- In F3/27 onward, we expect module earnings to improve as sales of RF modules rise.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: Positive
Secular Growth: Positive
View descriptions of Risk Rewards Themes [here](#)

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Nichicon: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	175.8	169.7	187.7	200.6	207.4	Cash flow from operating activities	18.3	8.2	10.2	16.0	18.7
Costs of Sales	147.4	139.7	150.7	157.7	162.0	EBIT	8.5	7.2	11.0	14.8	16.0
Gross Profit	28.3	30.0	37.0	42.9	45.4	Depreciation	8.5	8.1	8.5	9.4	10.3
SG&A	23.1	23.5	26.3	28.3	29.6	Changes in Receivables	3.2	-1.2	-4.0	-3.5	-1.9
Operating Income	5.2	6.5	10.7	14.6	15.8	Change in inventory	3.7	1.8	-2.7	-1.3	-0.1
Non-operating income/expenses	2.3	1.9	0.3	0.2	0.2	Changes in Payables	-2.2	-6.0	2.5	1.9	0.9
Recurring Profit	7.5	8.3	11.0	14.8	16.0	Other	-3.3	-1.7	-5.0	-6.1	-6.6
Extraordinary income	1.0	-1.1	0.0	0.0	0.0	Cash flow from investing activities	-8.4	-6.6	-10.6	-11.6	-12.1
Profit before income taxes	8.5	7.2	11.0	14.8	16.0	Purchase of PP&E	-10.7	-7.1	-12.0	-13.0	-13.5
Income taxes	2.1	0.4	2.8	4.5	4.0	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
(Effective tax rate)	25.3	5%	25%	30%	31%	Other	2.3	0.5	1.4	1.4	1.4
Profit attributable to non-controlling interests	0.5	0.5	0.5	0.5	0.5	Cash flow from financing activities	-14.3	-3.8	16.8	-3.3	-4.1
Net Profit	5.9	6.3	7.7	9.8	10.6	Net Change in Borrowings	2.2	-0.8	2.3	-0.6	-1.4
Capex	11.1	7.6	12.0	13.0	13.5	Purchase of treasury shares	-1.6	0.0	-5.0	0.0	0.0
Depreciation & amortization	8.5	8.1	8.5	9.4	10.3	Cash dividend paid	-2.5	-2.4	-2.5	-2.6	-2.7
R&D	7.2	7.3	7.5	8.0	8.3	Other	-12.5	-0.6	22.0	0.0	0.0
EBITDA	13.7	14.5	19.2	24.0	26.1	Effect of exchange rate changes	0.5	1.2	2.2	-0.2	0.0
ROE	5.3%	5.5%	6.4%	7.8%	8.0%	Changes in cash and cash equivalents	-3.9	-1.1	18.7	0.0	2.5
No. of average shares (mn)	68.3	67.1	66.7	65.3	65.3	Cash and cash equivalents at end of the period	25.5	24.4	43.1	43.1	45.6
EPS (JPY)	86.0	94.0	115.5	150.1	162.3	FCF	10.0	1.6	-0.4	3.4	6.6
BPS (JPY)	1,644.3	1,780.9	1,868.6	1,975.4	2,095.7	Sales by Product					
DPS (JPY)	36.0	38.0	39.0	41.0	43.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	151.6	150.0	156.0	155.0	155.0	Capacitor	99.2	102.7	116.7	120.8	123.7
Balance Sheet						Aluminum capacitor	79.3	82.0	90.1	93.8	95.6
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Film capacitor	19.8	20.7	25.6	27.0	28.1
Total assets	192.6	195.6	224.9	233.1	240.6	NECST	76.6	67.0	72.0	79.8	83.7
Current assets	108.3	107.7	133.0	137.8	142.2	Applied system and equipment	7.3	7.2	8.8	9.4	10.0
Cash and cash equivalents	25.5	24.5	43.1	43.1	45.6	EV	5.6	3.5	2.7	2.6	2.6
Accounts receivable	44.8	47.4	51.4	55.0	56.8	Power source	26.3	21.7	18.9	19.0	19.0
Inventories	33.2	32.3	35.0	36.3	36.4	Storage	35.6	32.4	39.5	46.7	50.0
Other financial assets	4.7	3.5	3.5	3.5	3.5	Others	1.9	2.2	2.2	2.2	2.2
Fixed assets	84.3	87.9	91.9	95.3	98.3	Capacitor	-7%	4%	13%	4%	2%
Tangible fixed assets	53.1	52.5	56.7	60.3	63.5	Aluminum capacitor	-9%	3%	10%	4%	2%
Intangible fixed assets	1.9	2.1	1.9	1.7	1.6	Film capacitor	-19%	6%	21%	13%	5%
Other non-current assets	29.3	33.2	33.2	33.2	33.2	NECST	2%	-13%	8%	11%	5%
Current liabilities	51.6	47.2	50.0	51.8	52.3	Applied system and equipment	6%	-1%	22%	7%	7%
Short term borrowings	9.2	10.9	11.1	11.0	10.6	EV	-31%	-37%	-24%	-2%	-2%
Accounts payable	28.5	23.3	25.6	27.5	28.5	Power source	11%	-17%	-13%	0%	0%
Other current liabilities	13.9	13.0	13.1	13.3	13.3	Storage	4%	-9%	22%	18%	7%
Fixed liabilities	27.2	24.8	48.9	48.4	47.5	Others	-20%	15%	0%	0%	0%
Long term borrowings	15.8	13.3	15.3	14.8	13.9	OP by Product					
Other fixed liabilities	11.5	11.6	33.6	33.6	33.6	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net assets	113.8	123.5	125.9	132.9	140.8	Capacitor	1.6	4.6	6.8	9.6	10.3
Total liabilities and equity	192.6	195.6	224.9	233.1	240.6	Aluminum capacitor	0.5	3.1	6.1	8.7	9.4
						Film capacitor	1.0	1.5	0.7	0.9	0.9
						NECST	3.7	1.9	4.0	5.0	5.5
						Applied system and equipment	1.1	0.8	1.3	1.0	1.1
						EV	-0.3	-0.8	-0.9	-0.8	-0.8
						Power source	0.5	0.1	-0.1	0.0	0.0
						Storage	2.2	1.5	3.6	4.8	5.1
						Others/Adjustment	0.2	0.2	0.1	0.0	0.0
						(CPM)	1.6%	4.5%	5.9%	7.9%	8.4%
						Capacitor	0.7%	3.8%	6.8%	9.3%	9.9%
						Aluminum capacitor	5.1%	7.2%	2.6%	3.2%	3.2%
						Film capacitor	4.8%	2.8%	6.5%	6.3%	6.5%
						NECST	14.4%	11.3%	14.7%	10.6%	10.6%
						Applied system and equipment	-5.4%	-21.3%	-34.8%	-29.4%	-29.4%
						EV	1.9%	0.3%	-0.3%	0.1%	0.2%
						Power source	6.2%	4.7%	9.1%	10.2%	10.3%
						Storage					

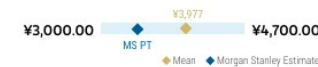
Risk Reward – Nichicon (6996.T)

Strong growth led by AI server capacitors and household batteries

PRICE TARGET ¥3,400

Our price target is ¥3,400, based on a DCF valuation (risk-free rate 3.0%; beta 1.0; risk premium 3.2%; WACC 4.8%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥180; our price target equates to P/E 22.7x our F3/28e EPS of ¥150.1, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE

¥7,400

Continuing strong growth for AEC & NECST sales

OP and EPS forecasts of ¥10.7bn/¥115.5 in F3/27, ¥18.2bn/¥188.4 in F3/28, ¥25.4bn/¥264.9 in F3/29, ¥32.3bn/¥338.5 in F3/30, and ¥40.7bn/¥427.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥425. The price equates to 39.3x our F3/28e EPS of ¥150.1 and 1.7x BPS of ¥1,975.4.

BASE CASE

¥3,400

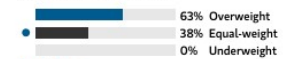
Value-added products drive earnings expansion

OP and EPS forecasts of ¥10.7bn/¥115.5 in F3/27, ¥14.6bn/¥150.1 in F3/28, ¥15.8bn/¥162.3 in F3/29, ¥16.7bn/¥173.1 in F3/30, and ¥17.6bn/¥182.2 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥180. Our PT equates to 22.7x our F3/28e EPS of ¥150.1 and 1.7x BPS of ¥1,975.4.

EQUAL-WEIGHT THESIS

The equity story is centered on increased sales of high value-added, high-capacity capacitors and the stable growth of the NECST business. From F3/27, we expect the capacitor business to expand, driven primarily by large-size products for AI servers and film capacitors for xEV applications, while competition in general-purpose products is likely to remain intense amid rivalry with Chinese players. We expect NECST business to deliver stable growth, supported by the wider adoption of residential energy storage systems.

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Electric Vehicles: Positive
New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Nippon Chemi-con: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	122.7	136.8	153.8	164.9	157.5	Cash flow from operating activities	-0.5	7.6	4.7	12.7	13.0
Costs of Sales	99.0	112.4	122.9	122.7	123.6	EBIT	0.6	3.5	5.5	6.6	7.9
Gross Profit	23.7	24.4	30.9	32.2	33.9	Depreciation	7.1	7.3	7.2	6.9	6.9
SG&A	20.0	21.1	23.4	23.6	24.1	Changes in Receivables	2.7	-4.3	-3.6	-0.2	-0.6
Operating Income	3.7	3.4	7.5	8.6	9.8	Change in Inventory	-3.8	3.1	-4.6	0.6	0.3
Non-operating income/expenses	-2.2	-1.3	-2.0	-2.0	-1.9	Changes in Payables	-1.0	-0.5	1.1	0.1	0.2
Recurring Profit	1.6	2.1	5.5	6.6	7.9	Other	-6.1	-1.5	-0.9	-1.2	-1.7
Extraordinary income	-1.0	1.4	0.0	0.0	0.0	Cash flow from investing activities	-8.8	-5.3	-5.7	-6.2	-6.3
Profit before income taxes	0.6	3.5	5.5	6.6	7.9	Purchase of PP&E	-0.3	-4.4	-5.1	-5.6	-5.7
Income taxes	0.5	1.1	1.8	2.0	2.4	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
(Effective tax rate)	88%	31%	33%	30%	30%	Other	-0.4	-0.9	-0.6	-0.6	-0.6
Profit attributable to non-controlling interests	0.0	0.1	0.0	0.0	0.0	Cash flow from financing activities	-11.9	-6.2	0.4	-5.6	-4.6
Net Profit	0.0	2.4	3.7	4.6	5.5	Net Change in Borrowings	-10.6	-4.6	3.5	-4.6	-3.5
Capex	7.6	5.9	5.7	6.2	6.3	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	7.1	7.3	7.2	6.9	6.9	Cash dividend paid	-0.2	-0.6	-1.1	-1.1	-1.1
R&D	4.2	3.9	4.4	4.4	4.5	Other	-1.2	-1.1	-2.0	0.0	0.0
EBITDA	10.8	10.7	14.7	15.5	16.7	Effect of exchange rate changes	0.7	1.3	4.3	-0.3	0.0
ROE	0.1%	4.0%	5.7%	6.7%	7.6%	Changes in cash and cash equivalents	-21.4	-2.6	3.6	0.6	2.2
No of average shares (mn)	38.9	34.4	27.4	27.4	27.4	Cash and cash equivalents at end of the period	23.9	21.3	24.9	25.5	27.6
EPS (JPY)	1.0	68.8	134.1	167.9	200.8	FCF	-10.2	2.3	-1.0	6.5	6.7
BPS (JPY)	2,631.2	2,594.1	2,457.4	2,574.1	2,735.2	Sales by Product					
DPS (JPY)	0.0	20.0	20.0	20.0	20.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	152.6	150.8	156.1	155.0	155.0	Capacitor	118.0	131.8	149.2	150.4	153.0
Balance Sheet						Aluminum capacitor	106.1	120.0	137.4	138.6	141.2
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	DLCAP	3.8	3.6	3.4	3.3	3.3
Total assets	162.7	166.0	175.5	174.2	175.3	Other capacitor	2.5	2.6	2.5	2.5	2.5
Current assets	86.6	87.6	99.4	99.6	102.0	Capacitor material	5.7	5.6	6.0	6.0	6.0
Cash and cash equivalents	23.9	21.3	24.9	25.5	27.6	Mechanical part	3.4	3.2	2.9	2.9	2.9
Accounts receivable	28.9	29.2	32.8	33.1	33.6	Other	1.2	1.8	1.7	1.7	1.7
Inventories	33.1	31.2	35.8	35.2	35.0	(YOY)					
Other financial assets	0.8	5.8	5.8	5.8	5.8	Capacitor	-19%	12%	13%	1%	2%
Fixed assets	76.1	78.4	76.1	74.6	73.3	Aluminum capacitor	-20%	13%	14%	1%	2%
Tangible fixed assets	48.3	47.5	46.0	45.3	44.7	DLCAP	-28%	-4%	-7%	-1%	0%
Intangible fixed assets	2.3	2.6	1.7	0.9	0.2	Other capacitor	-28%	4%	-4%	-1%	0%
Other non-current assets	25.4	28.3	28.3	28.3	28.3	Capacitor material	24%	-1%	7%	-1%	0%
Current liabilities	60.6	58.5	62.0	60.7	59.8	Mechanical part	-2%	-8%	-9%	-1%	0%
Short term borrowings	36.7	34.0	37.1	35.6	34.5	Other	-21%	51%	-9%	-1%	0%
Accounts payable	13.1	8.7	9.8	9.9	10.1						
Other current liabilities	10.8	15.1	15.2	15.2	15.2						
Fixed liabilities	45.4	44.4	45.5	42.4	40.0						
Long term borrowings	38.1	35.9	37.0	33.8	31.5						
Other fixed liabilities	7.3	8.5	8.5	8.5	8.5						
Net assets	96.7	63.1	68.0	71.2	75.6						
Total liabilities and equity	162.7	166.0	175.5	174.2	175.3						

Risk Reward – Nippon Chemi-Con (6997.T)

Near/mid-term targets present high hurdles

PRICE TARGET ¥2,400

Our price target is ¥2,400, based on a DCF valuation (risk-free rate 3.0%; beta 1.1; risk premium 3.2%; WACC 4.4%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥190; our price target equates to P/E 14.3x our F3/28e EPS of ¥167.9, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

RISK REWARD CHART



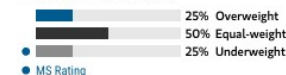
Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

UNDERWEIGHT THESIS

The equity story is earnings growth driven by rising AEC sales for industrial equipment and xEVs. While demand for value-added capacitors used in AI servers and xEVs should support further growth, we see a high hurdle to achieving current MTP targets. The company issued ¥9bn of C/D-class shares in June 2026, and investors should consider dilution risk from the conversion of D-class shares into common stock. The issuance also requires annual dividend payments to DBJ of ¥540mn through June 2029 and ¥720mn thereafter. We also note the potential for additional costs related to litigation involving the company's Singapore subsidiary.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: Negative

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥6,000	BASE CASE	¥2,400	BEAR CASE	¥1,300
Strong demand growth for AI/DC aluminum capacitors		Recovery continues, but cost & dilution concerns		Earnings remain subdued at the trough	
OP and EPS forecasts of ¥7.5bn/¥134.1 in F3/27, ¥10.6bn/¥215.4 in F3/28, ¥14.8bn/¥317.6 in F3/29, ¥17.2bn/¥379.7 in F3/30, and ¥19.5bn/¥438.1 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥410. The price equates to 27.9x our F3/28e EPS of ¥215.4 and 2.3x BPS of ¥2,621.6.		OP and EPS forecasts of ¥7.5bn/¥134.1 in F3/27, ¥8.6bn/¥167.9 in F3/28, ¥9.8bn/¥200.8 in F3/29, ¥10.4bn/¥219.1 in F3/30, and ¥10.3bn/¥219.1 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥190. The price equates to 14.3x our F3/28e EPS of ¥167.9 and 0.9x BPS of ¥2,574.1.		OP and EPS forecasts of ¥7.5bn/¥134.1 in F3/27, ¥7.8bn/¥149.7 in F3/28, ¥8.4bn/¥167.9 in F3/29, ¥8.2bn/¥167.9 in F3/30, and ¥7.3bn/¥146.0 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥120. The price equates to 8.7x our F3/28e EPS of ¥149.7 and 0.5x BPS of ¥2,555.9.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

KOA: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	64.1	72.3	89.5	94.2	100.1	Cash flow from operating activities	8.1	9.1	11.6	10.3	11.4
Costs of Sales	46.0	50.4	62.5	64.8	67.2	EBIT	1.3	5.3	9.3	5.6	7.4
Gross Profit	18.2	21.9	27.0	29.4	32.9	Depreciation	5.9	7.1	7.2	7.3	7.5
SG&A	17.0	18.2	21.3	22.6	24.4	Changes in Receivables	0.3	-1.6	-3.8	-1.0	-1.3
Operating Income	1.2	3.6	5.7	6.8	8.5	Change in inventory	0.7	0.5	-0.8	-0.2	-0.4
Non-operating income/expenses	0.1	1.6	0.6	-1.2	-1.1	Changes in Payables	-0.5	-0.9	0.8	0.2	0.3
Recurring Profit	1.2	5.2	6.3	5.6	7.4	Other	-0.5	-1.3	-1.1	-1.6	-2.2
Extraordinary income	0.0	0.0	3.0	0.0	0.0	Cash flow from investing activities	-23.9	-7.1	-4.1	-6.6	-7.0
Profit before income taxes	1.3	5.3	9.3	5.6	7.4	Purchase of PP&E	-25.1	-7.5	-4.0	-6.5	-6.9
Income taxes	1.0	1.3	1.3	1.7	2.3	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
(Effective tax rate)	79%	25%	14%	30%	31%	Other	1.1	0.4	-0.1	-0.1	-0.1
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Cash flow from financing activities	11.3	-0.5	-5.2	-1.0	-3.7
Net Profit	0.3	4.0	8.0	3.9	5.1	Net Change In Borrowings	13.2	0.7	-3.4	1.0	-1.8
Capex	24.2	5.9	4.1	6.6	7.0	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	5.9	7.1	7.2	7.3	7.5	Cash dividend paid	-1.8	-1.1	-1.8	-1.9	-1.9
R&D	3.6	3.6	3.5	3.7	3.9	Other	-0.1	-0.1	0.0	0.0	0.0
EBITDA	7.0	10.7	12.9	14.1	16.0	Effect of exchange rate changes	0.2	1.1	-6.8	-0.2	0.0
ROE	0.3%	4.7%	9.0%	4.4%	5.6%	Changes in cash and cash equivalents	-4.4	2.6	-4.5	2.6	0.7
No of average shares (mn)	34.1	37.1	37.1	37.1	37.1	Cash and cash equivalents at end of the period	26.6	29.0	24.5	27.1	27.8
EPS (JPY)	7.6	106.4	214.1	105.0	137.3	FCF	-15.8	2.0	7.5	3.7	4.4
BPS (JPY)	2,291.0	2,385.1	2,369.8	2,418.7	2,506.0	Sales by customer region					
DPS (JPY)	40.0	32.0	63.0	45.0	52.5	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	152.6	151.1	156.4	155.0	155.0	Japan	16.5	18.3	22.2	23.3	24.5
Balance Sheet						North America	11.0	11.5	14.1	14.0	14.4
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Asia	24.5	28.7	37.1	40.7	44.7
Total assets	141.4	151.6	148.4	151.4	153.1	Europe	12.1	13.8	16.1	16.2	16.6
Current assets	57.5	62.7	62.8	66.6	69.0	(YoY)					
Cash and cash equivalents	26.6	29.0	24.5	27.1	27.8	Japan	-9%	11%	21%	5%	5%
Accounts receivable	13.8	16.0	19.8	20.8	22.1	North America	-3%	5%	22%	0%	3%
Inventories	13.4	13.5	14.2	14.5	14.8	Asia	5%	17%	29%	10%	10%
Other financial assets	3.6	4.3	4.3	4.3	4.3	Europe	2%	14%	17%	1%	2%
Fixed assets	83.9	88.8	85.6	84.7	84.1	Sales by customer region					
Tangible fixed assets	71.5	74.5	71.3	70.5	69.8	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Intangible fixed assets	3.2	3.0	2.9	2.9	2.9	Automotive	34.7	37.4	-	-	-
Other non-current assets	9.2	11.3	11.3	11.3	11.3	Industrial	7.1	8.5	-	-	-
Current liabilities	15.9	24.8	23.2	22.1	21.1	Home appliance	3.1	3.2	-	-	-
Short term borrowings	3.3	9.9	7.5	6.2	4.9	Communication	4.5	5.6	-	-	-
Accounts payable	5.0	3.2	4.0	4.2	4.5	Power supply	3.2	4.0	-	-	-
Other current liabilities	7.6	11.7	11.7	11.7	11.7	Others	11.7	13.5	-	-	-
Fixed liabilities	47.3	38.1	37.1	39.4	38.9	(YoY)					
Long term borrowings	41.2	35.5	34.5	36.8	36.3	Automotive	4%	8%	-	-	-
Other fixed liabilities	6.1	2.6	2.6	2.6	2.6	Industrial	-21%	20%	-	-	-
Net assets	78.1	88.6	88.0	89.8	93.1	Home appliance	1%	5%	-	-	-
Total liabilities and equity	141.4	151.6	148.4	151.4	153.1	Communication	12%	26%	-	-	-
						Power supply	10%	26%	-	-	-
						Others	-9%	16%	-	-	-

Risk Reward – KOA (6999.T)

Earnings recovery continues centered on AI/DCs but is milder than MTP targets

PRICE TARGET ¥2,400

Our price target is ¥2,400, based on a DCF valuation (risk-free rate 3.0%; beta 1.2; risk premium 3.2%; WACC 5.3%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥165; our price target equates to P/E 22.9x our F3/28e EPS of ¥105.0, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

RISK REWARD CHART



EQUAL-WEIGHT THESIS

The equity story centers on growing demand for value-added resistors used in AI/data center and xEV applications. While competition from Taiwanese suppliers is likely intensifying in the more commoditized automotive resistor market, we believe earnings should continue to recover, supported by rising demand for high-value products such as shunt resistors for AI/data center applications. We expect current MTP OP target to be difficult to achieve, given higher fixed costs associated with capacity expansion. We also assume that the pass-through of higher precious metal costs will continue, albeit gradually.

Risk Reward Themes

New Data Era: Positive
Pricing Power: Negative

View descriptions of Risk Rewards Themes [here](#).

BULL CASE	¥4,100	BASE CASE	¥2,400	BEAR CASE	¥1,500
Achieves medium-term earnings targets					
OP and EPS forecasts of ¥5.7bn/¥214.1 in F3/27, ¥7.9bn/¥129.3 in F3/28, ¥10.8bn/¥183.1 in F3/29, ¥12.8bn/¥220.8 in F3/30, and ¥14.8bn/¥258.5 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥260. The price equates to 31.7x our F3/28e EPS of ¥129.3 and 1.7x BPS of ¥2,442.9.					
Mild growth led by AI/DC resistors continues					
OP and EPS forecasts of ¥5.7bn/¥214.1 in F3/27, ¥6.8bn/¥105.0 in F3/28, ¥8.5bn/¥137.3 in F3/29, ¥9.2bn/¥150.8 in F3/30, and ¥9.9bn/¥164.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥165. Our PT equates to 22.9x our F3/28e EPS of ¥105.0 and 1.0x BPS of ¥2,402.7.					
Earnings remain sluggish as sales growth slows					
OP and EPS forecasts of ¥5.7bn/¥214.1 in F3/27, ¥5.7bn/¥88.9 in F3/28, ¥6.1bn/¥96.9 in F3/29, ¥5.6bn/¥88.9 in F3/30, and ¥4.8bn/¥75.4 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥75. The price equates to 16.9x our F3/28e EPS of ¥88.9 and 0.6x BPS of ¥2,402.5.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Oct 6 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley MUFG Securities Co., Ltd. and its affiliates (collectively, "Morgan Stanley").

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The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Sota Harashima; Shoji Sato.

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Within the last 12 months, Morgan Stanley has received compensation for investment banking services from IRISO Electronics, Kyocera, Mabuchi Motor, Nidec, Niterra.

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Global Stock Ratings Distribution

(as of September 30, 2026)

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Coverage Universe			Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

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Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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INDUSTRY COVERAGE: Electronic Components

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/07/2026)
Shoji Sato		
ALPS ALPINE (6770.T)	E (10/06/2026)	¥2,228
Hamamatsu Photonics (6965.T)	E (06/17/2026)	¥2,473
Ibiden (4062.T)	O (08/04/2026)	¥12,270
Kyocera (6971.T)	E (06/25/2020)	¥3,827
Mabuchi Motor (6592.T)	E (11/03/2022)	¥1,786
Minebea Mitsumi (6479.T)	E (10/23/2025)	¥3,847
Murata Manufacturing (6981.T)	O (11/26/2025)	¥8,683
Nidec (6594.T)	NR (09/05/2025)	¥2,165

Niterra (5334.T)	O (01/17/2024)	¥4,008
Taiyo Yuden (6976.T)	E (09/16/2026)	¥11,460
TDK (6762.T)	O (08/02/2022)	¥3,530
Sota Harashima		
CMK (6958.T)	E (02/28/2025)	¥825
Daishinku (6962.T)	U (06/17/2026)	¥900
Hirose Electric (6806.T)	O (07/10/2024)	¥25,795
IRISO Electronics (6908.T)	E (08/02/2022)	¥2,883
Japan Aviation Electronics Industry (6807.T)	E (01/17/2024)	¥2,777
KOA (6999.T)	E (06/17/2026)	¥2,512
Meiko Electronics (6787.T)	O (09/29/2026)	¥26,900
Nichicon (6996.T)	E (11/10/2021)	¥2,742
Nihon Dempa Kogyo (6779.T)	E (03/07/2024)	¥3,635
Nippon Chemi-Con (6997.T)	U (09/20/2024)	¥2,355

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* Historical prices are not split adjusted.